The spacetime Penrose inequality and enclosing area

OpenAI

September 27, 2026

Abstract

We prove the sharp spacetime Penrose inequality for smooth one-ended initial-data exteriors in every spatial dimension $n \geq 3$, under the dominant energy condition, weak future trapping, the stated differentiated decay and positive enclosing area. These hypotheses force the ADM energy-momentum to be future timelike. Area is the infimum over full enclosing cuts in the original metric. The proof prepares the asymptotic end, then couples a graph deformation with a conformal change to control scalar curvature, energy and the area of every enclosing cut. A minimizing enclosure reduces the comparison to a Riemannian Penrose inequality. In dimensions three and four, two metric derivatives and one tensor derivative suffice, and both positive enclosing area and future timelikeness follow from the hypotheses.

2020 Mathematics Subject Classification. Primary 53C21; Secondary 83C05, 35J62.

Keywords. Spacetime Penrose inequality, enclosing area, initial data, graph deformation, nonlinear elliptic equations.

Contents

1 The inequality and its geometric quantity **3**
- 1.1 Data, area and statements . . . 3
- 1.2 How the proof controls curvature, mass and area . . . 5
- 1.3 From time symmetry to spacetime data . . . 6

2 The prepared geometric comparison **7**
- 2.1 Prepared data and the geometric output . . . 7
- 2.2 The Riemannian input and the energy bound . . . 8
- 2.3 Returning to the original data . . . 9

3 Preparing an end with zero momentum **10**
- 3.1 Conformal changes and a vacuum model with matching charges . . . 11
- 3.2 Joining the original and model ends . . . 13
- 3.3 Changing the vacuum slice to zero momentum . . . 16
- 3.4 Repairing the annular constraint error . . . 16
- 3.5 Comparing every full enclosing cut . . . 18
- 3.6 A strict scalar perturbation . . . 18
- 3.7 Increasing the energy at fixed momentum . . . 22

4 A scalar identity that controls area and mass **23**
4.1 Filling the boundary . . . 24
4.2 The conformal variable and the graph height . . . 24
4.3 The scalar identity . . . 25
4.4 Coercivity with polynomial constants . . . 28
4.5 The penalized system and the lower barrier . . . 29
4.6 Scalar curvature on solutions and height derivatives . . . 31
4.7 The penalty flux and its ADM normalization . . . 32

5 Solving the filled equations **33**
5.1 Height comparison and the boundary lower bound . . . 35
5.2 A polynomial upper bound before ellipticity . . . 36
5.3 A scalar gradient estimate with a measurable axial coefficient . . . 39
5.4 A bounded scalar equation with quadratic gradient growth . . . 46
5.5 Regularity of the coupled fixed points . . . 48
5.6 The trace equation for an arbitrary trial . . . 50
5.7 A compact map and the two homotopies . . . 52
5.8 Exhaustion, end decay, and the energy flux . . . 53

6 Height separation and a weak minimizing enclosure **55**
6.1 A negative height on the whole filling . . . 56
6.2 Conformal area increase and exterior scalar curvature . . . 57
6.3 Perimeter minimization and detachment from a distance obstacle . . . 58
6.4 The exterior, its full boundary, and its area . . . 62

7 Weakly asymptotically flat data in three dimensions **64**
7.1 Initial data, normals, and enclosing area . . . 64
7.2 The one-ended exterior inequality . . . 66
7.3 A positive lower bound for every enclosing cut . . . 66
7.4 A strict conformal direction . . . 67
7.5 Compactness of enclosing minimizers . . . 71
7.6 Schwarzschild reference charges . . . 73
7.7 Moments and compact linear corrections . . . 74
7.8 Bending to a rest slice and repairing the deficit . . . 77
7.9 Approximation by strict data . . . 80
7.10 Future timelikeness from the exterior inequality . . . 81
7.11 Complete data with finitely many ends . . . 81

8 Weak asymptotics in four dimensions **83**
8.1 Initial data and enclosing cuts . . . 84
8.2 Positivity and removal of other ends . . . 85
8.3 Conformal changes and an energy-raising shell . . . 87
8.4 Matching the charges before making a rest end . . . 88
8.5 Rest-end replacement and comparison of all cuts . . . 89
8.6 Strict trapping and a positive energy margin . . . 94
8.7 Applying the prepared theorem and returning to the original end . . . 96
8.8 The general maximal DEC energy inequality . . . 98

1 The inequality and its geometric quantity

The Penrose inequality asks how much mass is required to surround a trapped region. Its origin is a test of cosmic censorship: a collapsing configuration should settle to a black hole, radiation should carry energy away, and the horizon area should not decrease. Penrose's null-shell argument made this comparison concrete [39]. An initial-data theorem must express it using only a Riemannian metric, a second fundamental form and a specified enclosing area.

The choice of area is part of the problem. A surface outside an apparent horizon can have smaller area than the horizon itself. Ben-Dov gave spherically symmetric examples satisfying the dominant energy condition for which the bound using the apparent horizon's own area fails [4]. Carrasco and Mars found a different obstruction for generalized apparent horizons: their counterexamples already have an outer area-minimizing generalized horizon [12]. Thus a mass–area statement must specify both its trapping condition and its enclosing class. We use the infimum of the areas of all full cuts separating the boundary from infinity. Every component counts, including any part coincident with the boundary. The mass is the Lorentz-invariant length of the ADM energy-momentum vector. These choices lead to the following precise formulation.

1.1 Data, area and statements

Definition 1.1 (Initial-data exterior)**.** Let $n \geq 3$ and let Ω be a smooth connected orientable n-manifold with nonempty compact smooth boundary S. The metric g and symmetric covariant two-tensor K are smooth up to S, and g is complete with S included. Outside a compact set, Ω is a single coordinate end $\{x \in \mathbb{R}^n : |x| > R_0\}$. Thus there are no additional ends hidden in the compact-complement condition.

Write R_g for scalar curvature and ∇ for the connection of g. Write $\tau = \operatorname{tr}_g K$. Our constraint normalization is

$$2\mu = R_g + (\operatorname{tr}_g K)^2 - |K|_g^2, \qquad J_i = \nabla^j (K_{ij} - (\operatorname{tr}_g K) g_{ij}). \tag{1.1}$$

The dominant energy condition is $\mu \geq |J|_g$. We require $\mu, |J|_g \in L^1(\Omega, dV_g)$. Along S the unit normal ν points into Ω, toward the end. Its future expansion is

$$\theta_+ = H_S + \operatorname{tr}_S K, \qquad H_S = \operatorname{div}_S \nu, \qquad \operatorname{tr}_S K = (g^{ij} - \nu^i \nu^j) K_{ij}.$$

The boundary is weakly future trapped when $\theta_+ \leq 0$ on every component. No condition is imposed on the other null expansion.

For coordinate tensors, $T = O_j(r^{-a})$ means $|\partial^\alpha T| \leq C_\alpha r^{-a-|\alpha|}$ for $|\alpha| \leq j$, where $r = |x|$. The decay orders will be specified in each theorem. Set $k = n - 1$ and $\omega = \omega_{n-1} = |\mathbb{S}^{n-1}|$. The Laplacian is $\Delta = \operatorname{div} \nabla$ and symmetrization includes a factor $1/2$. We require the finite ADM limits

$$E = \frac{1}{2(n-1)\omega_{n-1}} \lim_{R \to \infty} \int_{|x|=R} (\partial_j g_{ij} - \partial_i g_{jj}) n_\delta^i \, dA_\delta, \tag{1.2}$$

$$P_i = \frac{1}{(n-1)\omega_{n-1}} \lim_{R \to \infty} \int_{|x|=R} (K_{ij} - (\operatorname{tr}_g K) g_{ij}) n_\delta^j \, dA_\delta. \tag{1.3}$$

Here n_δ and dA_δ are Euclidean. If $E > |P|_\delta$, define $m = (E^2 - |P|_\delta^2)^{1/2}$.

Definition 1.2 (Full enclosing cuts)**.** A *full enclosing cut* is the entire intrinsic manifold boundary $\Gamma = \partial D$ of a connected smooth codimension-zero submanifold-with-boundary $D \subset \Omega$. We require

D to be closed in Ω, its manifold interior to lie in $\operatorname{int}\Omega$, and D to contain the whole sufficiently distant end. Its full boundary must be compact, smooth, embedded and two-sided. In particular, $D = \Omega$ is admissible and has boundary S. Define

$$A_{\min,g}(S) = \inf_D \operatorname{Area}_g(\partial D). \tag{1.4}$$

The use of intrinsic boundary ensures that coincident portions of S are counted. A cut need not be connected, trapped or minimal. The infimum need not be attained.

Theorem 1.3 (The enclosing-area Penrose inequality)**.** *Let (Ω^n, g, K) be an initial-data exterior as in Definition 1.1, with integrable constraint densities, finite ADM limits, dominant energy condition and weakly future trapped boundary. Suppose*

$$g - \delta = O_6(r^{-q}), \qquad K = O_5(r^{-1-q}), \qquad \frac{n-2}{2} < q < n-2,$$

and assume $E > |P|_\delta$ and $A_{\min,g}(S) > 0$. Then

$$m \geq \frac{1}{2}\left(\frac{A_{\min,g}(S)}{\omega_{n-1}}\right)^{\frac{n-2}{n-1}}. \tag{1.5}$$

The coefficient and exponent are sharp in each dimension $n \geq 3$.

Thus the enclosing-area form of the spacetime Penrose conjecture holds in this class. Neither outermostness nor outer area minimization of S is assumed. The second fundamental form is arbitrary, and no spin, vacuum or interior-topology hypothesis is imposed. The time-symmetric Schwarzschild–Tangherlini exterior of mass $m > 0$ has $A_{\min,g}(S) = \omega_{n-1}(2m)^{(n-1)/(n-2)}$ and attains equality.

Corollary 1.4 (Automatic future timelikeness in the strong-decay class)**.** *Assume all hypotheses of Theorem 1.3 except the inequality $E > |P|_\delta$. In particular, retain the one-ended exterior, the O_6/O_5 decay with $(n-2)/2 < q < n-2$, and $A_{\min,g}(S) > 0$. Then $E > |P|_\delta$, and (1.5) holds.*

The proof is given in Section 2, using the energy-increasing conformal perturbation of Lemma 3.6.

Theorem 1.5 (Weaker decay in dimensions three and four)**.** *Let $n = 3$ or $n = 4$. Let (Ω^n, g, K) be the smooth one-ended exterior of Definition 1.1, with integrable constraint densities, finite ADM limits, dominant energy condition and weakly future trapped boundary. Assume*

$$g - \delta = O_2(r^{-q}), \qquad K = O_1(r^{-1-q}), \qquad q > \frac{n-2}{2}.$$

Then $A_{\min,g}(S) > 0$, $E > |P|_\delta$, and (1.5) holds.

Here positive enclosing area and timelikeness are both conclusions. The decay assumptions above also suffice for $\operatorname{tr}_g K$. The two derivative regimes concern the original data. After preparation, both lead to ends with estimates at every fixed derivative order, allowing one common deformation theorem to prove the numerical bound.

There are also designated-end formulations for finitely many ends, with different area classes in dimensions three and four. In the three-dimensional complete-data formulation the chosen outer region may retain other ends; in four dimensions its connected outer domain excludes their distant tails. Sections 7 and 8 give their precise definitions and proofs. Their reduction to the one-ended inequality uses a one-sided comparison between the two enclosing infima. The one-ended statements above isolate the common numerical construction. Equality for the original data requires additional horizon assumptions and a separate variational argument.

1.2 How the proof controls curvature, mass and area

The proof has three geometric stages: prepare an end with zero momentum, construct a graph and conformal deformation on a smooth filling, and locate a minimizing enclosure in the original exterior. The enclosing area must survive both metric changes. End preparation (Section 3) replaces the distant end by a vacuum reference slice with matching charges, bends that slice to rest, and repairs the constraints while comparing every full enclosing cut. Its output has compactly supported K, strict dominant energy condition and strict boundary trapping. In dimensions three and four the weak-decay preparation provides the same differentiated output. The three-dimensional enclosing infima converge to the original one; in four dimensions the available lower comparison is sufficient. Each finite-stage datum enters the common deformation theorem.

Fix a prepared exterior (Ω, g_0, K_0), with energy E_0 and enclosing area $A_0 = A_{\min,g_0}(S)$. For each small $\varepsilon > 0$, the deformation theorem constructs a family of smooth metrics indexed by a large parameter N. Each metric has a minimizing frontier lying in the original exterior and separating its inner boundary from infinity, with area at least $e^{-(n-1)\varepsilon}A_0$, nonnegative scalar curvature outside that frontier, and energy at most E_0 plus an error tending to zero as $N \to \infty$. The Riemannian numerical inequality therefore gives

$$E_0 \geq \frac{1}{2}\left(\frac{e^{-(n-1)\varepsilon}A_0}{\omega_{n-1}}\right)^{\frac{n-2}{n-1}}$$

after $N \to \infty$. Letting $\varepsilon \downarrow 0$ proves the prepared inequality. The end-preparation limit follows these two numerical limits.

The scalar identity in Section 4 connects the curvature, energy and area estimates. We fill the inner boundary smoothly and solve for a graph height and a conformal factor on the resulting manifold without boundary. The conformal factor has a lower bound, protecting the area of every enclosing cut. The curvature identity separates the dominant energy term, coercive squares and a divergence. In contrast to the zero-divergence coupling, the divergence has a specified nonzero source term; its integral controls the change of mass. Compared with the Schoen–Yau and Bray–Khuri identities [41, 8], the trace equation and the divergence source are modified to enforce these simultaneous controls.

Section 5 proves global solvability for this coupled system. Height comparison and a barrier argument establish the lower conformal bound. Testing the scalar divergence equation and applying a Sobolev inequality on the graph, based on Michael–Simon [34], yield the complementary upper estimate before uniform ellipticity is available. Together these bounds control graph distortion and make the equations uniformly elliptic for each fixed N. Separate scalar regularity estimates treat the trace equation, whose distinguished direction is the solution's gradient, and the divergence equation, which has quadratic gradient growth. They supply the control needed for continuation and exhaustion.

The filling removes inner boundary flux, but a minimizing enclosure might enter the filled region. The height argument in Section 6 prevents this by a further conformal change, large deep inside the filling and small near infinity. Two-sided density estimates force the minimizing frontier away from the interior obstacle and into the band where scalar curvature has the correct sign. The detached enclosure therefore lies where the curvature and full-area estimates can be applied together.

For dimensions three through seven, Bray–Lee's numerical theorem supplies the Riemannian endpoint. In higher dimensions the detached frontier can have a singular set of dimension at

most $n-8$. The required endpoint is the minimizing-frontier theorem, including its conformal flow, separation, smoothing and limiting arguments. We use the complete companion argument of [37]; Section 2 quotes the precise mathematical input. Its numerical application uses the stated second-order asymptotic decay and weak-boundary hypotheses. The endpoint thus requires control of both the singular frontier and the asymptotically flat end.

Section 2 states the geometric output and assembles the numerical proof before these constructions. Sections 7 and 8 supply the weak-decay preparations and the distinct designated-end transfers. The companion *Boundary graph deformations for the spacetime Penrose inequality* [36] studies what changes when one keeps an actual inner boundary, including its signed flux, or retains a decaying tensor in a maximal-data construction. It provides alternative analytic tools. The companion *Equality and rigidity in the spacetime Penrose inequality* [38] returns to the original g and K through variation, a causal adjoint system and static classification.

1.3 From time symmetry to spacetime data

When $K = 0$, the dominant energy condition becomes $R_g \geq 0$ and a marginal boundary is minimal. Geroch introduced the connected-surface Hawking-mass monotonicity underlying smooth inverse mean curvature flow [20]. Jang and Wald related this method to the Penrose bound under the assumption that a classical flow runs from the inner boundary to asymptotically round spheres [26]; Huisken and Ilmanen recount these precursors in [24]. Two different flow methods then established the three-dimensional Riemannian inequality. Huisken and Ilmanen's weak inverse mean curvature flow allows jumps while preserving the Hawking mass comparison; its bound applies to each connected component of an outermost minimal boundary [24]. Bray's conformal flow changes the metric, preserving the horizon area and making mass nonincreasing, and proves the sharp bound for the total area of a possibly disconnected horizon [6]. Bray and Lee extended that method through dimension seven [9]. Their numerical theorem requires no spin assumption; the additional spin hypothesis in the published statement belongs to its equality conclusion.

Above dimension seven, minimizing frontiers may be singular. Bi and Zhu extend conformal flow to this setting. Their revised theorem treats almost-minimizing Caccioppoli boundaries with minimal regular part, outermost minimizing enclosure and a metric extending with the stated regularity across the boundary [5]. Their work supplies the ancestry of the separation, mass–capacity and convergence arguments used for the Riemannian endpoint here. The exact version we need allows all components of a locally minimizing and outer-minimizing frontier; its full proof is part of the Riemannian argument described below.

Nonzero K introduces a different difficulty: neither scalar curvature nor the boundary mean curvature has the required Riemannian sign. Jang introduced a graph equation for extending Geroch's positive-energy argument beyond time symmetry [25]. Schoen and Yau's minimal-surface argument established the three-dimensional Riemannian positive-mass theorem [40]. Their spacetime proof combined Jang's equation with conformal deformation and a treatment of graph blow-up to prove positive energy for general three-dimensional initial data [41]. The Penrose problem asks more of the deformation: a lower bound for enclosing area must survive along with the mass comparison. Malec and Ó Murchadha explained the obstruction to obtaining both controls from the direct classical-Jang and conformal-deformation scheme, even in spherical symmetry [32].

In spherical symmetry, Malec and Ó Murchadha proved the inequality for maximal data, and Hayward obtained it from monotonicity of the Misner–Sharp energy in untrapped regions without that maximality restriction [31, 23]. Bray and Khuri introduced a warped graph and

a generalized Schoen–Yau scalar-curvature identity [7, 8]. On solutions of the generalized Jang equation, the identity separates the nonnegative dominant-energy contribution and square terms from a divergence. It does not by itself solve the equations needed to control that divergence. They gave a new proof in spherical symmetry and proposed coupled systems for the general problem. Han and Khuri established existence and boundary blow-up for prescribed warping factors with suitable boundary and asymptotic behavior [22]. Jaracz later constructed spherical data for which one particular coupling, generalized Jang with zero divergence, has no smooth radial solution with the required positive warp and asymptotics [27]. That obstruction concerns the specified coupling and radial solution class.

Sharp spacetime results are also known beyond the spherical three-dimensional case. Bryden, Khuri and Sormani proved the sharp inequality and Schwarzschild rigidity in spherical symmetry in every spatial dimension [11]. Khuri and Kunduri proved a sharp spacetime inequality for $\mathrm{SU}(\ell+1)$-invariant data of dimension $2(\ell+1)$; their result for the full cohomogeneity-one class is Riemannian [29].

For nonsymmetric three-dimensional asymptotically flat data, Allen, Bryden, Kazaras and Khuri proved a universal invariant-mass bound with a suboptimal constant under their asymptotic hypotheses, including extra decay of $\mathrm{tr}_g K$ [2]. For each end, their statement uses at least one outermost collection of future or past marginal components and its least enclosing area. Theorems 1.3 and 1.5 give the sharp coefficient under the original-data hypotheses above, using full enclosing cuts for any weakly future trapped boundary.

Other nonsymmetric results concern different hypotheses and area quantities. Ellithy's inequality bounds the total area of outermost future apparent-horizon sections under a quasi-final-state condition on the future development, negative ingoing expansion, and a piecewise smooth horizon tube with finitely many area-nondecreasing jumps [19]. Dong's pure-trace theorem gives the bound for each connected generalized-horizon component [18]; his two-convex theorem assumes a connected outermost past apparent horizon [17]. In the latter class the stated asymptotic restrictions force zero ADM linear momentum, as noted in [17]. These area and tensor assumptions are distinct from the full enclosing cuts and arbitrary second fundamental forms in our statements. Mars surveys the underlying alternatives in formulating the initial-data Penrose problem [33].

2 The prepared geometric comparison

The deformation produces a Riemannian exterior while controlling two quantities: its energy from above and its full boundary area from below. These estimates give the numerical inequality for prepared data. End preparation then transfers that inequality to the original data.

2.1 Prepared data and the geometric output

Definition 2.1 (Prepared exterior)**.** Let (Ω^n, g_0, K_0) be a smooth connected orientable one-ended exterior with nonempty compact smooth boundary S, complete with S included and with compact complement of its Euclidean coordinate end. Use the constraint and ADM conventions of (1.1)–(1.3). Let ϱ be a smooth positive function on the closed exterior, equal to the coordinate radius on a distant tail. Require that, for some $0<\beta<1$ and $c>0$,

$$\begin{gathered}\operatorname{supp} K_0 \text{ is compact}, \qquad g_0-\delta = O_d(r^{2-n}) \text{ for every fixed } d,\\ R_{g_0}=O(r^{-n-\beta}), \qquad \mu_0-|J_0|_{g_0}\geq c\varrho^{-n-\beta}, \qquad H_S+\mathrm{tr}_S K_0<0 \text{ on every component of } S.\end{gathered} \tag{2.1}$$

The constraint densities are integrable. The ADM momentum is $P_0 = 0$; write E_0 for the energy. Finally the full-cut infimum, with every component and all contact counted as in Definition 1.2, is $A_0 = A_{\min,g_0}(S) > 0$.

The construction in Sections 4–6 fixes one prepared exterior, denoting its data by g, K, its energy by E and its enclosing area by a. Constants in that construction may depend on the chosen datum.

The output below may have a singular frontier. For an enclosing-set exterior, length distance is computed by rectifiable paths in the closed exterior and may equal $+\infty$. Completeness means completeness on each finite-distance equivalence class; every frontier point and the full perimeter remain included.

Theorem 2.2 (The geometric deformation)**.** *Fix a prepared exterior and $0 < \epsilon < 1$. For all sufficiently large integers N there is a connected enclosing-set exterior X_N, complete with its compact frontier Σ_N included, in a smooth Riemannian ambient metric $\widetilde{g}_N$. It has one Euclidean coordinate end with compact complement, and*

$$\begin{aligned} &R_{\widetilde{g}_N} \geq 0, \quad R_{\widetilde{g}_N} \in L^1(X_N), \quad \widetilde{g}_N - \delta = O_2(r^{2-n}), \\ &R_{\widetilde{g}_N} = O(r^{-n-\beta'}) \quad \textit{for some } \beta' > 0, \\ &E(\widetilde{g}_N) \leq E_0 + P_N(e^{-N\epsilon} + e^{-N\epsilon/2}), \\ &\mathcal{H}^{n-1}_{\widetilde{g}_N}(\Sigma_N) \geq e^{-(n-1)\epsilon} A_0. \end{aligned} \tag{2.2}$$

Here P_N is a polynomial in N, with coefficients depending on the fixed datum and ϵ. The metric is smooth through the frontier. The whole frontier is outer minimizing and locally perimeter minimizing as the boundary of its filled side. It is smooth embedded minimal away from a compact singular set of Hausdorff dimension at most $n-8$; for $3 \leq n \leq 7$ the singular set is empty. Its area counts every component.

The proof is completed in Section 6. The analytic construction in Sections 4 and 5 takes place on a smooth filling and yields the energy estimate before any frontier is selected. The height argument then gives a detached minimizing frontier and the lower bound for its full area.

2.2 The Riemannian input and the energy bound

The Riemannian input is Theorem 1.1 of the companion *Conformal flow and the Riemannian Penrose inequality with minimizing frontiers* [37], in the following form.

> Let X^n be a connected complete enclosing-set exterior, including its compact frontier Σ, with a metric γ smooth through that frontier. Suppose there is one Euclidean coordinate end with compact complement, $\gamma - \delta = O_2(r^{-\tau})$ with $\tau > (n-2)/2$, and $R_\gamma \geq 0$, $R_\gamma \in L^1$, and $R_\gamma = O(r^{-n-\beta})$ for some $\beta > 0$. Suppose its full frontier is outer minimizing and locally perimeter minimizing as the boundary of the filled side, allowing compactly supported variations on either side. Its regular part is a smooth embedded minimal hypersurface. Its singular set has Hausdorff dimension at most $n-8$. Then
>
> $$E(\gamma) \geq \frac{1}{2}\left(\frac{\mathcal{H}^{n-1}_\gamma(\Sigma)}{\omega_{n-1}}\right)^{(n-2)/(n-1)}.$$
>
> The assertion includes zero frontier area and imposes neither a spin condition nor connectedness of the frontier.

The smooth numerical theorem of Bray–Lee [9, Theorem 1.4] is an alternative in dimensions three through seven. In higher dimensions the companion retains the full singular-frontier argument. Its conformal-flow ancestry includes Bi–Zhu's version 2 proof of area preservation before nesting, and its singular separation arguments [5]. The independent smooth positive-mass input is Brendle–Wang [10, Corollary 1.6], applied there to constructed smooth complete boundaryless approximants with strictly positive scalar curvature and the required differentiated end. Those stronger hypotheses hold on the approximants constructed in the companion proof.

Theorem 2.3 (The prepared energy inequality)**.** *Every exterior in Definition 2.1 satisfies*

$$E_0 \geq \frac{1}{2}\left(\frac{A_0}{\omega}\right)^{(n-2)/(n-1)}. \tag{2.3}$$

Proof. Fix the datum and ϵ. The quoted Riemannian theorem applies to the exterior X_N furnished by Theorem 2.2. Its metric is smooth through the compact frontier; its closure is complete and its open exterior is connected. The end has compact complement and decay exponent $n-2 > (n-2)/2$. Its scalar curvature is nonnegative and integrable with the required pointwise decay. The whole frontier is outer minimizing and locally perimeter minimizing, hence has minimal regular part and singular dimension at most $n-8$. These are the hypotheses of the Riemannian input. We obtain

$$E_0 + P_N(e^{-N\epsilon} + e^{-N\epsilon/2}) \geq E(\widetilde{g}_N) \geq \frac{1}{2}\left(\frac{\mathcal{H}^{n-1}_{\widetilde{g}_N}(\Sigma_N)}{\omega}\right)^{(n-2)/(n-1)} \geq \frac{1}{2}\left(\frac{e^{-(n-1)\epsilon}A_0}{\omega}\right)^{(n-2)/(n-1)}.$$

As $N \to \infty$, the exponential factors dominate the fixed polynomial. The subsequent limit $\epsilon \downarrow 0$ gives the stated inequality. The exhaustion has already produced a smooth solution at each fixed N; only its numerical bounds enter these two limits. Since $A_0 > 0$, this also proves $E_0 > 0$. □

2.3 Returning to the original data

Deduction of Theorem 1.3. Proposition 3.1, proved in the next section, supplies prepared data with energies $E_j \to m = \sqrt{E^2 - |P|^2}$ and enclosing infima $A_j \geq (1 - o(1))A_{\min,g}(S)$. Apply Theorem 2.3 to each prepared datum, completing all its fixed-data limits first. Then

$$m \geq \frac{1}{2}\left(\frac{A_{\min,g}(S)}{\omega}\right)^{(n-2)/(n-1)}.$$

Each preparation index has its own completed fixed-data comparison.

For sharpness, let $M > 0$, $r_M = (2M)^{1/(n-2)}$, and take the static Schwarzschild–Tangherlini slice

$$g = \frac{dr^2}{1 - 2M/r^{n-2}} + r^2\sigma_{n-1}, \quad K = 0, \qquad r \geq r_M.$$

Proper distance from the horizon makes this a smooth complete metric with minimal compact boundary. Its scalar curvature is zero, and its ADM vector is $(M, 0)$. Radial projection to the sphere of radius r_M has $(n-1)$-Jacobian at most one. More explicitly the pullback of that sphere's area form is closed with comass at most one; Stokes' theorem on an end truncation of any full outer domain shows that its integral over the entire cut is ωr_M^{n-1}. Thus every full cut has area at least that number, while the boundary itself attains it. Consequently $A_{\min} = \omega(2M)^{(n-1)/(n-2)}$ and equality holds. □

The same numerical bound forces future timelikeness once a positive enclosing area is retained under an energy-raising conformal change.

Proof of Corollary 1.4. If instead $E \le |P|_\delta$, choose $\varepsilon_j > 0$ tending to zero and put

$$\lambda_j = \frac{|P|_\delta + \varepsilon_j - E}{2} > 0.$$

Lemma 3.6, proved in the final subsection of Section 3, gives, separately for each j, data satisfying the same one-ended strong-decay, constraint and trapping hypotheses, with

$$E_j = |P|_\delta + \varepsilon_j, \qquad P_j = P, \qquad A_{\min,g_j}(S) \ge A_{\min,g}(S) > 0.$$

Their ADM vectors are future timelike, so Theorem 1.3 gives

$$\sqrt{(|P|_\delta + \varepsilon_j)^2 - |P|_\delta^2} \ge \frac{1}{2}\left(\frac{A_{\min,g}(S)}{\omega_{n-1}}\right)^{\frac{n-2}{n-1}} > 0.$$

The right side is fixed and the left side tends to zero, a contradiction. The theorem applies separately to each datum, with the end-preparation parameters allowed to depend on j. Thus $E > |P|_\delta$, and Theorem 1.3 applies to the original data. □

The proofs for weak original decay appear in Sections 7 and 8. Their reductions reach Definition 2.1 by different means: three-dimensional enclosing areas converge fully, whereas the four-dimensional rest replacement gives the one-sided comparison needed by the numerical argument. The latter is followed by strictification at fixed replacement radius, with that strictification removed before the radius tends to infinity.

3 Preparing an end with zero momentum

Put $k = n - 1$ and $p = n - 2 = k - 1$. A radius function ϱ below is smooth and positive on the closed exterior and equals the Euclidean radius on a sufficiently distant part of its coordinate end. Constants in the end-preparation argument may depend on the original data and on its fixed future-timelike ADM vector, with no uniform estimate as $E - |P| \downarrow 0$. The final subsection gives an independent conformal perturbation that increases the energy at fixed momentum. It applies without a timelike assumption and is used in Section 2 to prove automatic timelikeness.

We construct the data of Definition 2.1 while comparing every full cut of the new metric with a full cut of the original metric.

Proposition 3.1 (Reduction to a prepared exterior)**.** *Let (Ω, g, K) be a smooth one-ended exterior as in Definition 1.1, with compact complement of its coordinate end and with*

$$g - \delta = O_6(r^{-q}), \qquad K = O_5(r^{-1-q}), \qquad \frac{n-2}{2} < q < n - 2.$$

Assume integrable constraint densities satisfying $\mu \ge |J|_g$, finite ADM limits satisfying $E > |P|$, and $\theta_+(S) \le 0$ on every boundary component. Suppose that the full-cut infimum of Definition 1.2 is $a = A^g_{\min}(S) > 0$, and put $m = (E^2 - |P|^2)^{1/2}$. There are prepared data (g_j, K_j) on the same exterior, with one fixed $\beta \in (0,1)$, numbers $\epsilon_j \downarrow 0$, and proper diffeomorphisms $\Phi_j : \Omega \longrightarrow \Omega$ equal to the identity near S, such that

$$P_j = 0, \qquad E_j \longrightarrow m, \qquad \Phi_j^* g \le (1+\epsilon_j) g_j, \qquad A^{g_j}_{\min}(S) \ge (1+\epsilon_j)^{-k/2} a. \tag{3.1}$$

The positive constant c_j in (2.1) and the end coordinates may depend on j.

We first match the original energy and momentum with a boosted vacuum end. A compact annular correction joins the two ends with a constraint error whose total scale tends to zero. Within the vacuum region we then bend the slice to a static one, giving zero momentum. A positive conformal factor repairs the small annular error without decreasing the metric. A proper radial comparison map protects every full enclosing cut during the change of slice. For each already fixed replacement, a second scalar perturbation makes the energy condition and future trapping strict while changing the energy by a quantity tending to zero. The proof of Proposition 3.1 is completed after these constructions.

An asymptotic change to a rest frame is also used by Allen, Bryden, Kazaras and Khuri in the invariant-mass step of [2, Section 5]. Here the end is first replaced by an explicit vacuum model, and the proper comparison map in Proposition 3.1 controls every full cut during the replacement and bending.

3.1 Conformal changes and a vacuum model with matching charges

The conformal identities below explain how a scalar function can improve the constraint inequality and the boundary expansion at the same time. We then compute the vacuum model's charges directly in the ADM normalization used here.

For a smooth function s set $g_s = e^{2s}g$ and $K_s = e^sK$. If $|v|_g \le 1$, the corresponding vector in the new unit ball is $v_s = e^{-s}v$. The conformal connection formula and scalar curvature formula give

$$\begin{aligned} e^{2s}(\mu_s + J_s(v_s)) &= \mu + J(v) - k\Delta_g s - \frac{kp}{2}|ds|_g^2 + kK(\nabla s, v), \\ e^s\theta_{+,s} &= \theta_+ + k\partial_\nu s. \end{aligned} \tag{3.2}$$

Here the normal ν points into the exterior. The constituent transformation formulas are $\tau_s = e^{-s}\tau$, $|K_s|_{g_s}^2 = e^{-2s}|K|_g^2$, and

$$R_{g_s} = e^{-2s}(R_g - 2k\Delta_g s - kp|ds|_g^2).$$

Writing $\pi = K - \tau g$, contraction of $\Gamma(g_s)_{ij}^l - \Gamma(g)_{ij}^l = s_i\delta_j^l + s_j\delta_i^l - g_{ij}s^l$ yields $J_{s,i} = e^{-s}(J_i + kK_{ij}\nabla^j s)$. The tangential trace of K scales by e^{-s} and $H_s = e^{-s}(H + k\partial_\nu s)$, proving both identities. In particular, if $s = 2p^{-1}\log U$, $U > 0$, the entire added term in the first identity is

$$\frac{2k}{pU}(-\Delta_g U + K(\nabla U, v)). \tag{3.3}$$

Lemma 3.2 (Charges of a boosted Schwarzschild–Tangherlini end)**.** *With $p = n-2$ and $k = n-1$, consider the Schwarzschild–Tangherlini spacetime of geometric mass $M > 0$. Let b, y be static time and isotropic spatial coordinates. Make the Lorentz change*

$$b = \gamma(T + vx^1), \qquad y^1 = \gamma(x^1 + vT), \qquad y^a = x^a \ (2 \le a \le n), \qquad \gamma = (1-v^2)^{-1/2}, \tag{3.4}$$

where $0 \le v < 1$. The induced data on $T = 0$, oriented by the future normal, have

$$E = \gamma M, \qquad P_1 = \gamma M v, \qquad P_a = 0 \ (a > 1). \tag{3.5}$$

In the coordinates (3.4), the homogeneous leading metric perturbation is the spatial part of (3.7), and the homogeneous leading second fundamental form is (3.9), with the quantities defined in (3.6). Their remainders are $O_d(r^{-2p})$ and $O_d(r^{-2p-1})$, respectively, for every fixed derivative order d.

Proof. We compute the leading energy and momentum fluxes, then integrate them over coordinate spheres. The remainder estimates show that these leading terms determine the ADM charges.

Energy flux. The isotropic metric is

$$-\left(\frac{1-M/(2|y|^p)}{1+M/(2|y|^p)}\right)^2 db^2+\left(1+\frac{M}{2|y|^p}\right)^{4/p}|dy|^2.$$

Its leading perturbation of Minkowski space is $2M|y|^{-p}(db^2+p^{-1}|dy|^2)$. On $T=0$ put

$$\rho^2=\gamma^2(x^1)^2+\sum_{a=2}^{n}(x^a)^2,\qquad f=\rho^{-p},\qquad A=\gamma^2(v^2+p^{-1}),\quad B=p^{-1},\quad C=kp^{-1}\gamma^2 v. \tag{3.6}$$

The nonzero relevant leading components of the spacetime perturbation H are

$$\mathsf{H}_{11}=2MAf,\qquad \mathsf{H}_{aa}=2MBf,\qquad \mathsf{H}_{01}=2MCf. \tag{3.7}$$

Moreover

$$\partial_1 f=-p\gamma^2x^1\rho^{-n},\qquad \partial_a f=-px^a\rho^{-n},\qquad \partial_T f=-p\gamma^2vx^1\rho^{-n}.$$

Thus the energy flux vector $Q_i=\partial_j\mathsf{H}_{ij}-\partial_i\mathsf{H}_{jj}$ satisfies

$$Q_1=-2MkB\partial_1 f=2Mk\gamma^2x^1\rho^{-n},\qquad Q_a=-2M(A+pB)\partial_a f=2Mp(A+1)x^a\rho^{-n}.$$

Since $p(A+1)=k\gamma^2$, these expressions give the pointwise identity

$$Q_i=2Mk\gamma^2x_i\rho^{-n}. \tag{3.8}$$

Momentum flux. The chosen sign of the second fundamental form gives, to first order,

$$K_{ij}=\tfrac{1}{2}(\partial_T\mathsf{H}_{ij}-\partial_i\mathsf{H}_{0j}-\partial_j\mathsf{H}_{0i}).$$

Direct substitution gives

$$\begin{aligned} &K_{11}=M\gamma^4v(2p+1-pv^2)x^1\rho^{-n}, && K_{aa}=-M\gamma^2vx^1\rho^{-n},\\ &K_{1a}=Mk\gamma^2vx^a\rho^{-n}, && K_{ab}=0\quad(a\neq b). \end{aligned} \tag{3.9}$$

Their Euclidean trace is $M\gamma^4v(p+v^2)x^1\rho^{-n}$. Consequently, for the leading Euclidean trace reversal $\pi_{ij}=K_{ij}-(\operatorname{tr}_\delta K)\delta_{ij}$,

$$\pi_{1j}=Mk\gamma^2vx_j\rho^{-n},\qquad \pi_{a1}=Mk\gamma^2vx^a\rho^{-n},\qquad \pi_{ab}=-Mk\gamma^4vx^1\rho^{-n}\delta_{ab}. \tag{3.10}$$

Using the exact normal and the exact metric in the trace reversal changes these formulas by $O_d(r^{-2p-1})$. The energy remainder has first derivatives of this same order. Their sphere integrals are $O(r^{-p})$, and therefore vanish for every $n\geq 3$.

Integration. The ellipsoid $\{|\operatorname{diag}(\gamma,1,\dots,1)x|<1\}$ has volume $\omega/(n\gamma)$. Polar integration therefore gives

$$\int_{S^{n-1}}|\operatorname{diag}(\gamma,1,\dots,1)\theta|^{-n}\,d\theta=\frac{\omega}{\gamma}. \tag{3.11}$$

Integrating (3.8) and (3.10) and dividing by $2k\omega$ and $k\omega$ gives $E=M\gamma$ and $P_1=M\gamma v$. The integrand of a transverse momentum row is a constant times $x^1x^a\rho^{-n}/r$, whose integral vanishes by reflection. □

For the original charges we now fix $M=m$, $v=|P|/E$, and rotate the first axis to the direction of P when $P\neq 0$. Lemma 3.2 gives a model end $(g^{\mathrm{b}},K^{\mathrm{b}})$ with exactly the same charges. The parameter v is fixed in all limits that follow.

3.2 Joining the original and model ends

A direct cutoff need not preserve the dominant energy condition. Our immediate task is to remove its leading constraint error by compactly supported corrections. The obstructions to a compact correction are finitely many moments. Charge matching removes their constant limits, and integrability makes the remaining first moments small enough after rescaling.

Localized constraint deformation and the finite-dimensional obstruction from the adjoint kernel are central to the gluing methods of Corvino–Schoen [14] and Chruściel–Delay [13]. We construct the replacement directly for possibly nonvacuum data without parity conditions or convergent center of mass and angular momentum. The construction must preserve both the energy inequality and every full enclosing cut. On Euclidean tensors define

$$\mathsf{S}(h) = \partial_i\partial_j(h_{ij} - (\operatorname{tr}_\delta h)\delta_{ij}), \qquad \mathsf{D}(b)_i = \partial_j(b_{ij} - (\operatorname{tr}_\delta b)\delta_{ij}). \tag{3.12}$$

Their formal adjoint kernels relevant to compact support are, respectively, affine scalar functions and Euclidean Killing fields.

Lemma 3.3 (Compact right inverses)**.** *Let* S, D *be the Euclidean operators in* (3.12)*. Let* $\mathcal{A} \subset \mathbb{R}^n$ *be a bounded connected Euclidean annulus and fix a compact subset of its interior. Smooth scalar sources supported in that subset with zero constant and first moments have smooth symmetric solutions of* $\mathsf{S}(h) = f$ *supported in a larger fixed compact subset of* $\mathcal{A}$*. Smooth vector sources with zero integrals against every Euclidean Killing field have similarly supported symmetric solutions of* $\mathsf{D}(b) = F$*. For* $1 < a < \infty$ *and integers* $d \geq 0$*, these solutions may be chosen linearly with estimates*

$$\|h\|_{W^{d+2,a}} \leq C_{d,a}\|f\|_{W^{d,a}}, \qquad \|b\|_{W^{d+1,a}} \leq C_{d,a}\|F\|_{W^{d,a}}.$$

For arbitrary sources the same conclusions hold after subtracting fixed smooth moment bumps, whose coefficients are bounded by the absolute values of the respective moments.

Proof. A compact scalar divergence inverse gives the scalar tensor equation by two successive inversions. For the vector equation, a correction of the skew part preserves the divergence and makes the tensor symmetric.

The scalar divergence inverse. On a connected open set, a mean-zero smooth function supported in a fixed compact set has a compactly supported vector primitive, with one Sobolev derivative gained. The support and Sobolev mapping statements are given, for example, by the regularized Bogovskii operator in [15, Section 3.1, Equation (3.13), Theorem 3.2, Corollaries 3.3–3.4, Remark 3.5, and Proposition 4.1(ii)]. To see the support property locally, on a ball choose a smooth unit-integral function supported in a smaller concentric ball and use the Bogovskii integral

$$(\mathcal{B}f)(x) = \int f(y)(x-y)\int_1^\infty \eta(y + t(x-y))t^{n-1}\,dt\,dy.$$

Differentiation in distributions gives $\operatorname{div}\mathcal{B}f = f - \eta\int f = f$; the segments occurring in the integral remain in the ball. For compact input support and compact $\operatorname{supp}\eta$, their union is compact in that ball. The kernel is of order $1-n$, and its differentiated kernel has the cancellation of a Calderon–Zygmund kernel. The resulting $W^{d,a}$ estimate is the usual boundedness of that singular integral and its commutators with derivatives. A finite cover of the prescribed support by balls compact in $\mathcal{A}$ can be connected by a finite chain of overlapping such balls. A partition of unity decomposes the source, and unit-integral bumps in the overlaps transfer the means along the chain. Each resulting ball source has mean zero; the sum of its ball primitives proves the stated inverse on $\mathcal{A}$, with fixed support and constants.

The scalar tensor equation. For a scalar f with its affine moments zero, the divergence inverse gives $\partial_i v_i = f$. Compact support gives $\int v_i = -\int z_i f = 0$, so a second application, to each v_i, gives $\partial_j T_{ij} = v_i$. Then $Q = (T + T^{\mathsf{t}})/2$ has $\partial_i \partial_j Q_{ij} = f$. The tensor $h = Q - k^{-1}(\mathrm{tr}_\delta Q)\delta$ has trace reversal Q, so $\mathsf{S}(h) = f$ with the asserted two-derivative gain.

The symmetric divergence equation. For a vector source F, the translation moments permit the row-by-row solution of $\partial_j B_{ij} = F_i$. The rotation moments give

$$0 = \int (z_i F_j - z_j F_i) = \int (B_{ij} - B_{ji}).$$

Put $A_{ij} = (B_{ij} - B_{ji})/2$ and solve $\partial_l D_{ijl} = -A_{ij}$, choosing $D_{ijl} = -D_{jil}$. Set

$$E_{ij} = \partial_l (D_{ijl} - D_{ilj} - D_{jli}), \qquad C_{ij} = B_{ij} + E_{ij}.$$

Index interchange gives $E_{ij} - E_{ji} = -2A_{ij}$ and $\partial_j E_{ij} = 0$: the first two double derivatives cancel after interchanging j, l, and the last vanishes by skew symmetry in those indices. Thus C is symmetric and $\partial_j C_{ij} = F_i$. Taking $b = C - k^{-1}(\mathrm{tr}_\delta C)\delta$ proves the assertion, including the derivative count.

Removing moments. Take a nonnegative smooth bump supported in an open ball of $\mathcal{A}$. The Gram matrix of the affine functions, or of the vector Killing fields, with this weight is positive definite: a polynomial in either finite-dimensional space vanishing on that ball vanishes identically. Multiplying the basis elements by the bump and the inverse Gram matrix produces fixed dual moment bumps. Subtraction removes all moments with the stated bound on coefficients. □

The preceding inverses reduce the joining problem to estimates for the moments that they cannot remove. We use the following annuli and a smooth cutoff χ. For L large, set

$$\begin{aligned} &\mathcal{A} = \{z \in \mathbb{R}^n : 1 < |z| < 4\}, \qquad 0 \le \chi \le 1, \\ &\chi = 1 \text{ near } |z| = 1, \qquad \chi = 0 \text{ near } |z| = 4, \\ &\chi_L(x) = \chi(x/L) \text{ on } \mathcal{A}_L = \{L < |x| < 4L\}. \end{aligned} \tag{3.13}$$

Sublinear growth of the first moments is sufficient to make all obstructions small after rescaling.

Lemma 3.4 (Sublinear first moments)**.** *Assume the hypotheses of Proposition 3.1. Choose the boosted data $(g^{\mathrm{b}}, K^{\mathrm{b}})$ of Lemma 3.2 with $M = (E^2 - |P|^2)^{1/2}$ and $v = |P|/E$, rotating the first axis to the direction of P when $P \neq 0$, so that their ADM charges match those of (g, K). Use the operators in (3.12) and the annuli and cutoff in (3.13). Let $h = g - \delta$, $h^{\mathrm{b}} = g^{\mathrm{b}} - \delta$, and $b = K$, $b^{\mathrm{b}} = K^{\mathrm{b}}$ on their common coordinate tail. Put $H = h - h^{\mathrm{b}}$ and $B = b - b^{\mathrm{b}}$. Define the scalar and vector commutators on $\mathcal{A}_L$ by*

$$C_L = \mathsf{S}(\chi_L H) - \chi_L \mathsf{S}(H), \qquad F_L = \mathsf{D}(\chi_L B) - \chi_L \mathsf{D}(B),$$

and their rescaled versions on $\mathcal{A}$ by

$$\widehat{C}_L(z) = L^2 C_L(Lz), \qquad \widehat{F}_L(z) = L^2 F_L(Lz).$$

The flat sources $\mathsf{S}(H)$ and $\mathsf{D}(B)$ are integrable. For every fixed affine scalar function $a(z)$ and every fixed Euclidean Killing field $Y(z)$,

$$\int_{\mathcal{A}} a \widehat{C}_L \, dz = o(L^{2-n}), \qquad \int_{\mathcal{A}} Y^i \widehat{F}_{L,i} \, dz = o(L^{2-n}).$$

Proof. Integrability controls the first moments on expanding annuli; matching charges controls the constant moments. We estimate both in the original coordinates before rescaling.

Integrability and first moments. Expansion of the constraint formulas about $(\delta, 0)$ gives

$$\mathsf{S}(h) = 2\mu + O(r^{-2q-2}), \qquad \mathsf{D}(K) = J + O(r^{-2q-2}). \tag{3.14}$$

The remainder estimates follow, respectively, from hD^2h, $(Dh)^2$, and K^2, and from hDK and $Dh\,K$. The volume densities of g and δ are uniformly comparable on the tail. Hence integrability of μ, J, together with $2q > n-2$, gives the claim for the original data. The model is vacuum with faster decay, so its flat sources are integrable too.

For any integrable tail function f, even without an integrable first absolute moment,

$$\int_{R_0<r<4L} r|f|\,dx = o(L). \tag{3.15}$$

Indeed, after division by L, split the integral at a fixed large radius R_1. The inner integral divided by L tends to zero, whereas the outer integral divided by L is at most $4\int_{r>R_1}|f|$, which tends to zero as $R_1 \to \infty$.

Scalar fluxes. Write $Q = H - (\mathrm{tr}_\delta H)\delta$. For an affine function $a(x)$ define its Green flux by

$$\mathcal{B}_a(R) = \int_{S_R} \left(a\,\partial_j Q_{ij}\nu_\delta^i - (\partial_i a) Q_{ij}\nu_\delta^j\right) dA_\delta.$$

Integration by parts between spheres yields $\mathcal{B}_a(R) - \mathcal{B}_a(R_0) = \int_{R_0<r<R} a\,\mathsf{S}(H)$. Charge matching gives $\mathcal{B}_1(R) \to 0$, and (3.15) gives $\mathcal{B}_{x^a}(R) = o(R)$. For the scalar commutator C_L, supported where the cutoff varies, one consequently has

$$\int_{L<r<4L} aC_L = -\mathcal{B}_a(L) - \int_{L<r<4L} \chi_L a\,\mathsf{S}(H). \tag{3.16}$$

This is $o(1)$ for $a = 1$ and $o(L)$ for $a = x^a$.

Momentum fluxes. Put $Q_{ij} = (K - K^{\mathrm{b}})_{ij} - \mathrm{tr}_\delta(K - K^{\mathrm{b}})\delta_{ij}$. For a Euclidean Killing field Y, symmetry of Q gives $Q_{ij}\partial_j Y_i = 0$. Its Green flux is thus simply $\int_{S_R} Y_i Q_{ij}\nu_\delta^j$. For constant Y its limit is zero by equality of the momenta; replacing geometric trace reversal by Euclidean trace reversal changes the sphere flux by $O(R^{n-2-2q}) \to 0$. For rotational Y the same Green identity and (3.15) give $o(R)$. The identity (3.16) holds for F_L with this flux and pairing.

Rescaling. Under $x = Lz$ the fields are $h(Lz)$ and $LK(Lz)$, and both flat sources are multiplied by L^2. A constant moment is therefore multiplied by L^{2-n} and a first moment by L^{1-n}. The preceding $o(1)$ and $o(L)$ bounds prove the same $o(L^{2-n})$ bound for all rescaled moments. □

The two lemmas give a joining construction whose remaining constraint error tends to zero at the required scale. Under $x = Lz$, the cutoff χ interpolates the scaled fields on $\mathcal{A}$ between the original and model fields. The errors in $(2\mu, J)$ relative to the cutoff interpolation of the exact constraint densities equal $\widehat{C}_L$ and $\widehat{F}_L$, respectively, plus $O(L^{-2q})$ in scaled units. The commutators are supported in the fixed compact subset where the cutoff varies. By Lemma 3.3, compact corrections cancel these commutators after their moment bumps have been removed. The commutators have $W^{1,a}$ norm $O(L^{-q})$, for any fixed $a > n$, because the available asymptotic derivatives include those of order three for g and two for K. Sobolev embedding in dimension n therefore gives corrections of size $O(L^{-q})$ in $C^{2,\alpha}$ for the scaled metric and in $C^{1,\alpha}$ for the scaled second fundamental form, where $0 < \alpha < 1 - n/a$.

These are smooth compact corrections; only their first few norms need be bounded uniformly in L. Their nonlinear contribution is still $O(L^{-2q})$. The fixed smooth moment bumps contribute $o(L^{2-n})$ by Lemma 3.4. Thus, returning to physical units, the new data $(\widetilde{g}_L, \widetilde{K}_L)$ agree with the original for $r \le L$ and with the model for $r \ge 4L$, and satisfy

$$\widetilde{\mu}_L - |\widetilde{J}_L|_{\widetilde{g}_L} \ge -d_L L^{-n}\, \mathbf{1}_{\{L<r<4L\}}, \qquad d_L \longrightarrow 0. \tag{3.17}$$

To check the unit balls in this assertion, the interpolated metric, the original metric, and δ differ by $O(L^{-q})$. The scaled original J is $O(L^{-q})$. Transporting a unit vector between these balls thus changes its pairing with the interpolated J by $O(L^{-2q})$. The cutoff is between zero and one, so the original DEC and the vacuum model give the claimed sign after precisely the errors already estimated.

3.3 Changing the vacuum slice to zero momentum

We have joined the original data to a vacuum end with matching charges. The only possible failure of the dominant energy condition is bounded by (3.17). Changing the slice entirely within the vacuum end removes the momentum while preserving the vacuum constraints there.

In the model coordinates, $T = 0$ is the graph $b = vy^1$. Choose a smooth cutoff ζ equal to one initially and zero finally, with

$$\left|\frac{d\zeta}{d\log|y|}\right| \le \eta_v, \qquad v(1+\eta_v) < 1.$$

This can be done by taking a sufficiently long, but fixed, interval in $\log(|y|/L)$. Start it outside $|y| = 4\gamma L$, where the joined data are exactly the model. Replace the graph by $b = vy^1\zeta(\log(|y|/L))$ there. Its Euclidean slope is at most $v(1+\eta_v) < 1$; the Schwarzschild coefficients tend uniformly to their flat values on the transition as $L \to \infty$. Thus this is a smooth spacelike hypersurface for large L, and its induced data are exactly vacuum throughout the bending region. They agree with $b = 0$ beyond $|y| = C_v L$, for a fixed C_v. For $v = 0$ no bending is necessary.

The coordinates $y = \mathrm{diag}(\gamma, 1, \dots, 1)x$ parametrize the whole construction outside the unchanged core. The resulting metric and second fundamental form, denoted $(\bar{g}_L, \bar{K}_L)$, satisfy, in this region,

$$c_v\delta \le \bar{g}_L \le C_v'\delta, \qquad |D\bar{g}_L| + |\bar{K}_L| \le C_v'/r. \tag{3.18}$$

On the joining annulus the stronger small perturbation estimates hold; in the bending region these bounds follow by differentiating the graph and its normal. The tensor $\bar{K}_L$ is compactly supported, and the only possible DEC deficit is (3.17).

3.4 Repairing the annular constraint error

The slice is now static near infinity and its second fundamental form has compact support. It remains to repair the annular deficit. By (3.3), it is enough to build a positive function whose negative Laplacian dominates the drift $|\bar{K}_L|\,|dU_L|$ and the deficit. The function must be constant on the original core, so the boundary expansion keeps its sign.

We construct a nonnegative radial function $f_L(s)$, $s = |y|/L$, with bounded derivatives on every fixed transition interval, so that

$$U_L = 1 + e_L L^{-p} f_L(|y|/L), \qquad e_L \downarrow 0, \tag{3.19}$$

repairs that deficit. Choose a smooth nonnegative χ_* equal to one on the radial interval containing the full joining annulus and supported in a slightly larger interval. Begin before that interval with $z_L = 0$, and solve

$$z_L' = C_* z_L + \chi_*, \qquad z_L = -f_L'.$$

Continue this equation through the bending interval. The interval and C_* are fixed for the chosen boost. Thus z_L and its derivatives are bounded independently of large L. The radial function has $|ds|^2_{\bar{g}_L} \geq cL^{-2}$, $|\Delta_{\bar{g}_L} s| \leq CL^{-2}$, and $|\bar{K}_L||ds|_{\bar{g}_L} \leq CL^{-2}$ there, by (3.18). Consequently

$$\begin{aligned} -\Delta_{\bar{g}_L} U_L - |\bar{K}_L|\,|dU_L|_{\bar{g}_L} &= e_L L^{-p}(z_L'|ds|^2 + z_L \Delta s - |\bar{K}_L| z_L |ds|) \\ &\geq c' e_L L^{-n} \chi_* \end{aligned} \tag{3.20}$$

when C_* is sufficiently large. Where $z_L = 0$ the factor is constant, so the original constraints are unchanged up to their positive conformal scaling on the compact part.

This construction supplies the required sign through the joining and bending regions. To control the ADM charge, we continue it into the static tail using its radial flux.

In the static part set $a_L(s) = 1 + M/(2(Ls)^p)$. The radial flux factor is

$$q_L(s) = s^k a_L(s)^2 z_L(s).$$

The equation just used makes $q_L' > 0$ at the beginning of this static part, after increasing C_* if necessary. Extend q_L smoothly so that it stays nondecreasing and is ultimately a positive constant $q_{L,\infty}$. For example, continue its positive derivative over a short interval and then multiply that derivative by a smooth cutoff that becomes zero. This preserves the previous function on an overlap, and all resulting constants remain bounded independently of large L. Define $z_L = q_L/(s^k a_L^2)$ afterwards and $f_L(s) = \int_s^\infty z_L(t)\,dt$ everywhere. Then $f_L \geq 0$, is constant before the transition, and has a finite value there. The static radial Laplacian satisfies

$$\Delta_{\bar{g}_L} f_L = -L^{-2} a_L^{-2n/p} s^{-k} q_L'(s) \leq 0.$$

It is zero on the ultimate tail, where $\bar{K}_L = 0$.

Take $e_L \to 0$ sufficiently slowly that $d_L/e_L \to 0$. Set

$$g_L^* = U_L^{4/p} \bar{g}_L, \qquad K_L^* = U_L^{2/p} \bar{K}_L.$$

The positive contribution in (3.3) dominates the deficit (3.17) by (3.20), so DEC holds everywhere for all sufficiently large L. Since U_L is constant on the original core, the future boundary expansion keeps its weak sign. Moreover $U_L \geq 1$, so this repair increases the metric as a quadratic form.

On the static tail, $f_L(s) = q_{L,\infty} p^{-1} s^{-p} + O(L^{-p} s^{-2p})$. It follows that the coefficient of $|y|^{-p}$ in $U_L - 1$ is $e_L q_{L,\infty}/p$. Computed in the asymptotically Euclidean rest coordinates y, the energy is

$$E_L^* = M + \frac{2 e_L q_{L,\infty}}{p} = M + o(1), \qquad P_L^* = 0. \tag{3.21}$$

The normalization also follows from radial harmonicity: on this tail $a_L U_L$ is a radial Euclidean harmonic function with limit one, because $a_L^{4/p}\delta$ is scalar flat and U_L is harmonic for that metric. Thus the ultimate metric is exactly an isotropic Schwarzschild metric with the mass in (3.21). The new constraints are integrable, since the repair source is compactly supported and the ultimate end is vacuum.

3.5 Comparing every full enclosing cut

We have obtained zero momentum, energy tending to m, and the dominant energy condition. The numerical transfer also requires a lower bound for the infimum over *all* full cuts. The metric need not approach the original metric uniformly on the whole end during the change of slice. The following comparison supplies what is needed for all enclosing areas. In the original x coordinates the metrics g_L^* have a uniform lower bound $c_0\delta$ throughout the region $r \geq L$, where $c_0 > 0$ depends on the fixed boost. Choose a fixed $0 < \lambda < 1$ with $\lambda^2 < c_0/2$.

Let $h_L : [0,\infty) \to [0,\infty)$ be smooth and increasing, equal to r for $r \leq \sqrt{L}$, with $\lambda \leq h_L' \leq 1$, and with $h_L' = \lambda$ for $r \geq 2\sqrt{L}$. Set $\Phi_L(r\theta) = h_L(r)\theta$ on the end, and extend by the identity on the core. The derivative eigenvalues in the Euclidean radial and tangential directions are h_L' and h_L/r, respectively. They are at most one; for $r \geq L$ they are at most $\lambda + O(L^{-1/2})$, since $h_L(r) = \lambda r + O(\sqrt{L})$. Also $h_L \to \infty$, so Φ_L is a proper diffeomorphism of the entire exterior onto itself.

For $r \leq \sqrt{L}$ the map is the identity and $g_L^* \geq g$. For $\sqrt{L} \leq r \leq L$, both g at x and g at $\Phi_L(x)$ equal $\delta + o(1)$ uniformly, and g_L^* is a conformal multiple, at least one, of g. The derivative bound therefore gives $\Phi_L^* g \leq (1 + o(1))g_L^*$ there. For $r \geq L$ one instead has

$$\Phi_L^* g \leq (\lambda^2 + o(1))\delta \leq g_L^*.$$

Combining the regions proves

$$\Phi_L^* g \leq (1 + o(1))g_L^*. \tag{3.22}$$

If $\Gamma = \partial D$ is any full enclosing cut for g_L^*, then $\Phi_L(D)$ is connected, closed, has interior in $\operatorname{int}\Omega$, and contains the entire distant end. Its full intrinsic boundary is $\Phi_L(\Gamma)$, including every component and every portion on S. Restricting (3.22) to each k-dimensional tangent plane and taking determinants gives

$$\operatorname{Area}_{g_L^*}(\Gamma) \geq (1 + o(1))^{-k/2} \operatorname{Area}_g(\Phi_L(\Gamma)) \geq (1 - o(1))a. \tag{3.23}$$

Since this estimate holds for each admissible cut, it passes directly to the infimum, whether or not that infimum is attained.

3.6 A strict scalar perturbation

The use of a spacetime Poisson equation for conformal strictification has a direct precedent in Jaracz's three-dimensional construction [28, Theorem 1.1 and Section 3.1, preprint version 1]. The conformal scaling below is its dimension-dependent counterpart. Here a negative inner normal derivative also makes trapping strict; the smoothed drift, prescribed decay, and charge and full-cut estimates are established for the present exterior class.

At this stage the inequalities may still be non-strict. We solve a scalar equation with positive source and negative inner normal derivative. The source gives a quantitative strict energy margin; the normal derivative makes the future expansion strictly negative. We state the equation on the original decay class as well as on a static tail, so the resulting scalar perturbation is available independently of the preceding replacement.

Lemma 3.5 (Strict energy and boundary inequalities)**.** *Use the constraint, ADM and boundary-normal conventions of Definition 1.1, with $k = n - 1$. Let Ω be a smooth connected exterior, complete with its nonempty compact smooth boundary S included, and with exactly one Euclidean coordinate end outside a compact set. Let ϱ be smooth and positive on the closed exterior,*

equal to the coordinate radius on a sufficiently distant tail. Let (g, K) be smooth up to S, with $g - \delta = O_6(r^{-q})$ and $K = O_5(r^{-1-q})$ for some $q > (n-2)/2$. Choose $0 < \beta < \min(1, q)$ and a smooth nonnegative function $B \geq |K|_g$ with $B = O_5(r^{-1-q})$. Then the problem

$$-\Delta_g \phi = B\sqrt{|d\phi|_g^2 + \varrho^{-2k}} + \varrho^{-n-\beta}, \qquad \partial_\nu \phi = -1 \text{ on } S, \qquad \phi \longrightarrow 0 \text{ at infinity} \tag{3.24}$$

has a smooth positive solution, unique among decaying smooth solutions. It satisfies

$$\begin{aligned} \phi = O_6(r^{2-n}), \qquad & \frac{-\Delta_g \phi}{r^{-n-\beta}} \longrightarrow 1, \\ \mathfrak{f}_\phi := \lim_{R\to\infty} \int_{S_R} \partial_{\nu_g} \phi \, dA_g \qquad & \\ = -\operatorname{Area}_g(S) - \int_\Omega (B\sqrt{|d\phi|^2 + \varrho^{-2k}} + \varrho^{-n-\beta}) \, dV_g < 0. & \end{aligned} \tag{3.25}$$

If K is compactly supported, B may be chosen compactly supported. If in addition $g - \delta = O_d(r^{2-n})$ for all d, the solution has this decay through every derivative order.

Suppose also that the ADM limits of (g, K) exist and are finite, that the data satisfy DEC, and that $\theta_+(S) \leq 0$ on every boundary component. For all sufficiently small $\delta > 0$,

$$g_\delta = e^{2\delta\phi} g, \qquad K_\delta = e^{\delta\phi} K$$

satisfy

$$\mu_\delta - |J_\delta|_{g_\delta} \geq c_\delta \varrho^{-n-\beta}, \qquad \theta_{+,\delta} < 0, \qquad E_\delta = E - \frac{\delta}{\omega}\mathfrak{f}_\phi, \qquad P_\delta = P. \tag{3.26}$$

Here $g_\delta \geq g$, and their ratio tends uniformly to one as $\delta \downarrow 0$.

Proof. We solve the equation on bounded truncations by continuation. Bounds independent of the outer radius close the continuation and permit exhaustion; the resulting decay and flux estimates then give the strict inequalities and the charge formulas.

1. Continuation on a truncation. Write $a = \varrho^{-k}$ and $b = \varrho^{-n-\beta}$. The positive regularizing term a^2 makes the right side smooth as a function of $d\phi$, with first gradient derivative of norm at most B. Choose a large coordinate sphere enclosing S and a compact core. On a larger truncation Ω_R impose $\phi = 0$ on S_R, and multiply B in the equation by $t \in [0, 1]$. The inner Neumann and outer Dirichlet boundary components are disjoint smooth hypersurfaces.

At $t = 0$ this mixed Laplace problem is solvable. Indeed its weak formulation is coercive on H^1 functions of zero outer trace by Poincare's inequality; the Neumann datum is a bounded boundary functional by the trace theorem. Lax–Milgram gives the weak solution and local boundary regularity gives a smooth solution. The linearization at any solution for $0 \leq t \leq 1$ is

$$-\Delta_g u - tB\frac{\langle d\phi, du\rangle_g}{\sqrt{|d\phi|^2 + a^2}}, \qquad \partial_\nu u = 0 \text{ on } S, \quad u = 0 \text{ on } S_R.$$

It has a bounded drift, is uniformly elliptic, and has no zeroth order term. Its homogeneous kernel is zero by the strong maximum principle and the Hopf boundary lemma: a nonzero maximum or minimum cannot occur in the interior or on a zero-Neumann component [21, Chapter 3]. The same conclusion follows for weak limits with merely bounded drift. The drift is a compact perturbation of the mixed Laplace isomorphism in Hölder spaces, so its index is zero and this injectivity gives surjectivity. The implicit function theorem gives openness of the continuation set.

2. Bounds independent of the truncation. The estimates needed for closedness and exhaustion begin with nonnegativity of every solution. An interior negative minimum contradicts $-\Delta\phi > 0$. At a minimum on S the Hopf lemma requires a negative outward derivative, whereas the outward derivative of Ω_R is $-\partial_\nu\phi = 1$. The outer boundary value is zero.

For $0 < \alpha < \min(\beta, q)$ and sufficiently large r_0, the function

$$W(r) = r^{-p}(1 - r^{-\alpha}), \qquad r \geq r_0,$$

is positive and satisfies

$$-\Delta_g W - B|dW|_g \geq cr^{-n-\alpha}. \tag{3.27}$$

In fact $-\Delta_\delta W = \alpha(p+\alpha)r^{-n-\alpha}$, the metric error is $O(r^{-n-q})$, and $B|dW| = O(r^{-n-q})$. Since $b+Ba = O(r^{-n-\beta})$, a sufficiently large multiple $C(A+1)W$, where $A = \sup_{S_{r_0}} \phi$, is a supersolution outside r_0, dominates ϕ on S_{r_0}, and is nonnegative on S_R. Here we used $\sqrt{|d\phi|^2 + a^2} \leq |d\phi| + a$ and the fact that this gradient dependence is Lipschitz. Subtracting a supersolution therefore gives a scalar linear inequality with bounded drift, to which comparison applies. Hence

$$0 \leq \phi(x) \leq C(1 + A)r^{-p} \quad (r_0 \leq r \leq R), \tag{3.28}$$

uniformly in R and t.

We claim that the suprema of all these solutions are uniformly bounded. Otherwise divide a sequence by its suprema $M_i \to \infty$. The bound (3.28) ensures that the normalized functions have a positive supremum on a fixed compact region. Local W^{2,a_0} estimates, for any fixed $a_0 > n$, apply uniformly, including the inner Neumann boundary: the equation has right side bounded by a fixed multiple of $|d\phi| + 1$, and first derivative interpolation absorbs the gradient term in the second derivative estimate. The same argument applies after division by M_i. Compactness gives a nonnegative, nonzero limit u with $u \to 0$ at infinity and

$$-\Delta_g u = t_* B|du|, \qquad \partial_\nu u = 0.$$

If the outer radii instead have a finite limit, the limit has zero data at that limiting sphere. In either case its positive maximum is attained. The strong maximum principle, or the Hopf lemma at S, contradicts that maximum. These principles apply because $B|du|$ equals a bounded measurable drift paired with du, defining that drift to be zero where $du = 0$. This proves the uniform supremum estimate.

3. Exhaustion and uniqueness. The W^{2,a_0} estimates now give uniform $C^{1,\gamma}$ bounds on fixed patches. Their right sides are then Hölder continuous, and the scalar Schauder estimates give $C^{2,\gamma}$ bounds. Differentiation and the strictly positive regularizer a give all higher local bounds. For each fixed truncation these estimates close continuation to $t = 1$. Letting $R \to \infty$ gives a smooth solution of (3.24); the tail bound gives the specified limit. It is strictly positive by the same minimum principle. The difference of two decaying solutions has zero Neumann data and solves a homogeneous equation with bounded drift, by integrating the gradient derivative of the square root along the segment between their gradients. Its positive and negative extrema are excluded as above, proving uniqueness.

4. End decay and flux. On dyadic end annuli rescale $x = Lz$ and $\Phi(z) = L^p\phi(Lz)$. The bound (3.28) bounds Φ. The rescaled equation has uniformly controlled metric coefficients and the form

$$-\Delta_{g(Lz)}\Phi = O_5(L^{-q})\sqrt{|d\Phi|^2_{g(Lz)} + |z|^{-2k}} + L^{-\beta}|z|^{-n-\beta},$$

where the coefficient denoted $O_5(L^{-q})$ is nonnegative and smooth. The same interpolation, scalar estimates and differentiation give uniform bounds for six derivatives of Φ. In the compactly supported B case the equation is just a Poisson equation on the distant end, and all available metric derivative orders give corresponding derivative bounds. This proves the asserted decay. It also gives $B\sqrt{|d\phi|^2 + r^{-2k}} = O(r^{-n-q})$, so $\beta < q$ proves the Laplacian ratio in (3.25).

The source is integrable. Integrating the equation on Ω_R, remembering that its inner outward normal is $-\nu$, gives

$$\int_{S_R} \partial_{\nu_g}\phi \, dA_g = -\operatorname{Area}_g(S) - \int_{\Omega_R} \left(B\sqrt{|d\phi|^2 + a^2} + b\right) dV_g.$$

This proves the claimed finite negative flux.

5. Strict inequalities and charges. Finally $|d\phi|^2 \le Cb$ globally: on the end the quotient has order $r^{-(n-2-\beta)}$, which is bounded because $\beta < 1 \le n-2$, and on the core b is positive. The equation (3.24) implies, for every unit-ball vector v, $-\Delta\phi + K(\nabla\phi, v) \ge b$. In (3.2) with $s = \delta\phi$, for sufficiently small fixed $\delta > 0$ the negative square term is at most half the positive $k\delta b$ term. This proves the strict DEC margin, and $e^{\delta\phi}\theta_{+,\delta} = \theta_+ - k\delta < 0$. The conformal energy vector changes by $-2k\delta\, d\phi + o(r^{1-n})$. The geometric and Euclidean fluxes of $d\phi$ agree in the limit, since their difference is $O(r^{-q})$ after integration. Division by $2k\omega$ gives the energy formula in (3.26) exactly. The momentum trace reversal is multiplied by $e^{\delta\phi}$; its flux change is $O(r^{-q}) \to 0$. Positivity and boundedness of ϕ give the last metric statements. □

For any datum in Lemma 3.5, the same construction gives a quantitative infinitesimal statement at points where the dominant energy condition is saturated. Write $\mathfrak{C}_\delta(v) = 2\big(\mu_\delta + J_\delta(e^{-\delta\phi}v)\big)$. At an active ray, meaning $|v|_g \le 1$ and $\mu + J(v) = 0$, differentiation of (3.2) gives

$$\mathfrak{C}_0'(v) = 2k\big(-\Delta\phi + K(\nabla\phi, v)\big) \ge 2k\varrho^{-n-\beta}. \tag{3.29}$$

Its quotient by $r^{-n-\beta}$ tends to $2k$ along any active rays escaping to infinity: the two terms involving B and K are $O(r^{-n-q})$. At a marginal boundary the expansion derivative of the same direction is exactly $-k$.

Proof of Proposition 3.1. For the repaired data (g_L^*, K_L^*), Lemma 3.5 provides the strict perturbation. Choose B compactly supported and $\delta_L \downarrow 0$ so that $\delta_L\|\phi_L\|_\infty \to 0$ and $\delta_L|\mathfrak{f}_{\phi_L}| \to 0$. This choice is made after the entire replacement at that L, so the constants in this solve may depend on L. Its scalar curvature has the required decay, since the ultimate initial end is scalar flat and

$$R_{e^{2\delta_L\phi_L}g_L^*} = e^{-2\delta_L\phi_L}\big(2k\delta_L r^{-n-\beta} - kp\delta_L^2|d\phi_L|^2\big)$$

there. Compact support of K_L^* is preserved, and all metric derivative orders have the stated asymptotics. The strict margin and expansion are those of Lemma 3.5. The core remains smooth and the tail is uniformly Euclidean comparable, so completeness with S included is preserved. The constraints are integrable on the core and have integrable decay on the tail.

By (3.21) and the choice of δ_L, the final energy tends to $M = m > 0$, and the final momentum is zero. Strictification only increases the metric, so (3.22) and the full-cut comparison remain valid; in particular, every resulting full-cut infimum is positive. Pass to a sequence along which the comparison errors are bounded by numbers $\epsilon_j \downarrow 0$. These data satisfy Definition 2.1 and every assertion in (3.1). □

3.7 Increasing the energy at fixed momentum

The reduction above uses a fixed original future-timelike ADM vector. The following independent conformal change applies without a timelike assumption: it raises the energy by a prescribed amount, fixes the momentum, and increases every full-cut area. These are the properties used in the proof of Corollary 1.4 in Section 2.

Lemma 3.6 (Prescribed energy increase in the strong-decay class)**.** *Assume all hypotheses of Theorem 1.3 except possibly $E > |P|_\delta$, and put $p = n-2$. For every $\lambda > 0$ there is a smooth function $U_\lambda \geq 1$, constant near S, such that*

$$g_\lambda = U_\lambda^{4/p} g, \qquad K_\lambda = U_\lambda^{2/p} K$$

are smooth complete one-ended data with the same O_6/O_5 decay and q, integrable constraint densities, the dominant energy condition and weakly future trapped boundary, and satisfy

$$E_\lambda = E + 2\lambda, \qquad P_\lambda = P, \qquad A_{\min,g_\lambda}(S) \geq A_{\min,g}(S) > 0.$$

Proof. We construct a radial conformal factor with a nonnegative contribution to the energy condition, then compute its energy and momentum fluxes.

Conformal signs. For $U > 0$ put $a = U^{2/p}$ and $(g_U, K_U) = (a^2 g, aK)$. Substitution of $s = 2p^{-1}\log U$ into (3.2) gives, as scalar and covector identities,

$$a^2 \mu_U = \mu - \frac{2k}{pU}\Delta_g U, \qquad a J_U = J + \frac{2k}{pU} K(\nabla U, \cdot).$$

Indeed, the scalar terms combine as $-k\Delta s - \frac{kp}{2}|ds|^2 = -\frac{2k}{pU}\Delta U$. Taking the new-metric covector norm introduces another factor a^{-1}, so

$$a^2(\mu_U - |J_U|_{g_U}) \geq \mu - |J|_g + \frac{2k}{pU}(-\Delta_g U - |K|_g |dU|_g). \tag{3.30}$$

The boundary identity is

$$a\theta_{+,U} = \theta_+ + \frac{2k}{pU}\partial_\nu U.$$

Thus a factor constant near S preserves its expansion sign, and $-\Delta_g U \geq |K|_g|dU|_g$ preserves the dominant energy condition.

The radial factor. Choose $0 < \xi < \min(q, 1)$. In a sufficiently distant part of the coordinate end set

$$T(r) = r^{-p} - r^{-p-\xi}.$$

There $T > 0$ and $T' < 0$. Since $p = n-2$, $\Delta_\delta r^{-\alpha} = \alpha(\alpha + 2 - n) r^{-\alpha-2}$. The original metric and tensor estimates therefore give

$$-\Delta_g T - |K|_g |dT|_g = \xi(p+\xi) r^{-n-\xi} + O(r^{-n-q}) > 0$$

after increasing the radius. Choose a nondecreasing smooth switch β from zero to one in this region, constant near its endpoints, and put

$$G(r) = -\int_r^\infty \beta(t) T'(t)\, dt.$$

Extend G constantly through the interior. Then $G \geq 0$, it is constant near S, and $G = T$ sufficiently far out. Because $dG = \beta\, dT$,

$$-\Delta_g G - |K|_g |dG|_g = \beta(-\Delta_g T - |K|_g |dT|_g) - \beta' T' |dr|_g^2 \geq 0.$$

Set $U_\lambda = 1 + \lambda G$. The inequality (3.30) and the boundary formula prove the required signs for every fixed $\lambda > 0$, without a smallness condition.

Geometry and decay. The factor is bounded for each fixed λ and is at least one, so completeness with S included is preserved. The manifold and its unique coordinate end are unchanged. On the tail $G = O_j(r^{-p})$ at every fixed derivative order. Since $q < p$, the added metric tail is faster than the original decay, and

$$g_\lambda - \delta = O_6(r^{-q}), \qquad K_\lambda = O_5(r^{-1-q}).$$

Moreover,

$$\Delta_g U_\lambda = O(r^{-n-\xi}) + O(r^{-n-q}), \qquad |K|_g |dU_\lambda|_g = O(r^{-n-q}).$$

These terms are integrable against $dV_g \asymp r^{n-1} dr\, d\omega$. The scalar and covector identities above, and the bounded conformal factors for fixed λ, show that the new constraint densities are integrable against dV_{g_λ}.

Charges and enclosing areas. Write $f_\lambda = U_\lambda^{4/p}$. Then

$$f_\lambda = 1 + \frac{4\lambda}{p} G + O_1(r^{-2p}).$$

In the metric flux of (1.2), the leading change is $-k\partial_i(f_\lambda - 1)$. Terms involving $g - \delta$ or its first derivatives have integrated flux $O(R^{-q})$, and the nonlinear remainder has flux $O(R^{-p})$. Since

$$\lim_{R\to\infty} \int_{S_R} \partial_r G\, dA_\delta = -p\omega_{n-1},$$

the current normalization gives

$$E_\lambda - E = -\frac{2\lambda}{p\omega_{n-1}} \lim_{R\to\infty} \int_{S_R} \partial_r G\, dA_\delta = 2\lambda.$$

If $\pi = K - (\mathrm{tr}_g K)g$, its transformed tensor is exactly $\pi_\lambda = U_\lambda^{2/p}\pi$. The additional momentum flux in (1.3) is $O(R^{-q})$, because $U_\lambda^{2/p} - 1 = O(r^{-p})$ and $\pi = O(r^{-1-q})$. Thus $P_\lambda = P$, and the new ADM limits are finite.

Finally, the admissible domains in Definition 1.2 are unchanged. For their entire intrinsic boundaries,

$$\mathrm{Area}_{g_\lambda}(\partial D) = \int_{\partial D} U_\lambda^{2k/p}\, dA_g \geq \mathrm{Area}_g(\partial D).$$

This comparison includes every component and boundary-coincident portion. Taking the infimum proves the claimed full-cut area inequality. □

4 A scalar identity that controls area and mass

Throughout this section the data satisfy Definition 2.1. We seek a metric $\widehat{g} = e^{2t}(g + l(t)^2 df \otimes df)$ with $t > -\epsilon$, so that $\widehat{g} \geq e^{-2\epsilon} g$ controls the loss of full-cut area. The normalized height $h = \eta f$ will separate the filling from the enclosure; weighted coercivity will control its derivatives. We

solve for the graph height f and the modified conformal variable Z in (4.2). The scalar divergence equation for Z controls the penalty's contribution to the ADM energy. Section 5 constructs smooth solutions by exhaustion at fixed ϵ, N; the later limit in N uses only the explicit estimates. All identities below hold in every dimension $n = k + 1 \geq 3$.

4.1 Filling the boundary

A smooth filling removes the inner boundary from the divergence equation. We impose the energy condition only on the original exterior; the height will later separate the added region from the Riemannian comparison region.

The original exterior has compact complement of its coordinate end. Cut it at a large coordinate sphere, take an orientation-reversed copy of the resulting compact manifold, and cap the copied spherical face by a ball. The resulting compact manifold has boundary a copy of the *whole* S. Gluing along that boundary therefore fills all components of S simultaneously; no individual component is required to bound. Smooth collars perform the gluing. Initially extend g, K across S on a fixed short collar. Strict negativity of the expansions of the signed-distance leaves persists there by continuity. Beyond that collar interpolate the metric to a product, and subtract a smooth nonnegative multiple of the metric from K. Start this subtraction while the strict inequality still holds and make the multiple large before performing the remaining interpolation. Since the transition is fixed and compact, its leaf mean curvatures and the other tangential traces are bounded. Thus the subtraction can ensure

$$H + \operatorname{tr}_{\text{leaf}} K < -c_0 < 0$$

through every transition. Arrange $K = -L_0 g$ on the ensuing product necks and on the remaining cap, with L_0 a sufficiently large fixed constant. The neck length at each boundary component may be chosen between $c(1 + N)$ and $C(1 + N)$, increasing C when the height ramp in Section 6 is chosen.

Denote the resulting complete manifold without boundary by $\mathcal{M}_N$, and continue to write g, K for the extended fields. They are unchanged on the original exterior. All the local metric models belong to a fixed finite family, apart from translation along product cylinders. Consequently their injectivity radii are bounded below, every prescribed finite collection of local geometric bounds is bounded independently of N, and the enlarged core has volume and covering number $O(1 + N)$. Extend the end radius to a smooth positive function ϱ, uniformly comparable to 1 on that core. Hence weighted integrals over the core are bounded by a polynomial in N.

4.2 The conformal variable and the graph height

The lower bound for the conformal variable controls area, while the trace and divergence equations will control scalar curvature and mass. We first choose the cutoff and height scales, then define the two unknowns and the tensors entering those equations.

In the scalar construction, $p = p(t)$ is the cutoff defined below; the dimensional exponent is written $n - 2 = k - 1$. Fix $0 < \epsilon < 1$. Let $\vartheta \in C^\infty(\mathbb{R})$ be nonincreasing, equal to 1 on $(-\infty, 0]$ and to 0 on $[1, \infty)$, with values in $[0, 1]$. For an integer $N > \max(k, 2/\epsilon)$ put

$$\begin{aligned} p(t) &= N\vartheta(Nt), \qquad l(t) = \exp\left(\int_0^t p(s)\, ds\right), \\ \ell &= e^{-N\epsilon}, \qquad \eta = \ell^{3/2}, \qquad m_0(t) = \vartheta\big(N(t + \epsilon)\big). \end{aligned} \tag{4.1}$$

In particular $l = e^{Nt}$ for $t \le 0$, $l \le e$ everywhere, and $\sup |p^{(j)}| \le C_j N^{j+1}$ for each fixed j.

For smooth unknowns f, Z, define t implicitly by

$$Z = t + \frac{1}{2k} \log D, \qquad D = 1 + l(t)^2 |df|_g^2. \tag{4.2}$$

For fixed df the derivative of the right side with respect to t is $(k+pv)/k \ge 1$, where $v = 1 - D^{-1}$. Its limits as t tends to $\pm\infty$ are $\pm\infty$. Thus this equation defines t globally and smoothly as a function of $(Z, |df|^2)$, over the full range $-\infty < t < \infty$.

The vector w measures the graph slope, and d and χ are the axial weights in the inverse graph metric and the trace equation. The tensor L combines the rescaled Hessian of f with K, while its weighted trace F will be prescribed by the normalized height h. The weight u and vector field V enter the divergence equation. Using g to identify vectors and covectors, set

$$\begin{aligned}
&w = lD^{-1/2}\nabla f, \qquad a_w = |w|, \qquad v = a_w^2, \qquad d = D^{-1} = 1 - v,\\
&\chi = \frac{kd}{k+pv}, \qquad A_d = I - w \otimes w, \qquad A_\chi = I - \frac{k+p}{k+pv} w \otimes w,\\
&H^f = lD^{-1/2} \operatorname{Hess}_g f, \qquad L = K + H^f, \qquad F = \operatorname{tr}_{A_\chi} L, \qquad h = \eta f,\\
&U = l\sqrt{D}, \qquad u = e^{(k-1)t} U, \qquad V = u A_\chi (k \nabla Z + K(w, \cdot)),\\
&\bar g = g + l^2 df \otimes df, \qquad \widehat g = e^{2t} \bar g, \qquad s_p = p + \frac{k-1}{2}.
\end{aligned} \tag{4.3}$$

The notation $\operatorname{tr}_A B$ means $A^{ij} B_{ij}$, and $|\alpha|_A^2 = A^{ij} \alpha_i \alpha_j$. The tensors A_d, A_χ have transverse eigenvalue 1 and axial eigenvalues d, χ, respectively. More explicitly,

$$0 < \frac{k}{k+N} d \le \chi \le d \le 1, \qquad k\, dZ = (k + pv)\, dt + H^f(w, \cdot). \tag{4.4}$$

The second equality follows by differentiating (4.2): $\frac{1}{2} d \log D = pv\, dt + H^f(w, \cdot)$. Every tensor in (4.3) is smooth at $df = 0$. We use axis coordinates below to display these already defined tensors; at gradient zeros their coefficients are given by the same smooth formulas.

In this and the next two sections $\mathcal{P}_N$ denotes a polynomial in $1 + N$, whose coefficients may depend on the fixed prepared data, n, and ϵ. Increasing such a polynomial finitely many times does not change this convention. An estimate with an arbitrary constant depending on fixed N will be explicitly identified as such; it cannot be substituted into a later $\mathcal{P}_N$ estimate.

4.3 The scalar identity

The model for this calculation is the Schoen–Yau scalar-curvature identity for Jang graphs [41], and more specifically its warped generalization in [8, Identity 9, p. 580]. Our conformal variable, gradient-aligned trace, and redistributed divergence are different. We derive their exact identity before imposing the coupled equations; its algebra and their global solvability are separate steps.

For symmetric tensors define the bilinear form and trace reversal

$$\mathsf{B}(B, C) = \langle B, C \rangle_g - (\operatorname{tr}_g B)(\operatorname{tr}_g C), \qquad P_B = B - (\operatorname{tr}_g B) g.$$

At a nonzero gradient let $e = \nabla f / |df|$, and decompose L and dt relative to $e^\perp \oplus \mathbb{R} e$ as

$$L = \begin{pmatrix} T & M \\ M^\intercal & j \end{pmatrix}, \qquad dt = (y, x), \qquad T^0 = T - \frac{\operatorname{tr} T}{k} I_{e^\perp}.$$

Here $\operatorname{tr} T = F - \chi j$. At $df = 0$ one may choose any unit axis.

Proposition 4.1 (Scalar identity)**.** *For every smooth f, Z and every smooth g, K, with the quantities defined in (4.1)–(4.3), the following identity holds, without a field equation or energy hypothesis:*

$$\frac{1}{2}e^{2t}R_{\widehat{g}} = \mu + J(w) + \mathcal{T} + w(F) - F\tau - u^{-1}\operatorname{div}_g V, \tag{4.5}$$

where $2\mu = R_g + \tau^2 - |K|^2$, $J = \operatorname{div}_g P_K$, and

$$\begin{aligned}\mathcal{T} = \frac{1}{2}|T^0|^2 + |M + s_p a_w y|^2 + \left(ks_p - \frac{(k-1)^2}{4}v\right)|y|^2 + ks_p dx^2 \\ - \frac{k-1}{2k}(F - \chi j)^2 + \chi j^2 - \chi F j + F^2 + 2s_p \chi j a_w x.\end{aligned} \tag{4.6}$$

The quantity $\mathcal{T}$ is smooth, including where $df = 0$.

Proof. We first compute the curvature using stationary and static coordinates for the same Lorentz metric. The conformal law then isolates a divergence; rewriting that divergence in terms of V produces the trace derivative and the quadratic form.

Stationary curvature. On $\mathbb{R}_z \times \mathcal{M}_N$ consider the stationary Lorentz metric

$$\mathbf{G} = -l^2(dz - df)^2 + \bar{g} = -l^2 dz^2 + 2l^2 dz\, df + g.$$

The constant-z slices have lapse U, shift $Uw = l^2\nabla f$, future unit normal $\mathbf{n} = U^{-1}\partial_z - w$, and second fundamental form

$$b = -U^{-1}\operatorname{sym}\nabla(Uw) = -H^f - p(dt \otimes w + w \otimes dt). \tag{4.7}$$

This sign follows directly from $\partial_z g = 2Ub + \mathcal{L}_{Uw}g = 0$ and the prescribed future-normal convention. Write $B_0 = \operatorname{tr} b$. Contracted Gauss and the normal lapse equation give

$$\mathbf{R} + 2\mathrm{Ric}(\mathbf{n}, \mathbf{n}) = R_g - \mathsf{B}(b, b), \qquad \mathrm{Ric}(\mathbf{n}, \mathbf{n}) = -\mathbf{n}(B_0) - |b|^2 + U^{-1}\Delta_g U.$$

Since B_0 is stationary, $\mathbf{n}(B_0) = -w(B_0)$, and taking the trace in (4.7) gives $\operatorname{div}(Uw) = -UB_0$. Substitution proves

$$\mathbf{R} = R_g + \mathsf{B}(b, b) - 2U^{-1}\operatorname{div}(\nabla U + B_0 Uw). \tag{4.8}$$

Graph curvature. In coordinates $z' = z - f$, the same metric is the static warped product $-l^2(dz')^2 + \bar{g}$. Its only mixed connection terms are $\mathbf{\Gamma}^{z'}_{iz'} = \partial_i \log l$ and $\mathbf{\Gamma}^{i}_{z'z'} = l\bar{g}^{ij}\partial_j l$; contracting their curvature contributions yields $\mathbf{R} = R_{\bar{g}} - 2l^{-1}\Delta_{\bar{g}} l$. Also $\bar{g}^{-1} = A_d$, $dV_{\bar{g}} = \sqrt{D}\, dV_g$, and therefore

$$l^{-1}\Delta_{\bar{g}} l = U^{-1}\operatorname{div}((U/l)A_d\nabla l).$$

Differentiating $U = l\sqrt{D}$ and using (4.4) proves the vector identity

$$\nabla U - (U/l)A_d\nabla l = -Ub(w, \cdot).$$

Consequently (4.8) becomes

$$\frac{1}{2}R_{\bar{g}} = \frac{1}{2}R_g + \frac{1}{2}\mathsf{B}(b, b) + U^{-1}\operatorname{div}(UP_b w). \tag{4.9}$$

Set $Q = K - b = L + p(dt \otimes w + w \otimes dt)$. For any symmetric C, the product rule and (4.7) give the contraction

$$U^{-1}\operatorname{div}(UP_C w) = (\operatorname{div} P_C)(w) - \mathsf{B}(C, b).$$

Expanding $\mathsf{B}(K-b,K-b)$ and applying this formula to $C=K$ transforms (4.9) into

$$\frac{1}{2}R_{\bar g}=\mu+J(w)+\frac{1}{2}\mathsf{B}(Q,Q)-U^{-1}\operatorname{div}(UP_Qw). \tag{4.10}$$

The terms involving K on the right cancel to $R_g/2-\mathsf{B}(b,b)/2+(\operatorname{div}P_b)(w)$.

Conformal change and divergence weight. The conformal scalar law in dimension $k+1$ is

$$\frac{1}{2}e^{2t}R_{\widehat g}=\frac{1}{2}R_{\bar g}-k\Delta_{\bar g}t-\frac{k(k-1)}{2}|dt|^2_{A_d}.$$

The volume formula gives

$$\Delta_{\bar g}t=U^{-1}\operatorname{div}(UA_d\nabla t)-p|dt|^2_{A_d}.$$

Changing the divergence weight from U to $u=e^{(k-1)t}U$ now yields

$$\begin{aligned}\frac{1}{2}e^{2t}R_{\widehat g}=\mu+J(w)+\frac{1}{2}\mathsf{B}(Q,Q)+(k-1)P_Q(w,dt)+ks_p|dt|^2_{A_d}\\ -u^{-1}\operatorname{div}\big(u(P_Qw+kA_d\nabla t)\big).\end{aligned} \tag{4.11}$$

In particular the last change of weight adds $k(k-1)|dt|^2_{A_d}$ to the preceding coefficient $k[p-(k-1)/2]$.

The trace and flux. The divergence in (4.11) is related to V by the following two first-order identities:

$$\begin{aligned}u(P_Qw+kA_d\nabla t)&=V-uFw,\\ u^{-1}\operatorname{div}(uw)&=\operatorname{tr}_gL-\tau+2s_pw(t)=F+(1-\chi)j-\tau+2s_pa_wx.\end{aligned} \tag{4.12}$$

For the first, P_Qw+kA_ddt has transverse and axial components

$$a_wM+(k+pv)y,\qquad -a_w\operatorname{tr}T+kdx,$$

whereas V/u has components $a_wM+(k+pv)y$ and χa_wj+kdx by (4.4). Their difference is Fw. For the second, changing the weight in $\operatorname{div}(Uw)=-U\operatorname{tr}b$ gives the stated expression. Thus the smooth invariant definition of the claimed quadratic form is

$$\mathcal{T}=\frac{1}{2}\mathsf{B}(Q,Q)+(k-1)P_Q(w,dt)+ks_p|dt|^2_{A_d}+F\big(\operatorname{tr}_gL+2s_pw(t)\big). \tag{4.13}$$

This proves smoothness and (4.5). To obtain its displayed block form, the blocks of Q are T, $M+pa_wy$, and $j+2pa_wx$. The transverse terms are

$$|M+pa_wy|^2+(k-1)a_w(M+pa_wy)\cdot y=|M+s_pa_wy|^2-\frac{(k-1)^2}{4}v|y|^2.$$

The terms containing Fa_wx cancel, while their remaining axial cross term is $2s_p\chi ja_wx$. Finally $|T|^2=|T^0|^2+(F-\chi j)^2/k$ gives (4.6).

Derivatives of the cutoff. The derivatives of p enter through the divergences. In the flux of (4.11) the part explicitly containing p is $p\{v\nabla t-w(t)w\}$; its divergence contributes $p'v|y|^2$. Thus the scalar side contains $-p'v|y|^2$. In (4.5) the redistribution of this term can also be checked directly. Holding t,dt,df,H^f,K fixed when taking the partial derivative in p, one has $\chi_p=-\chi v/(k+pv)$. The terms containing p' in $w(F)$ and $u^{-1}\operatorname{div}V$ are respectively

$$p'\chi_pa_wxj,\qquad p'\big(v|y|^2+\chi_pa_wxj\big).$$

Their difference is $-p'v|y|^2$, accounting for the cutoff derivative in the scalar identity. □

4.4 Coercivity with polynomial constants

The identity has isolated the divergence, but it is useful only if the remaining quadratic form controls the graph and conformal derivatives. The next proposition identifies the controlled axial weights. Its constants are polynomial in the large parameter; this is what will allow the exponentially small penalty flux to tend to zero.

Proposition 4.2 (Weighted coercivity)**.** *For $0 \le p \le N$ and $0 \le v < 1$ the quadratic form in (4.6) is nonnegative, and*

$$|T|^2 + |M|^2 + dj^2 + F^2 + |dt|^2_{A_d} \le C_k(1+N)^2\mathcal{T}. \tag{4.14}$$

At $dt = 0$ one has, with constants depending only on k,

$$c_k(|T^0|^2 + |M|^2 + F^2 + \chi j^2) \le \mathcal{T} \le C_k(|T^0|^2 + |M|^2 + F^2 + \chi j^2). \tag{4.15}$$

Proof. Completing the axial squares controls the weighted axial variables; the transverse square then controls M. The axial part of (4.6) is

$$\begin{aligned} &ks_p d\left(x + \frac{a_w j}{k+pv}\right)^2 + \frac{k+1}{2k}\left(F - \frac{\chi j}{k+1}\right)^2 + \frac{kdB_k(v)}{(k+pv)^2}j^2, \\ &B_k(v) = d\frac{k(k+2)}{2(k+1)} + v\frac{k+1}{2} = \frac{k(k+2)+v}{2(k+1)}. \end{aligned} \tag{4.16}$$

Indeed completing first the x square and then the F square leaves

$$\chi - \frac{k}{2(k+1)}\chi^2 - s_p\frac{\chi^2 v}{kd} = \frac{kdB_k(v)}{(k+pv)^2}.$$

The last coefficient is at least $c_k d/(1+N)^2$, and

$$ks_p - \frac{(k-1)^2}{4}v \ge \frac{k^2-1}{4} > 0.$$

These inequalities control $|T^0|^2, |y|^2, dj^2$ and the two completed squares. The resulting bound for M, obtained from $M + s_p a_w y$, has a factor at most $C_k(1+N)^2$. The bound for $\sqrt{d}\,x$ follows from the completed square and the already controlled $\sqrt{d}\,j$, since $a_w/(k+pv) \le 1/k$. The same bounds control F because $\chi^2 \le d$. The relation $\operatorname{tr} T = F - \chi j$ then controls $|T|^2$. This proves (4.14) with the stated single quadratic polynomial and with the factors d retained.

When $dt = 0$ the axial expression instead completes as

$$\frac{k+1}{2k}\left(F - \frac{\chi j}{k+1}\right)^2 + \chi\left(1 - \frac{k\chi}{2(k+1)}\right)j^2.$$

Its second parenthesis is bounded above and below by positive constants, and $\chi^2 \le \chi$. This proves (4.15) independently of N. □

4.5 The penalized system and the lower barrier

We now choose the equations using the signs in (4.5). Imposing $F = h = \eta f$ turns $w(F)$ into the nonnegative term $\eta a_w |df|$. We can assign a small fixed fraction of this term and of $\mathcal{T}$ to the divergence source while retaining the rest in the scalar-curvature formula. To enforce the lower bound for t, the source will also contain a negative term near $t = -\epsilon$. Its sign improves scalar curvature, but its integral can increase ADM energy. We therefore need both a pointwise lower-bound argument and a separate estimate for the weighted integral of that penalty.

Choose

$$\frac{n}{4} < b_1 < \min\left(n-2, \frac{n}{2}\right), \qquad \rho_0 = \varrho^{-n-\beta}, \quad \rho_1 = \varrho^{-2b_1}, \quad \rho = c_* \rho_0 > 0. \tag{4.17}$$

This interval is nonempty for every $n \geq 3$. Its three restrictions serve different parts of the argument: $b_1 < n-2$ permits the radial height comparison in Section 5; $2b_1 < n$ lets $v\rho_1$ dominate the end-curvature error at a minimum of t; and $4b_1 > n$ makes ρ_1^2 integrable in the flux estimate. Take c_* small enough that $\mu - |J| \geq 2\rho$ on the original exterior. The system to be solved is

$$\begin{aligned} F = h = \eta f, \qquad & \operatorname{div} V = u\Xi, \\ \Xi = \delta_0(\mathcal{T} + \eta a_w |df|) + \rho - m_0 C_N(\rho_0 + v\rho_1). & \end{aligned} \tag{4.18}$$

Here $\delta_0 > 0$ is a sufficiently small fixed number and C_N is a sufficiently large polynomial, specified by the next lemma. Substitution in the scalar identity retains $(1-\delta_0)(\mathcal{T} + \eta a_w |df|)$; the positive source ρ is at most half of the strict energy margin on the original exterior. The remaining term $-h\tau$ will be controlled on a small height band. The negative penalty is chosen to contradict the flux lower bound at a minimum with $t = -\epsilon$.

For a solution with $t > -\epsilon$, the penalty is supported in $-\epsilon < t < -\epsilon + N^{-1}$, where $\ell \leq l \leq e\ell$. Proposition 4.4 will use this small weight to bound the possible energy increase by a polynomial times $\ell + \ell^2/\eta$. The height-derivative estimate below instead uses η/ℓ. The choice $\eta = \ell^{3/2}$ makes both ratios ℓ^2/η and η/ℓ equal to $\ell^{1/2}$. Both ratios tend to zero faster than any polynomial in N at fixed ϵ.

On a large truncation the outer data are $f = Z = 0$; there is no inner boundary. For the existence argument we use two homotopies:

1. Replace K by aK, $0 \leq a \leq 1$, everywhere in these definitions, keeping the weights and constants fixed.

2. Keep $K = 0$ and multiply the whole expression Ξ by $s \in [0,1]$.

For a symmetric tensor B write

$$\mathcal{S}(B) = \frac{1}{2}\left\{(\operatorname{tr}_{A_d} B)^2 - |B|^2_{A_d \otimes A_d}\right\}. \tag{4.19}$$

Lemma 4.3 (Exclusion of an interior minimum at the lower bound)**.** *On the filled manifolds $\mathcal{M}_N$ of Section 4.1, use the quantities in (4.3), (4.6), and (4.19). There is $c_1 > 0$ depending on k alone such that, at any point with $dt = 0$,*

$$\mathcal{T} - \mathcal{S}(H^f) \geq c_1 \mathcal{T} - \mathcal{P}_N(vF^2 + |K|^2). \tag{4.20}$$

Suppose a smooth solution on a truncation with outer $f = Z = 0$ satisfies the trace equation and has $|h| \leq C$ on the core and $|h| \leq C\varrho^{-b_1}$ on the end, with constants independent of the homotopy

parameter and N. One may choose $0 < \delta_0 < \min(c_1, 1)/2$ and a polynomial C_N such that a solution of (4.18) for which $\min t = -\epsilon$ cannot attain this minimum of t at an interior point. This holds for both homotopies 1 and 2.

Proof. We first compare the quadratic form with the Gauss term at $dt = 0$. At an interior minimum, the curvature formula gives an upper bound which, together with that comparison, forces a lower bound for the flux. The penalty in the system contradicts this lower bound.

Quadratic comparison. At $dt = 0$, direct block contraction gives

$$\mathcal{S}(L) = \frac{k-1}{2k}(F - \chi j)^2 - \frac{1}{2}|T^0|^2 + dj(F - \chi j) - d|M|^2.$$

Subtracting from (4.6) gives the full difference, including its mixed term:

$$\begin{aligned}\mathcal{T} - \mathcal{S}(L) = |T^0|^2 + (1+d)|M|^2 + \frac{F^2}{k} \\ + \left(\frac{k-2}{k}\chi - d\right) Fj + \chi\left(1 + d - \frac{k-1}{k}\chi\right) j^2.\end{aligned} \tag{4.21}$$

If $(1+p)v \le \zeta_k$ with $\zeta_k > 0$ sufficiently small, then $|d-1| + |\chi - 1| \le C_k\zeta_k$. At $d = \chi = 1$ the expression is

$$|T^0|^2 + 2|M|^2 + \frac{(F-j)^2}{k} + j^2 = |L|^2.$$

Perturbation of this finite-dimensional positive form, and (4.15), therefore bound (4.21) below by a fixed positive multiple of $\mathcal{T}$.

In the other case $1/v \le (1+p)/\zeta_k$. The last coefficient in (4.21) is at least χ, since $\chi \le d$. The absolute value of its mixed coefficient is at most d. Hence Young's inequality bounds the difference below by

$$|T^0|^2 + |M|^2 + \tfrac{1}{2}\chi j^2 + \left(\frac{1}{k} - \frac{d^2}{2\chi}\right) F^2.$$

Here

$$\frac{d^2}{\chi} = \frac{d(k+pv)}{k} \le 1 + \frac{N}{k}.$$

It follows, again using (4.15), that the desired fixed multiple of $\mathcal{T}$ is obtained after subtracting at most $C_k(1+N)^2 vF^2/\zeta_k$.

Polarizing $\mathcal{S}$ in the $A_d \otimes A_d$ norm compares L with $H^f = L - K$. The preceding stationary coercivity gives

$$|L|^2_{A_d \otimes A_d} = |T|^2 + 2d|M|^2 + d^2 j^2 \le C_k(1+N)\mathcal{T}.$$

Moreover $|K|_{A_d \otimes A_d} \le |K|$. Therefore

$$|\mathcal{S}(L) - \mathcal{S}(L-K)| \le \zeta\mathcal{T} + C_{k,\zeta}(1+N)|K|^2$$

for every fixed $\zeta > 0$. Taking ζ below half the positive constant already obtained proves (4.20).

Curvature at a minimum. At an interior minimum of t, $dt = 0$ and $\operatorname{Hess}_g t \ge 0$. To estimate curvature from above, view $\bar{g}$ as the metric on the graph of f in the Riemannian warped product $g + l^2 dz^2$. At that point $dl = 0$ and $l^{-1}\operatorname{Hess} l = p \operatorname{Hess} t$. Its ambient scalar curvature is $R_g - 2p\Delta t$, its horizontal Ricci tensor is $\operatorname{Ric}_g - p \operatorname{Hess} t$, and its unit vertical Ricci component is $-p\Delta t$. The graph unit normal is $D^{-1/2}\partial_{z,\text{unit}} - w$ and its second form with this orientation is $-H^f$, which

has the same quadratic Gauss term as H^f. The Gauss equation followed by the conformal law consequently gives the minimum-point formula

$$\begin{aligned}\frac{1}{2}e^{2t}R_{\widehat{g}} &= \frac{1}{2}R_g - v\,\mathrm{Ric}_g(e,e) + \mathcal{S}(H^f)\\ &\quad - pv\,\mathrm{tr}_{e^\perp}\,\mathrm{Hess}\,t - k\,\mathrm{tr}_{A_d}\,\mathrm{Hess}\,t\\ &\le \frac{1}{2}R_g - v\,\mathrm{Ric}_g(e,e) + \mathcal{S}(H^f).\end{aligned} \tag{4.22}$$

When $v = 0$, terms using e vanish. In particular all $p'(dt)^2$ terms vanish at this point.

After solving (4.5) for the divergence, scalar curvature appears with a minus sign. Thus the upper bound (4.22) yields a lower bound whose quadratic part is $\mathcal{T} - \mathcal{S}(H^f)$. Applying (4.20) to this difference and using $F = \eta f$, which gives $w(F) = \eta a_w|df|$, yields

$$u^{-1}\,\mathrm{div}\,V \ge c_1\mathcal{T} + \eta a_w|df| - \mathcal{P}_N(\rho_0 + v\rho_1). \tag{4.23}$$

Here are the weights in this estimate. The curvature-independent constraint contribution is $\mu - R_g/2 = (\tau^2 - |K|^2)/2$. This, $J(w)$, $F\tau$, and the $|K|^2$ error are uniformly bounded on the enlarged core and vanish on the far end, so their absolute values are bounded by $C\rho_0$ pointwise. On the end, $v|\,\mathrm{Ric}_g\,| \le Cv\varrho^{-n} \le Cv\rho_1$ because $2b_1 < n$; the height comparison gives $vF^2 \le Cv\rho_1$. The same statements hold uniformly for aK, $0 \le a \le 1$. On the filling, the calculation uses the uniform geometric bounds established above.

Contradiction with the penalty. At $t = -\epsilon$ one has $m_0 = 1$. Subtracting the right side of (4.18) from the lower bound in (4.23) gives

$$(c_1 - \delta_0)\mathcal{T} + (1 - \delta_0)\eta a_w|df| + (C_N - \mathcal{P}_N)(\rho_0 + v\rho_1) - \rho.$$

Choose $C_N > \mathcal{P}_N + c_* + 1$. The expression is strictly positive, contradicting the equation. In the second homotopy stage the trace comparison gives $f = 0$: a positive maximum or negative minimum contradicts $\mathrm{tr}_{A_\chi} H^f = \eta f$. Thus $t = Z$, and at a minimum with $t = -\epsilon$ the left side of the divergence equation divided by u is $k\Delta t \ge 0$, whereas its right side is $s(\rho - C_N\rho_0) < 0$ for $s > 0$. At $s = 0$, the equation $\mathrm{div}(u\nabla Z) = 0$ with zero outer data gives $Z = 0$ by the maximum principle. This finishes the pointwise exclusion. □

The lemma applies to the boundary $\min t = -\epsilon$ of the moving degree domain. The uniform height comparisons in its hypotheses and the separate outer-boundary exclusion are proved in Section 5.

4.6 Scalar curvature on solutions and height derivatives

On the original exterior, substituting (4.18) into the identity gives

$$\begin{aligned}\frac{1}{2}e^{2t}R_{\widehat{g}} &= \mu + J(w) - \rho - h\tau + (1 - \delta_0)\mathcal{T} + (1 - \delta_0)\eta a_w|df|\\ &\quad + m_0 C_N(\rho_0 + v\rho_1).\end{aligned} \tag{4.24}$$

In particular a small fixed height band has a strict scalar margin: choose $b_3 > 0$ such that $b_3|\tau| \le \rho/2$ on the support of τ. Then $\mu - |J| - \rho - |h\tau| \ge \rho/2$ on $|h| \le b_3$. This is possible since τ is compactly supported and $\mu - |J| \ge 2\rho$ there.

The height derivatives satisfy

$$\begin{aligned} |dh|_{A_d} &= \frac{\eta|df|}{\sqrt{D}} \le \frac{\eta}{l}, \\ e^{2t}\Delta_{\widehat{g}} h &= \frac{\eta}{l\sqrt{D}}\left\{\operatorname{div} w + (k-1-p)w(t)\right\}, \\ \operatorname{div} w &= \operatorname{tr}_{A_d} H^f + pd\, w(t). \end{aligned} \tag{4.25}$$

The middle formula follows from $\sqrt{D}A_d\nabla h = (\eta/l)w$ and the conformal Laplacian law; the last follows by differentiating $w = lD^{-1/2}\nabla f$. For a solution with $t > -\epsilon$, $l \ge \ell$, and Proposition 4.2 therefore implies

$$|dh|_{A_d} \le \ell^{1/2}, \qquad |e^{2t}\Delta_{\widehat{g}} h| \le \mathcal{P}_N \ell^{1/2}(1+\sqrt{\mathcal{T}}). \tag{4.26}$$

Indeed $\operatorname{tr}_{A_d} H^f = \operatorname{tr} T + dj - \operatorname{tr}_{A_d} K$ is bounded by $\mathcal{P}_N(1+\sqrt{\mathcal{T}})$. The other potentially large term is $w(t) = a_w x$. The prefactor $D^{-1/2} = \sqrt{d}$ converts it to $a_w\sqrt{d}\,x$, precisely the weighted quantity controlled by (4.14). The factor $\sqrt{d}$ therefore remains in the bound. Section 6 uses these estimates to raise the metric inside a compact height band.

4.7 The penalty flux and its ADM normalization

Proposition 4.4 (Flux and ADM energy)**.** *Fix ϵ and N as in (4.1). Suppose a smooth global solution of (4.18) satisfies $t > -\epsilon$, $Z \le \mathcal{P}_N$, and on the coordinate end*

$$f = O_j(e^{-c_N r}), \qquad t = O_j(r^{2-n}), \qquad \Delta_g t = O(r^{-n-\beta})$$

for all required derivative orders, with constants allowed to depend arbitrarily on this fixed N. Then

$$\begin{aligned} \mathfrak{F}_N &:= \lim_{r\to\infty} \int_{S_r} g(V,\nu_g)\, dA_g = \int_{\mathcal{M}_N} u\Xi\, dV_g, \\ \widehat{E} - E &= -\frac{\mathfrak{F}_N}{k\omega}, \qquad \mathfrak{F}_N \ge -\mathcal{P}_N(\ell + \ell^{1/2}), \\ \widehat{E} &\le E + \mathcal{P}_N(\ell + \ell^{1/2}). \end{aligned} \tag{4.27}$$

The polynomial in the error uses only the fixed geometry, the weights, and the explicit parameters, not the arbitrary fixed-N derivative constants in the stated end estimates.

Proof. For the solutions used here, Theorem 5.1 supplies (5.3) for every integer $j \ge 0$ at fixed N. These estimates give the end hypotheses, the flux limit and its ADM normalization. The lower bound for the flux will then follow from the penalty band, independently of the constants in the end estimates.

Existence of the flux. For fixed N, the end hypotheses and (4.4) give

$$D - 1 = O_j(e^{-c'_N r}), \quad Z - t = O_j(e^{-c'_N r}), \quad V = k\nabla t + O_N(r^{3-2n}) + O(e^{-c'_N r}).$$

The notation O_N here permits arbitrary fixed-N constants. Since $N\epsilon > 2$ and $t \to 0$, $m_0 = 0$ sufficiently far out. There $\mathcal{T} = ks_p|dt|^2 + O(e^{-c'_N r})$, so every term in $u\Xi$ is integrable: the gradient term is $O_N(r^{2-2n})$, and $\rho = O(r^{-n-\beta})$. Smoothness gives integrability on the compact filled core. The divergence theorem on exhausting domains has no inner boundary and proves the first equality and existence of the flux limit. It also proves existence of $\lim \int_{S_r} \partial_{\nu_g} t$ directly from $\Delta_g t \in L^1$.

ADM normalization. The graph part $l^2 df^2$ contributes zero to the ADM flux by its exponential decay. The conformal perturbation is

$$(e^{2t} - 1)g = 2t\delta + O_1(r^{4-2n}).$$

The linear ADM integrand of $2t\delta$ is $-2k\,\partial_i t$; the remainder has integrated flux $O(r^{2-n}) \to 0$. Euclidean and g-normal fluxes differ by the same vanishing order. Thus

$$\widehat{E} - E = -\omega^{-1} \lim_{r\to\infty} \int_{S_r} \partial_{\nu_g} t \, dA_g = -\frac{\mathfrak{F}_N}{k\omega}.$$

The negative contribution. We bound the negative part of the integral independently of those end constants. It is supported in the penalty band $-\epsilon < t < -\epsilon + 1/N$, where $t < 0$ and $\ell \le l \le e\ell$. Put $\sigma = |df|$. Directly from the definitions, with constants independent of N on this band,

$$u \le C(\ell + \ell^2\sigma), \qquad uv \le C\ell^2\sigma, \qquad ua_w\sigma = e^{(k-1)t} l^2\sigma^2 \ge c\ell^2\sigma^2. \tag{4.28}$$

For the second estimate use $l\sqrt{D}\, v = l^3\sigma^2/\sqrt{1 + l^2\sigma^2} \le l^2\sigma$. The nonnegative terms $u\delta_0\mathcal{T}$ and $u\rho$ may be dropped, while the favorable gradient term absorbs the linear σ terms. More explicitly Young's inequality gives

$$CC_N\ell^2\sigma(\rho_0 + \rho_1) \le \tfrac{1}{2}\delta_0 c\eta\ell^2\sigma^2 + C'C_N^2 \frac{\ell^2}{\eta}(\rho_0^2 + \rho_1^2).$$

Consequently

$$\mathfrak{F}_N \ge -CC_N\ell \int \rho_0 - C'C_N^2 \frac{\ell^2}{\eta} \int (\rho_0^2 + \rho_1^2).$$

All integrals use dV_g on $\mathcal{M}_N$. They are bounded by $C(1 + N)$: on the end ρ_0 is integrable and $4b_1 > n$ makes ρ_1^2 integrable, while on the core the weights are uniformly bounded and its volume is $O(1 + N)$. Since $\ell^2/\eta = \ell^{1/2}$, this proves the polynomial flux estimate. For fixed $\epsilon > 0$ the resulting error tends to zero as $N \to \infty$. □

Finally, every solution with $t > -\epsilon$ satisfies the pointwise comparison

$$\widehat{g} \ge e^{-2\epsilon} g. \tag{4.29}$$

This comparison and the flux estimate apply to each smooth solution at fixed N. The limit $N \to \infty$ is taken in their explicit bounds; the metric comparison holds for every full cut before taking its area infimum.

5 Solving the filled equations

The metric comparison and the energy estimate from Section 4 become useful only after its two scalar equations have a smooth solution. We construct that solution here, on the filled manifold and at fixed prepared data. The boundary of a large truncation is a single coordinate sphere; both unknowns vanish there.

The first task is to keep the conformal variable above $-\epsilon$ and to bound Z without assuming uniform ellipticity. The height comparison and the minimum-point calculation give the lower bound. Testing the scalar divergence equation on the graph of f then gives a polynomial upper

bound for Z. Together these bounds control the slope and make both scalar equations uniformly elliptic for each fixed N.

We next prove the local estimates needed to pass from these bounds to smoothness. The gradient-aligned trace equation and the bounded natural-growth divergence equation are treated separately, as reusable scalar lemmas. Finally we solve the trace equation for every trial Z, use it to define a compact map, and obtain a fixed point by two homotopies. Exhausting the outer spheres gives the complete solution and the end estimates needed for its ADM flux.

Theorem 5.1 (Filled construction)**.** *Fix smooth prepared data satisfying (2.1) on a complete one-ended exterior with nonempty compact smooth boundary. In particular, K is compactly supported, $g-\delta = O_d(r^{2-n})$ for every fixed d, $R_g = O(r^{-n-\beta})$ for some $0<\beta<1$, the dominant-energy margin is bounded below by a positive multiple of $\varrho^{-n-\beta}$, and every boundary component is strictly future trapped. Fix $0<\epsilon<1$ and*

$$\frac{n}{4} < b_1 < \min\{n-2, n/2\}.$$

Use the filled manifolds $\mathcal{M}_N$, weights ρ_0, ρ_1, ρ, and definitions in (4.3). Choose $\delta_0 > 0$ and the polynomial C_N as in Lemma 4.3. For all sufficiently large integers $N > k = n-1$, there are smooth functions f, Z on $\mathcal{M}_N$ solving (4.18). The associated function t satisfies

$$-\epsilon < t \le Z \le P(N), \qquad \ell \le l(t) \le e, \qquad |f| + |df|_g \le \exp(P(N)), \qquad \ell = e^{-N\epsilon}. \tag{5.1}$$

Here and below P is a polynomial with coefficients depending only on the fixed prepared data, n, ϵ, and the fixed choices in the filling. The height $h = \eta f$, $\eta = \ell^{3/2}$, obeys

$$|h| \le C, \qquad |h(x)| \le Cr(x)^{-b_1} \quad \text{on the original end,} \tag{5.2}$$

where C is independent of N. At each fixed N, for every integer $j \ge 0$ there are constants $C_{N,j}$ and $a_N > 0$ such that on that end

$$|D^j f| \le C_{N,j} e^{-a_N r}, \qquad t, Z = O_j(r^{2-n}), \qquad \Delta_g t = O(r^{-n-\beta}). \tag{5.3}$$

The constants in (5.3), and higher derivative bounds on compact sets, may depend arbitrarily on fixed N.

The degree argument uses two homotopies. On a large outer truncation $\mathcal{M}_{N,R}$ impose $f = Z = 0$ on its sole boundary S_R. In the first homotopy replace K everywhere by $K_a = aK$, $0 \le a \le 1$, retaining the definitions of $\mathcal{T}, F, V$ with that replacement, and put

$$F = \eta f, \qquad \operatorname{div} V = u\Xi_a, \qquad \Xi_a = \delta_0(\mathcal{T} + \eta a_w |df|) + \rho - m_0 C_N(\rho_0 + v\rho_1). \tag{5.4}$$

The second homotopy keeps $K = 0$ and replaces Ξ_0 by $s\Xi_0$, $0 \le s \le 1$. The weights and C_N do not vary in either homotopy. The endpoint $(a, s) = (0, 1)$ is common to them. Until the arbitrary-trial problem is introduced, the estimates concern smooth fixed points on a truncation with $t \ge -\epsilon$. Fixed points with $\min t = -\epsilon$ will be excluded on the boundary of the degree domain. Solvability for arbitrary trials will be established separately, without that restriction.

5.1 Height comparison and the boundary lower bound

The trace equation controls the normalized height before any derivative estimate. Its end comparison also bounds the outer normal derivative; this will give $t > -\epsilon$ on the truncating sphere.

Lemma 5.2. *Use the prepared data and parameters of Theorem 5.1. There are constants C and r_1, independent of N, R and the homotopy parameters, such that every smooth pair (f, Z) on $\mathcal{M}_{N,R}$ with $f = 0$ on S_R, whose first component satisfies the trace equation $F = \eta f$ in (5.4) with $K_a = aK$, $0 \le a \le 1$, satisfies*

$$|\eta f| \le C, \qquad |\eta f| \le C(r^{-b_1} - R^{-b_1}) \quad (r_1 \le r \le R).$$

For each fixed N there is $R_{\min}(N)$ such that $R \ge R_{\min}(N)$ and $Z = 0$ on S_R imply $t > -\epsilon$ there. The same conclusions hold in the second homotopy, where in fact $f = 0$.

Proof. At a positive maximum of f, $df = 0$, $A_\chi = I$, and

$$\eta f = \operatorname{tr} K_a + l\Delta_g f \le \operatorname{tr} K_a.$$

The negative minimum gives the reverse bound. The tensor fields in the filling have a common pointwise bound, including along the product necks, so this proves the first assertion uniformly. If $K = 0$, both comparisons give $f = 0$.

On the end $K_a = 0$. At a point of contact with $H(r) = C(r^{-b_1} - R^{-b_1})$, considered as a test for $h = \eta f$, the two gradients agree. Thus the axial direction of A_χ is the radial direction. In the Euclidean metric the Hessian contraction is

$$A_\chi : \operatorname{Hess} H = Cb_1 r^{-b_1-2}((b_1+1)\chi - k).$$

The error from $g - \delta = O_2(r^{2-n})$ has absolute value at most $C' r^{2-n} C b_1 r^{-b_1-2}$, uniformly for $0 < \chi \le 1$. Since $b_1 + 1 < k$, the contraction is strictly negative for $r \ge r_1$, with r_1 independent of N. The height equation is

$$A_\chi : \operatorname{Hess} h = \frac{\eta\sqrt{D}}{l}\, h.$$

Consequently H and $-H$ are respectively an upper and a lower comparison function. For $R \ge 2r_1$, their constant C can be chosen to dominate the already established height bound on S_{r_1}. The maximum principle then proves the displayed end estimate.

Taking the one-sided derivative at the outer zero boundary gives

$$|df|_{S_R} \le C\eta^{-1} R^{-b_1-1}. \tag{5.5}$$

Tangential derivatives vanish there. For all real t we have $l(t) \le e$. The identity $Z = t + (2k)^{-1}\log(1 + l(t)^2|df|^2)$ therefore gives, on S_R,

$$0 \ge t \ge -\frac{1}{2k}\log(1 + e^2 C^2 \eta^{-2} R^{-2b_1-2}).$$

Choosing $R_{\min}(N)$ sufficiently large proves the assertion. □

Together, Lemmas 5.2 and 4.3 exclude every fixed point with $\min t = -\epsilon$. The height lemma supplies the bounds assumed by Lemma 4.3 and excludes the outer sphere by (5.5). That lemma then excludes an interior minimum in either homotopy. This argument concerns the boundary of the degree domain, where $\min t = -\epsilon$; solutions whose minima lie strictly below $-\epsilon$ remain outside its scope.

5.2 A polynomial upper bound before ellipticity

For $t \geq -\epsilon$, the implicit relation expresses the metric distortion directly in terms of Z. We therefore need an upper bound for this single scalar quantity. The proof first bounds its high-level mass, then combines a graph Sobolev inequality with exponential testing. The bound for Z must be polynomial in N: the enclosure argument in Section 6 needs metric distortion at most $\exp(P(N))$ while N increases.

Proposition 5.3. *Use the prepared data and parameters of Theorem 5.1. Every smooth solution (f, Z) on $\mathcal{M}_{N,R}$ of (5.4), $0 \leq a \leq 1$, or of its second homotopy with $K = 0$ and Ξ_0 replaced by $s\Xi_0$, $0 \leq s \leq 1$, with $f = Z = 0$ on S_R and $t \geq -\epsilon$, satisfies (5.1), with the same polynomial bounds for all $R \geq R_{\min}(N)$ and all homotopy parameters.*

Proof. Write $\sigma = |df|$, $B = K_a(w, \cdot)$, $A = A_\chi$, and

$$d\mu_D = \sqrt{D}\, dV_g, \qquad W = e^{(k-1)t} l, \qquad u = W\sqrt{D} = l e^{kZ-t}.$$

The lower bound for t implies

$$\ell \leq l \leq e, \quad t \leq Z, \quad D = e^{2k(Z-t)}, \quad \sigma \leq \ell^{-1} e^{k(Z+\epsilon)}. \tag{5.6}$$

It remains to bound Z from above.

High levels. On the support of m_0, the lower bound for t implies $-\epsilon \leq t \leq -\epsilon + N^{-1}$ and $l \asymp \ell$. In particular,

$$\eta a_w \sigma = \frac{\eta}{l}(\sqrt{D} - D^{-1/2}).$$

The weights are bounded above uniformly, and C_N is polynomial. It follows from $D = e^{2k(Z-t)}$ that there is a positive polynomial threshold $Z_0 = P(N)$ such that

$$\Xi_a \geq \delta_0 \mathcal{T} + \rho + \frac{\delta_0}{2} \eta a_w \sigma \quad \text{when } Z > Z_0. \tag{5.7}$$

For example, it suffices that $e^{kZ_0} \geq C(1+C_N)\eta^{-1}$, after increasing the fixed constant; its logarithm has polynomial size. Away from the support of m_0 the inequality is immediate.

Choose a smooth nondecreasing ζ that vanishes on $(-\infty, Z_0]$ and equals one on $[Z_0 + 1, \infty)$, with $0 \leq \zeta' \leq C$. Testing the divergence equation with $\zeta(Z)$ gives

$$\int \zeta(Z) u \Xi_a = -\int \zeta'(Z) u \{k|dZ|_A^2 + \langle B, dZ\rangle_A\} \leq C \int_{\{Z_0 < Z < Z_0+1\} \cap \operatorname{supp} K} u|K|^2.$$

The integration has no boundary term since $Z = 0 < Z_0$ on S_R. On the last strip $u = l e^{kZ-t} \leq \exp(P(N))$. The support of K consists of a fixed exterior compact set and the enlarged filling; it has volume $O(1 + N)$. Thus

$$\int_{\{Z > Z_0+1\}} u(\delta_0 \mathcal{T} + \rho + \tfrac{1}{2}\delta_0 \eta a_w \sigma)\, dV_g \leq \exp(P(N)). \tag{5.8}$$

This estimate holds for unbounded σ and controls the high levels of Z with the volume and slope weights needed on the graph. It supplies the initial integrability for an iteration on fixed base patches.

Fix $0 < \alpha \le k-1$, independent of N. Cover the enlarged core and a fixed extra end annulus by base coordinate patches of uniformly controlled size and geometry. The number of patches is $O(1+N)$, and ρ has a uniform positive lower bound on them. We claim that on each such patch Q,

$$\int_Q e^{\alpha Z}\, d\mu_D \le \exp(P(N)). \tag{5.9}$$

Below Z_0+1, (5.6) proves this directly. Above that level, when $D \le 2$ use

$$\frac{e^{\alpha Z}\sqrt{D}}{u} = l^{-1} e^{(\alpha-k+1)t} D^{\alpha/(2k)} \le \exp(P(N)).$$

When $D \ge 2$, use instead

$$u\eta a_w \sigma = \eta e^{(k-1)t}(D-1), \qquad e^{\alpha Z}\sqrt{D} = e^{\alpha t} D^{1/2+\alpha/(2k)}.$$

The ratio of the second expression to the first is at most $C\eta^{-1}e^{(\alpha-k+1)t} \le \exp(P(N))$ because $1/2+\alpha/(2k) \le 1$ and $t \ge -\epsilon$. The integral bound in (5.8) proves the claim.

Sobolev inequality on the product graph. We use the ordinary graph of f in $(Q \times \mathbb{R}, g + dz^2)$ so that its ambient geometry depends only on the fixed background patch. The warp $l(t)$ has no a priori derivative bound yet. We compare the ordinary graph's measure and inverse metric with $d\mu_D$ and A, and use $\mathcal{T}$ to control its mean curvature. The ordinary graph has measure and inverse metric

$$\sqrt{1+\sigma^2}\, dV_g, \qquad A_* = I - \frac{df \otimes df}{1+\sigma^2}.$$

The ratios of D and $1+\sigma^2$ lie between $\exp(-P(N))$ and $\exp(P(N))$, even when σ is unbounded. The axial eigenvalues d and $\chi = kd/(k+pv)$ differ by at most a factor $1+N/k$. Thus these graph norms compare with $d\mu_D$ and A with factors $\exp(P(N))$.

Its mean curvature is $(1+\sigma^2)^{-1/2} \operatorname{tr}_{A_*} \operatorname{Hess} f$. Write $K_a + H^f$ in its transverse and axial blocks. The transverse contraction is controlled by $|T|$, and the axial coefficient of A_* is at most $e^2 d$. Also $\sqrt{D}/(l\sqrt{1+\sigma^2}) \le \exp(P(N))$. Proposition 4.2 therefore gives

$$|H_{\text{graph}}| \le \exp(P(N))(1+\sqrt{\mathcal{T}}).$$

Extend the metric of a slightly larger fixed base patch to a closed manifold and use Nash's smooth isometric embedding theorem [35]; then take its product with the line. The additional second fundamental form is uniformly bounded. Such embeddings can be chosen from finitely many fixed models for the cap, transitions, and product necks. The Michael–Simon inequality [34], in the formulation of [42, Chapter 4, Section 6, Theorem 6.7], applied to compactly supported functions on this smooth n-dimensional graph, then gives, after its usual L^1 to L^2 conversion,

$$\left(\int |\varphi|^{2\kappa_n}\, d\mu_D\right)^{1/\kappa_n} \le \exp(P(N)) \int \left(|d\varphi|_A^2 + (1+\mathcal{T})\varphi^2\right) d\mu_D, \qquad \kappa_n = \frac{n}{n-2}. \tag{5.10}$$

To see the conversion explicitly, apply the L^1 inequality to $|\varphi|^{2(n-1)/(n-2)}$ and use Cauchy–Schwarz on its derivative and mean-curvature terms. The resulting common $L^{2n/(n-2)}$ factor cancels, with constants independent of the height and area of the graph.

Exponential testing. The graph inequality still contains $\mathcal{T}$ on its right side. The divergence equation controls that term together with the gradient of an exponential of Z; this is the estimate

that closes the iteration. Globally $\Xi_a \geq \delta_0 \mathcal{T} - P(N)$. Let ψ be a base cutoff compactly supported in Q and test the equation with $\psi^2 e^{2qZ}/W$. Put $\beta_W = d \log W = (k-1+p)dt$. Since $A \leq A_d$, scalar coercivity gives $|\beta_W|_A^2 \leq P(N)\mathcal{T}$. Integration by parts yields

$$\int \psi^2 e^{2qZ} \Xi_a \, d\mu_D = -\int e^{2qZ} (2q\psi^2 dZ - \psi^2 \beta_W + 2\psi d\psi) \cdot A(kdZ + B) \, d\mu_D.$$

Young's inequality bounds the $\beta_W \cdot A\, dZ$ term by $\delta_0 \mathcal{T}/8 + P(N)|dZ|_A^2$, the $\beta_W \cdot AB$ term by $\delta_0 \mathcal{T}/8 + P(N)$, and the $qB \cdot A\, dZ$ term by a fixed fraction of $kq|dZ|_A^2 + Cq$. The cutoff terms are bounded by a further fixed fraction of that gradient term and $P(N)(\psi^2 + |d\psi|^2)$. Consequently there is a polynomial $q_*(N) \geq \max\{1, \alpha\}$ such that, for $q \geq q_*(N)$,

$$\int e^{2qZ} \psi^2 (\mathcal{T} + q|dZ|_A^2) \, d\mu_D \leq P(N)(1+q) \int e^{2qZ} (\psi^2 + |d\psi|_g^2) \, d\mu_D. \tag{5.11}$$

Only dt occurs in this test; it does not differentiate p.

In (5.10), take $\varphi = \psi e^{qZ}$. The bound (5.11) controls its gradient and $\mathcal{T}$ terms. For nested base patches $Q' \Subset Q$ separated by a coordinate distance $b \leq 1$, this gives

$$\|e^Z\|_{2\kappa_n q, Q'} \leq \left[e^{P(N)}(1+q)^2 b^{-2}\right]^{1/(2q)} \|e^Z\|_{2q,Q}.$$

All norms here use $d\mu_D$. Iterate with $q_i = \kappa_n^i q_*$ and geometrically decreasing patch gaps. The sums of q_i^{-1} and $i q_i^{-1}$ converge, so

$$\sup_{Q'} e^Z \leq e^{P(N)} b^{-C} \|e^Z\|_{2q_*, Q}, \tag{5.12}$$

where C is independent of N.

Finally, interpolation with (5.9) bounds the norm on the right:

$$\|e^Z\|_{2q_*,Q} \leq \left(\sup_Q e^Z\right)^\theta e^{P(N)}, \qquad \theta = 1 - \frac{\alpha}{2q_*} < 1.$$

Choose expanding patches Q_j, all inside one fixed larger patch, with gaps comparable to 2^{-j}. The logarithms $M_j = \log \sup_{Q_j} e^Z$ satisfy $M_j \leq \theta M_{j+1} + P(N) + Cj$. Iteration gives

$$M_0 \leq \theta^m M_m + \sum_{j=0}^{m-1} \theta^j (P(N) + Cj) \leq \theta^m M_m + P(N).$$

The last polynomial is justified by $(1-\theta)^{-1} = 2q_*/\alpha \leq P(N)$. For each individual smooth solution the suprema on the largest patch are finite, so the first term tends to zero. We have proved $Z \leq P(N)$ on the enlarged core and the drift support. A larger maximum outside that set would have $K_a = 0$, $dZ = 0$, and, by (5.7), $\Xi_a > 0$; at the same point $u^{-1} \operatorname{div}(kuA\nabla Z) = kA :$ $\operatorname{Hess} Z \leq 0$, a contradiction.

For the second homotopy $f = 0$. A positive maximum above the penalty band has right side $s\rho > 0$ if $s > 0$, which is impossible. When $s = 0$ the solution is zero. This proves its upper bound as well. The estimates in (5.6) and the height bound prove the remaining bounds in (5.1). In particular,

$$d \geq e^{-2k(P(N)+\epsilon)}, \qquad \chi \geq \frac{k}{k+N} e^{-2k(P(N)+\epsilon)}.$$

The same bounds give positive upper and lower bounds for u at each fixed N. Thus both A_χ and kuA_χ are uniformly elliptic, with constants independent of the truncating radius and the homotopy parameter. These are the ellipticity bounds used in the following scalar estimates. □

5.3 A scalar gradient estimate with a measurable axial coefficient

The upper bound has made both scalar equations uniformly elliptic at each fixed N. Regularity still has to pass between them in a specific order. Dividing the trace equation by $l/\sqrt{D}$ gives

$$A_\chi : \operatorname{Hess}_g f = \frac{\sqrt{D}}{l}(\eta f - \operatorname{tr}_{A_\chi} K_a).$$

Its right side and Df are bounded, but its coefficients depend on Z, for which we have no continuity modulus. Schauder theory does not yet apply.

The gradient direction accounts for the matrix structure; the remaining unknown eigenvalue χ can be treated as merely measurable. The next lemma gives both a Hölder bound for Df and a local mass bound for $|D^2 f|^2$. This second conclusion is needed because the divergence source for Z contains $|D^2 f|^2$: the mass bound will give Hölder continuity of Z through the following scalar lemma. Once both Z and Df have a modulus, scalar Schauder estimates can begin. All constants in this regularity argument may depend arbitrarily on fixed N; the bounds needed as N increases were established above.

We prove the gradient estimate in two regimes. Strict rank-one absorption first gives $W^{2,s}$ control for some $s > 2$. A trace-reversed Hessian flux then shows that either the gradient norm drops on a smaller ball or the solution is close to an affine function with nonzero gradient. In the latter regime we freeze the direction while retaining the measurable axial coefficient; the axial derivative satisfies a scalar divergence equation. Iterating these two alternatives gives gradient continuity and the required Hessian mass bound.

Lemma 5.4 (Gradient-aligned rank-one equation)**.** *Let $n \geq 3$ and $0 < c \leq 1$. Let Q be a smooth metric coordinate ball, or a smooth half-ball with constant Dirichlet data on its face. Suppose that u is C^2 up to the face and satisfies almost everywhere*

$$A : \operatorname{Hess}_g u = G, \qquad A = \mathrm{id} - (1-\chi) e \otimes e, \qquad c \leq \chi \leq 1, \qquad e = \frac{\nabla_g u}{|\nabla_g u|_g} \quad \text{where } \nabla_g u \neq 0. \tag{5.13}$$

At a zero of the gradient, e may be any measurable unit vector. Assume $|\nabla_g u|_g + |G| \leq M$. On every smaller ball, including its portion of the Dirichlet face, there are estimates

$$[Du]_{C^\alpha} \leq C, \qquad \int_{B_r(x)\cap Q} |\operatorname{Hess}_g u|_g^2 \, dV_g \leq C r^{n-2+2\alpha}. \tag{5.14}$$

Here $\alpha \in (0,1)$ depends only on n, c, and C depends on n, c, M, the fixed coordinate geometry and the distance from the other chart boundaries. No modulus of continuity of χ enters these constants.

Proof. We use the scalar linear De Giorgi–Nash and weak Harnack estimates, the Krylov–Safonov interior estimate for strong nondivergence solutions, and the Euclidean Hessian estimates of Calderón and Zygmund; see [21]. The two structural arguments needed to apply these results are proved below. All coordinate norms of matrices use the Frobenius norm unless indicated otherwise.

1. A Hessian estimate with a strict absorption margin. Normalize the metric at the center of a chart. The coordinate principal matrix is

$$a^{ij} = g^{ij} - (1-\chi) e^i e^j.$$

If $g = I$, then $|I - a| = 1 - \chi \le 1 - c$. Consequently, on a sufficiently small chart, depending only on c and the metric modulus,

$$|I - a| \le 1 - \frac{c}{2}. \tag{5.15}$$

This remains true for the reflected metrics used below, since they are continuous and uniformly Lipschitz. The bound holds for arbitrary oscillation of e and χ.

Let $\mathcal{R} = D^2\Delta^{-1}$ on $\mathbb{R}^n$, with its matrix-valued range given the Frobenius norm. Plancherel's identity gives $\|\mathcal{R}\|_{L^2\to L^2} = 1$. Its L^4 norm is finite by the Calderón–Zygmund theorem. Interpolation therefore supplies $s_0 > 2$, depending only on n, c, such that

$$\|\mathcal{R}\|_{L^s\to L^s}(1 - c/2) < 1, \qquad 2 \le s \le s_0.$$

For a compactly supported v the identity $\Delta v = a : D^2 v + (I - a) : D^2 v$ now gives, by absorption,

$$\|D^2 v\|_{L^s} \le C\|a : D^2 v\|_{L^s}, \qquad s = 2, s_0.$$

Cutoffs, first-derivative interpolation and absorption on nested balls give the local version

$$\|D^2 v\|_{L^s(B_{r/2})} \le C\big(\|a : D^2 v\|_{L^s(B_r)} + r^{-2}\|v\|_{L^s(B_r)}\big), \qquad s = 2, s_0. \tag{5.16}$$

For completeness, let ψ be a smooth cutoff for the nested balls. Applying the compact-support estimate to ψv first produces an additional $(t - \rho)^{-1}\|Dv\|_{L^s(B_t)}$ on $B_\rho \Subset B_t$. The interpolation estimate on a slightly larger ball bounds this by $\varepsilon\|D^2 v\|_{L^s}$ plus $C\varepsilon^{-1}(t - \rho)^{-2}\|v\|_{L^s}$, with the scale factors absorbed into ε. Choose the coefficient of the first term smaller than the square of the geometric ratio of the successive gaps, and sum over increasing nested radii. The resulting geometric series gives (5.16). These estimates initially apply to the individually finite Sobolev norms of the functions under consideration and then, by approximation, to $W^{2,s}$ functions.

2. The exact flux and the constant-value reflection. Write $P = \nabla_g u$, $H = \operatorname{Hess}_g u$, and $q = |P|_g^2/2$. The equation implies $\Delta_g u = (1 - \chi)H(e, e) + G$. Since $\nabla_g q = H^\sharp P$, it gives the pointwise identity

$$A\nabla_g q - GP = \big(H^\sharp - (\Delta_g u)\,\mathrm{id}\big)P =: \mathcal{F}. \tag{5.17}$$

Both sides vanish when $P = 0$, so the identity is independent of the choice of axis at those points. For smooth u, commuting third derivatives gives

$$\operatorname{div}_g \mathcal{F} = |H|_g^2 - (\Delta_g u)^2 + \operatorname{Ric}_g(P, P). \tag{5.18}$$

For C^2 functions the same formula holds distributionally: approximate u in C^2 on compact subsets of each original smooth side. The flux and the right side converge uniformly, so integration against a smooth compactly supported test function passes to the limit. This argument establishes only the geometric divergence identity; the algebraic identity (5.17) is applied to the original solution, and no approximant is required to solve the equation. Neither identity differentiates χ.

Put $\gamma = 1 - (1 - c)^2 > 0$. Young's inequality, with the gap $\gamma/2$, gives

$$((1 - \chi)H(e, e) + G)^2 \le (1 - \gamma/2)|H|_g^2 + C_c G^2.$$

Thus, when $|P|_g \le 1$ and $s = 1 - |P|_g^2 \ge 0$,

$$\operatorname{div}_g(A\nabla_g s + 2GP) \le -\gamma|H|_g^2 + C_c G^2 + 2|\operatorname{Ric}_g|_g. \tag{5.19}$$

We record precisely how this inequality is used at the face. Subtract the constant boundary value and take Gaussian coordinates $g = dt^2 + h_{ab}(y,t)\,dy^a dy^b$ on $t \geq 0$. Extend u oddly and h evenly across $t = 0$, and extend χ evenly and G oddly. The reflected equation holds almost everywhere. Tangential derivatives of u vanish on the face, while the normal derivative agrees from both sides. Therefore u is C^1 after reflection and belongs locally to $W^{2,s}$ for every finite s individually. The metric is Lipschitz, smooth on either side, and s is a continuous $W^{1,2}$ function.

Let $J = \sqrt{\det g}$ and define the face mean curvature using the normal $+\partial_t$ by

$$\kappa(y) = \frac{1}{2} h^{ab}(y,0)\partial_t h_{ab}(y,0+), \qquad p(y) = u_t(y,0).$$

On the upper side, $H_{ab} = \frac{1}{2}(\partial_t h_{ab})p$ and hence $\mathcal{F}^t_+ = -\kappa p^2$. On the lower side the signs of these tangential Hessians reverse and $\mathcal{F}^t_- = \kappa p^2$. Since J is continuous across the face, the density-valued flux $J(A\nabla_g s + 2GP) = -2J\mathcal{F}$ has normal jump

$$j(y) = 4J(y,0)\kappa(y)p(y)^2.$$

Adding the coordinate vector

$$B_{\text{face}}(y,t) = -j(y)\mathbf{1}_{\{t>0\}}\partial_t \tag{5.20}$$

cancels that interface measure. Its coordinate divergence is $-j(y)\delta_{\{t=0\}}$: the vector has only a t component, so its divergence differentiates only in that direction. Its L^∞ norm is at most $C\|\kappa\|_\infty$ under the gradient normalization. All these statements are local, so the coefficient $j(y)$ only needs to be extended constantly in the normal direction over the chart.

On a normalized unit ball let the metric oscillation, Christoffel symbols, smooth-side curvature, face second fundamental form if present, and forcing be at most ε. The reflected and interior cases then both give the distributional inequality

$$\operatorname{div}(a_1 Ds + B_1) \leq -\gamma_1 |H|_g^2 + C\varepsilon, \qquad a_1^{ij} = Ja^{ij}, \qquad \|B_1\|_\infty \leq C\varepsilon, \tag{5.21}$$

where $B_1^i = 2JGP^i + B^i_{\text{face}}$ and $\gamma_1 > 0$ is fixed. The matrix a_1 is symmetric and uniformly elliptic with fixed constants. The face can be absent or can be a plane at any position in the ball.

3. Gradient-norm drop or approximation by a nonzero affine function. Fix $r_* \in (0, 1/32)$. For every $b_* > 0$ there are $k_* \in (r_*, 1)$ and $\varepsilon_* > 0$ with the following property. Suppose that $|\nabla_g u|_g \leq 1$ on B_1, the normalized geometric and forcing errors just listed are at most ε_*, and $u(0) = 0$. Then either

$$\sup_{B_{r_*}} |\nabla_g u|_g \leq k_*, \tag{5.22}$$

or there is an affine L such that

$$\frac{1}{2} \leq |DL| \leq 2, \qquad \sup_{B_{r_*}} |u - L| \leq b_* r_*. \tag{5.23}$$

Here is the compactness argument with its energy step. Otherwise choose a sequence with errors tending to zero, no approximation in (5.23), and gradient supremum tending to one on B_{r_*}. Its functions are uniformly Lipschitz and vanish at zero. The local estimate (5.16), applied to

$$a : D^2 u = G + a^{ij}\Gamma^\ell_{ij} u_\ell,$$

gives uniform local L^2 bounds for D^2u, hence for H and Ds. The weak Harnack inequality applies to the nonnegative weak supersolution s in (5.21) after discarding the negative Hessian term. It gives, for some $p_0 > 0$,

$$\left(\frac{1}{|B_{1/3}|}\int_{B_{1/3}} s^{p_0}\right)^{1/p_0} \le C\left(\inf_{B_{1/3}} s + \|B_1\|_\infty + \varepsilon\right) \longrightarrow 0.$$

The infimum tends to zero because $B_{r_*} \subset B_{1/3}$ and the gradient supremum there tends to one. Thus $s \to 0$ in measure locally.

To recover the Hessian energy, take a compactly supported cutoff ψ in $B_{1/3}$ and test the same inequality with $\psi^2(b-s)_+$, where $0 < b < 1$ is fixed. Its weak form is

$$\int (a_1 Ds + B_1)\cdot D\varphi \ge \gamma_1 \int |H|_g^2 \varphi - C\varepsilon \int \varphi, \qquad \varphi \ge 0.$$

Dropping the nonnegative Hessian term, expanding the test derivative and using Young's inequality proves

$$\int_{\{s<b\}} \psi^2 |Ds|^2 \le C_\psi b^2 + C_\psi \|B_1\|_\infty^2 + C_\psi b\|B_1\|_\infty + C_\psi \varepsilon b. \tag{5.24}$$

On a fixed compact subset, the integral of $|Ds|$ over $\{s < b\}$ has limsup at most Cb. Its integral over $\{s \ge b\}$ tends to zero by the uniform L^2 bound for Ds and convergence in measure of s. Letting $b \downarrow 0$ yields $Ds \to 0$ locally in L^1. Testing (5.21) with a fixed nonnegative smooth cutoff now yields $H \to 0$ locally in L^2. Since $\Gamma \to 0$ and Du is bounded, also $D^2u \to 0$ locally in L^2.

Uniform Lipschitz compactness produces a locally uniform limit. Its distributional Hessian vanishes, so it is affine. Poincaré's inequality and $D^2u \to 0$ give convergence of the gradients to its slope in local L^2. Since $s \to 0$ in measure and $g \to I$, that slope has norm one. This contradicts the failure of (5.23) and proves the dichotomy. Increasing k_* if necessary ensures $r_* < k_* < 1$.

Repeated gradient-norm drops will give decay directly. To handle the other alternative, we next improve approximation by a nonzero affine function. Its fixed slope determines the direction to freeze; the axial eigenvalue remains measurable throughout this argument.

4. The frozen-axis linear equation. Let e_0 be a fixed Euclidean unit vector. We first prove the quantitative statement

$$\|Y_0\|_{C^{1,\alpha_1}(B_{1/2})} \le C\|Y_0\|_{W^{2,2}(B_1)} \quad \text{if} \quad \Delta_{e_0^\perp} Y_0 + \chi \partial_{e_0}^2 Y_0 = 0 \text{ a.e. in } B_1, \tag{5.25}$$

for $Y_0 \in W^{2,2}$ and arbitrary measurable $c \le \chi \le 1$. Rotate so that $e_0 = \partial_n$ and put $w = \partial_n Y_0 \in W^{1,2}$. Differentiating the displayed equation only in the sense of distributions gives

$$\sum_{a<n} \partial_a^2 w + \partial_n(\chi \partial_n w) = 0, \quad \text{or} \quad \operatorname{div}\big((I - e_0\otimes e_0 + \chi e_0 \otimes e_0) Dw\big) = 0. \tag{5.26}$$

This is a scalar divergence equation for a $W^{1,2}$ weak solution with a measurable uniformly elliptic matrix. The De Giorgi–Nash estimate gives $w \in C^{\alpha_1}$ on smaller balls, for a fixed $0 < \alpha_1 < 1$, quantitatively in its L^2 norm. Testing with $\psi^2(w - w(x))$, where ψ is a cutoff on the ball, and using that Hölder bound gives, uniformly for balls in a smaller fixed ball,

$$\int_{B_r(x)} |Dw|^2 \le C r^{n-2+2\alpha_1}. \tag{5.27}$$

The ordinary Laplacian of Y_0 is $f = (1-\chi)\partial_n w$. By Cauchy–Schwarz,

$$\int_{B_r(x)} |f| \le C r^{n-1+\alpha_1}. \tag{5.28}$$

Here is why this controls every component of DY_0. Cut f off on an intermediate ball and form its Newton potential V. For two points at distance r in a smaller ball, the part of DV within distance $2r$ of either point is bounded by

$$C \sum_{j \ge 0} (2^{-j} r)^{1-n} (2^{-j} r)^{n-1+\alpha_1} \le C r^{\alpha_1}.$$

For distances $R \ge 2r$, the difference of the two gradient kernels is at most CrR^{-n}. Summing over dyadic annuli and using (5.28) gives

$$Cr \sum_{r \le R \le 1} R^{\alpha_1 - 1} \le C r^{\alpha_1}.$$

The farther part is bounded by the total L^1 norm of f and contributes $O(r)$. The same estimates show that V, DV are bounded. The difference $Y_0 - V$ is harmonic on the smaller ball, so interior harmonic estimates control its gradient and its gradient Hölder norm by its L^2 norm. Together with the initial $W^{2,2}$ bound this proves (5.25). Taking slightly nested balls in this argument yields the stated $B_{1/2}$ estimate.

5. Improvement near a nonzero affine function. Fix a universal gradient bound, say $|\nabla_g u|_g \le 4$. There are $\alpha_2 \in (0, \alpha_1)$, $\lambda \in (0, 1/8)$ and small $b_0, \delta > 0$, depending only on n, c, with the following property. If L is affine and

$$\frac{1}{4} \le |DL| \le 4, \qquad \|u - L\|_{L^\infty(B_1)} \le b, \quad 0 < b \le b_0, \qquad \|g - I\|_\infty + \|\Gamma\|_\infty + \|G\|_\infty \le b\delta, \tag{5.29}$$

then there is an affine L' such that

$$\sup_{B_\lambda} |u - L'| \le b\lambda^{1+\alpha_2}, \qquad |DL' - DL| \le Cb. \tag{5.30}$$

The metrics may again be smooth or reflected.

Set $Y = (u - L)/b$. Its coordinate equation is

$$a : D^2 Y = b^{-1}(G + a^{ij}\Gamma^\ell_{ij} u_\ell),$$

whose right side has absolute value at most $C\delta$. The local estimate (5.16) and interpolation give

$$\|Y\|_{W^{2,s_0}(B_{7/8})} \le C. \tag{5.31}$$

Put $e_0 = DL/|DL|$ and $a_0 = I - (1-\chi) e_0 \otimes e_0$. Since $Du = DL + bDY$ and $|DL| \ge 1/4$, the direction projections obey

$$\|a - a_0\|_{L^{s_0}(B_{7/8})} \le Cb, \qquad \|a - a_0\|_\infty \le C.$$

Indeed the map taking a nonzero vector to its unit-direction projection has the estimate $|\Pi(v) - \Pi(q)| \le C \min(1, |v-q|/|q|)$ when $q \ne 0$. At $v = 0$ any assigned unit projection satisfies the same estimate after enlarging C. The change from Du to its metric gradient contributes $O(\|g - I\|_\infty)$.

Hölder's inequality, with $1/2 = 1/s_0 + 1/p$ and interpolation of the bounded matrix difference in L^p, therefore gives

$$R := a_0 : D^2 Y, \qquad \|R\|_{L^2(B_{3/4})} \le C(\delta + b^\sigma), \quad \sigma = \min\{1, (s_0 - 2)/2\} > 0. \tag{5.32}$$

We remove this residual in $W^{2,2}$, without any Sobolev embedding into C^0. On the convex ball $\mathcal{B} = B_{3/4}$, integration by parts for a zero-trace function gives

$$\int_{\mathcal{B}} |D^2 v|^2 \le \int_{\mathcal{B}} |\Delta v|^2.$$

The omitted boundary term is the nonnegative integral of the sphere's mean curvature times $(\partial_\nu v)^2$; the identity extends by density to $W^{2,2} \cap W_0^{1,2}$. Thus the Dirichlet solution operator Δ_D^{-1} satisfies $\|D^2 \Delta_D^{-1} f\|_2 \le \|f\|_2$. The map

$$v \longmapsto \Delta_D^{-1}(R + (1 - \chi)\partial_{e_0}^2 v)$$

is a contraction, with factor at most $1 - c$, on $W^{2,2}(\mathcal{B}) \cap W_0^{1,2}(\mathcal{B})$ equipped with the Hessian norm. Its fixed point satisfies

$$a_0 : D^2 v = R, \qquad \|v\|_{W^{2,2}(\mathcal{B})} \le C\|R\|_2.$$

Consequently $Y_0 = Y - v$ satisfies the homogeneous frozen equation and has bounded $W^{2,2}(\mathcal{B})$ norm. By (5.25), $\|Y_0\|_{C^{1,\alpha_1}(B_{3/8})} \le C$.

Independently, Y has a uniform C^θ bound on $B_{1/2}$ by the Krylov–Safonov estimate applied to its unfrozen equation: the matrix a is measurable and uniformly elliptic, its source is bounded, and $\|Y\|_\infty \le 1$. The function Y is individually $W^{2,n}$ (also across the face), so the strong-solution version of that theorem applies. Hence $v = Y - Y_0$ has uniformly bounded C^θ norm on $B_{3/8}$. Its small L^2 norm now implies

$$\|v\|_{L^\infty(B_{1/3})} \le C\|v\|_2^{2\theta/(n+2\theta)} \le C(\delta + b^\sigma)^{2\theta/(n+2\theta)}. \tag{5.33}$$

To see the first inequality for small $\|v\|_2$, a value $|v(x)| = m$ forces $|v| \ge m/2$ on a ball of radius $(m/(2C))^{1/\theta}$; integrating its square gives the stated exponent.

Choose $\alpha_2 < \alpha_1$, then λ so small that $C\lambda^{1+\alpha_1} \le \frac{1}{2}\lambda^{1+\alpha_2}$, and finally δ, b_0 so small that the right side of (5.33) is at most $\frac{1}{2}\lambda^{1+\alpha_2}$. The affine function

$$L' = L + b(Y_0(0) + DY_0(0) \cdot x)$$

then satisfies (5.30).

6. Iteration, including its transition between the two alternatives. Decrease b_0 further so that $Cb_0/(1 - \lambda^{\alpha_2}) \le 1/8$, where C is the slope constant in (5.30). Apply the dichotomy with $b_* = b_0$. At each center, first normalize g there and choose a fixed sufficiently small radius R_0. Divide heights by KR_0, where $K \ge \max(1, \|\nabla_g u\|_\infty)$, and subtract the center value. On the rescaled unit ball the gradient is at most one. The forcing is multiplied by R_0/K, the connection and face second form by R_0, and smooth-side curvature by R_0^2. Metric oscillation from its center value is $O(R_0)$. Choose R_0 uniformly so that these errors satisfy both the dichotomy threshold and $b_0\delta$.

Denote this normalized function by u_0. After j successive norm drops, the normalized function is

$$u_j(x) = \frac{u_0(r_*^j x)}{r_*^j k_*^j}.$$

It again has gradient at most one. Its forcing has acquired the factor $(r_*/k_*)^j$, and its metric, connection and face errors have acquired at least the factor r_*^j. All required smallness conditions persist because $r_* < k_*$. If this branch continues forever, then

$$\sup_{B_{r_*^j}} |u_0 - u_0(0)| \le C r_*^j k_*^j = C(r_*^j)^{1+\alpha_d}, \qquad \alpha_d = \frac{\log k_*}{\log r_*} \in (0,1).$$

If the affine alternative first occurs after j norm drops, rescale its ball once by r_*, retaining its current gradient normalization. The resulting function U has a unit-ball affine error at most b_0 and an initial slope in $[1/2, 2]$. Iterate the affine improvement using

$$U_i(x) = \frac{U(\lambda^i x)}{\lambda^i}, \qquad b_i = b_0 \lambda^{i\alpha_2}.$$

Only constants are subtracted from the nonlinear unknown if a renormalization of its value is needed; the affine functions are used solely as comparison functions. In particular the original gradient-aligned equation remains its equation at every step. The relative error threshold remains valid: geometric and forcing errors acquire a factor λ^i, and

$$\lambda^i b_0 \delta \le b_i \delta \quad \text{because} \quad \alpha_2 < 1.$$

The gradient remains bounded by its entry normalization, and the affine slopes change by at most Cb_i. Their total change is at most $1/8$, so all slopes remain within $[1/4, 4]$, as required by (5.29).

Take $0 < \alpha \le \min(\alpha_d, \alpha_2)$. Before affine entry the constant approximations above have error $Cr^{1+\alpha}$. At entry the original radius is $r_e = r_*^{j+1}$ and the gradient normalization is $a_e = k_*^j$. At the subsequent radii $r = r_e \lambda^i$, the affine errors in the original u_0 are bounded by

$$C a_e r_e \lambda^{i(1+\alpha_2)} = C r^{1+\alpha} \left(a_e r_e^{-\alpha}\right) \lambda^{i(\alpha_2 - \alpha)} \le C r^{1+\alpha},$$

because $a_e r_e^{-\alpha} = k_*^{-1} r_e^{\alpha_d - \alpha} \le k_*^{-1}$. Thus the constants remain uniform across every possible entry scale. Intermediate radii are handled by restricting the approximation from the next larger radius, whose ratio is one of the two fixed numbers r_*^{-1} and λ^{-1}.

We have proved that at every center and every sufficiently small radius there is an affine approximation with uniform error $Cr^{1+\alpha}$. The elementary affine Campanato criterion gives the gradient estimate: comparison of the affine functions on concentric radii $r, r/2$ bounds their slope difference by Cr^α; summation identifies the limiting slope with Du at the center. Comparing the approximations at two centers in a common ball of radius comparable to their distance then bounds the difference of these limiting slopes by that distance to the power α. This proves the first estimate in (5.14), also on the reflected face, with constants uniform on the smaller original chart.

Finally let $L_x(y) = u(x) + Du(x) \cdot (y - x)$ in fixed coordinates. The gradient estimate gives $\|u - L_x\|_{L^\infty(B_{2r}(x))} \le Cr^{1+\alpha}$. Since

$$a : D^2(u - L_x) = G + a^{ij} \Gamma_{ij}^\ell u_\ell,$$

the local estimate (5.16) with $s = 2$ yields

$$\|D^2 u\|_{L^2(B_r(x))} \le C\big(r^{n/2} + r^{n/2 - 1 + \alpha}\big).$$

The bounded connection term converting $D^2 u$ to $\mathrm{Hess}_g u$ has L^2 norm $O(r^{n/2})$. Squaring the estimate and using $\alpha < 1$ proves the Hessian estimate in (5.14). Restricting the reflected estimate to the original half-ball proves the face version. □

5.4 A bounded scalar equation with quadratic gradient growth

The trace estimate now supplies the local mass control needed for the second equation. Its source has quadratic growth in DZ and also contains $|D^2 f|^2$. The latter need not yet have a pointwise bound uniform over solutions: the estimate $\int_{B_r} |D^2 f|^2 \le C r^{n-2+2\alpha}$ from the preceding lemma is the information available at this stage.

The following bounded-solution estimate uses this mass bound. Two exponential changes of variable absorb the quadratic DZ term, and the mass bound makes the remaining source potential small on short scales. Oscillation decay then gives a Hölder modulus for Z while its principal coefficients are still treated as measurable.

Lemma 5.5 (Natural-growth scalar estimate)**.** *Let $n \ge 3$ and let Q be a Euclidean coordinate ball, or a smooth flattened half-ball with constant Dirichlet value on its face. Suppose a bounded classical function Z satisfies weakly*

$$\begin{aligned} -\partial_i(A^{ij}\partial_j Z) = \partial_i B^i + S, \qquad \lambda I \le A \le \Lambda I, \qquad A = A^{\mathsf{T}}, \\ |B| \le C_0, \qquad |S| \le C_0(1 + |DZ|^2 + F_0), \end{aligned} \tag{5.34}$$

where A is measurable and $F_0 \ge 0$ is bounded for each individual solution. Assume, for some $a \in (0,1)$ and $r_0 > 0$, and every ball of radius $0 < r \le r_0$ with center in the coordinate patch, that the measure $\nu = (1 + F_0)\,dx$ obeys

$$\nu(B_r \cap Q) \le C_1 r^{n-2+a}. \tag{5.35}$$

Then Z has a uniform C^b bound on smaller patches, including the constant-Dirichlet face, for some $b \in (0, a)$. Its exponent and bound depend only on $n, \lambda, \Lambda, C_0, C_1, a, r_0$, the bound for $|Z|$, the fixed chart geometry and the distance from the remaining patch boundary. They use neither a modulus of continuity for A nor a uniform pointwise bound for F_0 or $D^2 Z$.

Proof. If a Dirichlet face is present, first subtract its constant boundary value. This changes neither the equation nor its source bounds. Flatten the face as $\{x_n = 0\}$ and write $J = \operatorname{diag}(1, \dots, 1, -1)$. For $x_n < 0$ extend the four quantities by

$$Z^e(x) = -Z(Jx), \quad A^e(x) = JA(Jx)J, \quad B^e(x) = -JB(Jx), \quad S^e(x) = -S(Jx). \tag{5.36}$$

The zero trace makes Z^e a $W^{1,2}$ function, and ellipticity is preserved. To verify the equation across the face, test the original equation on the upper half with $\varphi(x) - \varphi(Jx)$, which has zero trace. Changing variables in the reflected half gives precisely (5.34) for the extended quantities. Equivalently, the normal component of the full flux $ADZ + B$ agrees in the weak flux sense on the two sides. Thus the reflected equation has no measure on the face. Reflect the nonnegative density $1 + F_0$ evenly. The ball-mass bound changes by at most a fixed multiplicative factor. The reflected coefficients need not be smooth.

Exponential absorption and potentials. Recall $\nu = (1 + F_0)\,dx$, with the even extension just described where needed. We give the two-sign exponential argument for (5.34). Put $Q_\varepsilon = e^{\varepsilon L Z}$ for $\varepsilon \in \{-1, 1\}$, and let $\mathcal{L} = -\partial_i(A^{ij}\partial_j)$. The weak chain rule gives

$$\mathcal{L} Q_\varepsilon = \partial_i(\varepsilon L Q_\varepsilon B^i) - L^2 Q_\varepsilon B \cdot DZ + \varepsilon L Q_\varepsilon S - L^2 Q_\varepsilon ADZ \cdot DZ. \tag{5.37}$$

Indeed it follows directly by testing the equation for Z with $\varepsilon L Q_\varepsilon \varphi$. Young's inequality bounds the second term on the right by $\frac{1}{2}\lambda L^2 Q_\varepsilon |DZ|^2 + CL^2 Q_\varepsilon$. Taking $L \geq 2C/\lambda$ absorbs also the quadratic part of $\varepsilon L Q_\varepsilon S$. Since Z is bounded and L is now fixed, we obtain, for both signs,

$$\mathcal{L} Q_\varepsilon \leq \operatorname{div} B_\varepsilon + C\nu, \qquad B_\varepsilon = \varepsilon L Q_\varepsilon B, \qquad \|B_\varepsilon\|_\infty \leq C. \tag{5.38}$$

This uses no derivative of B, A, or the density of ν.

On a ball B_r in the resulting full patch, let v_ε solve

$$\mathcal{L} v_\varepsilon = \operatorname{div} B_\varepsilon + C\nu, \qquad v_\varepsilon|_{\partial B_r} = 0. \tag{5.39}$$

The density of ν is bounded for each individual classical solution, so this solve and the following tests can first be made in $W^{1,2}$. Only the uniform mass bound in (5.35) enters the estimate. The part with right side $\operatorname{div} B_\varepsilon$ has absolute value at most $Cr\|B_\varepsilon\|_\infty$. This is the scaled estimate for a bounded source in divergence form: testing by positive and negative level truncations gives $\int |D(v-k)_+|^2 \leq C\|B_\varepsilon\|_\infty^2 |\{v > k\}|$, and Sobolev level iteration on the unit ball proves the asserted bound after rescaling.

For the measure part, the scalar Dirichlet Green function of a uniformly elliptic divergence operator in dimension $n \geq 3$ satisfies $0 \leq G_r(x,y) \leq C|x-y|^{2-n}$. For bounded measurable coefficients this follows from the whole-space Green-function comparison [30, Section 7]: extend A uniformly elliptically off the ball, and compare the whole-space and zero-Dirichlet Green functions by the maximum principle. Dividing B_r into annuli about any $x \in B_r$ gives

$$\int_{B_r} G_r(x,y)\, d\nu(y) \leq C \sum_{j \geq 0} (2^{-j}2r)^{2-n} \nu\big(B_{2^{-j}2r}(x)\big) \leq C \sum_{j \geq 0} (2^{-j}2r)^a \leq Cr^a.$$

We work sufficiently far inside the coordinate patch that the mass bound applies to the balls in this sum. Consequently

$$\|v_\varepsilon\|_{L^\infty(B_r)} \leq E_r, \qquad E_r = C(r + r^a). \tag{5.40}$$

Oscillation decay. Here is the oscillation reduction, including the role of the two signs. Let $m = \inf_{B_r} Z$, $M = \sup_{B_r} Z$, and $\omega = M - m$. By (5.38) and (5.39), the functions

$$W_+ = e^{LM} - Q_+ + v_+ + E_r, \qquad W_- = e^{-Lm} - Q_- + v_- + E_r$$

are nonnegative weak supersolutions of $\mathcal{L}W \geq 0$ on B_r. At least one of the sets

$$\{Z \leq (m+M)/2\} \cap B_{r/2}, \qquad \{Z \geq (m+M)/2\} \cap B_{r/2}$$

has at least half the measure of $B_{r/2}$. On the first set $e^{LM} - Q_+ \geq c\omega$, and on the second $e^{-Lm} - Q_- \geq c\omega$, because the exponential and its inverse are uniformly Lipschitz on the fixed bounded range of Z. For the corresponding sign the weak Harnack inequality, with its fixed exponent $q > 0$, yields

$$c\omega \leq C \left(\frac{1}{|B_{r/2}|} \int_{B_{r/2}} W_\varepsilon^q \right)^{1/q} \leq C \inf_{B_{r/4}} W_\varepsilon.$$

Subtracting $v_\varepsilon + E_r$, whose absolute value is at most $2E_r$, and using the same Lipschitz comparison with Z, improves one of its two extremes. Hence, for a fixed $\theta \in (0,1)$,

$$\operatorname{osc}_{B_{r/4}} Z \leq (1-\theta) \operatorname{osc}_{B_r} Z + C(r + r^a). \tag{5.41}$$

Iteration gives a uniform C^b bound for any fixed

$$0 < b < \min\left\{a, \frac{-\log(1-\theta)}{\log 4}\right\}.$$

Applying the same conclusion to the reflected equation proves the estimate up to the constant Dirichlet face.

□

5.5 Regularity of the coupled fixed points

We apply the two local lemmas in a definite order. The trace equation first controls Df and the local mass of $|D^2 f|^2$. The scalar divergence lemma then controls the oscillation of Z. These two Hölder bounds permit scalar Schauder estimates, followed by an absorption of the remaining quadratic gradient term. Each step has constants uniform in the outer truncation at fixed N.

Proposition 5.6. *Use the prepared data and parameters of Theorem 5.1, and fix N. Smooth solutions (f, Z) of either homotopy problem in Proposition 5.3, on $\mathcal{M}_{N,R}$ with $f = Z = 0$ on S_R and $t \geq -\epsilon$, have bounds of every fixed order on coordinate balls and on outer Dirichlet half-balls of a sufficiently small fixed size. Their size and the constants may depend arbitrarily on N, but are independent of $R \geq R_{\min}(N)$ and of the homotopy parameter. In particular their $C^{1,\gamma}$ norms are bounded on each fixed truncation, for any chosen $0 < \gamma < 1$.*

Proof. 1. The trace estimate and divergence form. Fix N and a member of the homotopy. Proposition 5.3 gives uniform bounds for $|f|$, $|df|$, and $|Z|$, and uniform ellipticity on coordinate balls of a fixed small radius. The radius and constants may depend on N, but can be chosen independently of the outer truncation. Lemma 5.4, applied to the trace equation, gives some $a \in (0, 1)$ such that

$$[df]_{C^a} \leq C, \qquad \int_{B_r \cap \mathcal{M}_{N,R}} |D^2 f|^2 \, dx \leq C r^{n-2+2a}, \qquad 0 < r \leq r_0. \tag{5.42}$$

Here and below the boundary version uses a smooth half-ball with constant Dirichlet data; coordinate and covariant Hessians differ by a bounded quantity. Each occurrence of a ball in a mass estimate allows all centers in the coordinate patch, not just one fixed center.

The implicit relation defining t has derivative $1 + pv/k \geq 1$ with respect to t. On the bounded range now under consideration it therefore defines a smooth function of (x, Z, Df), whose relevant partial derivatives are bounded. In particular,

$$|Dt| \leq C(1 + |DZ| + |D^2 f|). \tag{5.43}$$

All the expressions in the equations are smooth before the block decomposition, including at $df = 0$. Since $\mathcal{T}$ is quadratic in H^f, dt, and K, (5.43) and the bounded zeroth- and first-order quantities imply the following coordinate form of the second equation:

$$-\partial_i(A^{ij}\partial_j Z) = \partial_i B^i + S, \qquad \lambda I \leq A \leq \Lambda I, \qquad |B| \leq C, \qquad |S| \leq C(1 + |DZ|^2 + |D^2 f|^2). \tag{5.44}$$

The matrix A is symmetric. With volume densities included, A represents kuA_χ, B represents $uA_\chi K(w, \cdot)$, and $S = -\sqrt{\det g}\, u\Xi$. The same bounds hold uniformly along both stages of the homotopy; scaling Ξ towards zero improves its absolute-value bound. The matrix A is still treated as measurable.

Apply Lemma 5.5 with $F_0 = |D^2 f|^2$. Its coefficient and source hypotheses are precisely (5.44). On patches of size at most one, (5.42) gives $\nu(B_r) \le C r^{n-2+a}$ for $\nu = (1+|D^2 f|^2)\,dx$. The lemma therefore gives a uniform C^b bound for Z for some $0 < b < a$, up to the constant outer Dirichlet face as well. Both Z and Df now have the continuity needed to improve the trace equation by Schauder theory.

2. Absorbing the quadratic gradient term. The coefficients of the trace equation are smooth functions of (x, Z, Df). The preceding estimate and (5.42) therefore give a uniform C^c modulus for those coefficients and the trace right side, with $c = \min(a, b) > 0$. Interior and Dirichlet Schauder estimates give

$$\|f\|_{C^{2,c}} \le C \tag{5.45}$$

on smaller patches of the same uniform type. This use of Schauder requires only the now established Hölder modulus of (Z, Df).

Expand the divergence equation on each original ball or half-ball. Since $A = A(x, Z, Df)$ and $B = B(x, Z, Df)$, their derivatives consist of bounded terms, terms linear in DZ, and terms linear in $D^2 f$. The bound (5.45) and the formula for Dt show that the expanded equation has the form

$$A^{ij} D_{ij} Z = H, \qquad |H| \le C(1 + |DZ|^2), \qquad \|A\|_{C^c} \le C. \tag{5.46}$$

The bound on the Hölder norm of A does not presuppose a bound for DZ. Coordinate lower-order terms are included in H.

Fix any finite $s > n$. Freezing the now Hölder principal matrix on sufficiently small balls gives the interior and flat-Dirichlet $W^{2,s}$ estimates with uniformly bounded coefficient in front of $\|H\|_s$. On nested patches $Q_r \subset Q_{2r}$ these give

$$\|D^2 Z\|_{s,Q_r} \le C_s \|DZ\|^2_{2s,Q_{2r}} + C_{s,r}. \tag{5.47}$$

The last constant includes the bounded height, the constant term in (5.46), and the scaled cutoff terms. For a ball or a fixed-shape half-ball the scaled interpolation inequality is

$$\|DZ\|^2_{2s,Q} \le C_s \operatorname{osc}_Q Z \, \|D^2 Z\|_{s,Q} + C_s r^{n/s-2} (\operatorname{osc}_Q Z)^2, \tag{5.48}$$

where Q has size r. Indeed apply the usual second-order interpolation inequality on the unit ball or half-ball after subtracting a constant and rescaling. On a half-ball, the constant Dirichlet value belongs to the closed range of Z, so it can be used for this subtraction. The constants depend only on the uniform chart shapes, s, and the dimension.

The absorption constant is independent of the number of patches in the truncation. For a fixed sufficiently small r, let

$$\mathcal{M}_r = \sup_x \|D^2 Z\|_{s,Q_r(x)}.$$

It is finite for each individual classical solution on a truncation. The uniform local geometry allows each $Q_{2r}(x)$ to be covered by at most J_0 patches of size r, with J_0 independent of the outer radius. Thus (5.47) and (5.48) imply

$$\mathcal{M}_r \le C_s J_0^{1/s} \sup_x \operatorname{osc}_{Q_{2r}(x)} Z \, \mathcal{M}_r + C_{s,r}.$$

Choose r small enough, using the uniform C^b bound already proved, that the displayed coefficient of $\mathcal{M}_r$ is at most $1/2$. This gives a uniform local $W^{2,s}$ bound up to the outer face, independently of the total number of patches. Sobolev embedding now bounds Z in $C^{1,d}$ for $d < 1 - n/s$.

3. Bootstrap. The trace equation consequently bounds f in $C^{3,d'}$ for some $d' > 0$. Its coefficients now have one Hölder derivative; the expanded second equation has Hölder right side, giving $Z \in C^{2,d'}$. Repetition yields bounds of every fixed order, since the background data are smooth. More explicitly, once $Z \in C^{j,d'}$ and $f \in C^{j+1,d'}$ for $j \geq 1$, the trace equation first gives $f \in C^{j+2,d'}$, and the divergence equation then gives $Z \in C^{j+1,d'}$. Shrinking the exponent once at the first step suffices. The argument applies initially to the $C^{2,\gamma}$ fixed points constructed below: their individual Hessian norms are already finite, and smoothness is the conclusion of the bootstrap. All constants in this paragraph may depend arbitrarily on fixed N and the chosen derivative order.

□

5.6 The trace equation for an arbitrary trial

The degree argument requires a map defined beyond the region $t > -\epsilon$. The next lemma supplies it. Its estimates are separate from Proposition 5.3: here a trial Z is prescribed, and no lower bound for its associated t is assumed.

Lemma 5.7. *Let $\mathcal{D}$ be a compact smooth n-dimensional Riemannian manifold with nonempty smooth boundary, with $n \geq 3$ and $k = n - 1$, and let K be a smooth symmetric tensor on $\mathcal{D}$. Fix $N > k$, $0 < \gamma < 1$, and a positive constant η. Use $p(t) = N\vartheta(Nt)$ and $l(t) = \exp(\int_0^t p(s)\,ds)$, with the smooth cutoff ϑ of (4.1), and define D, t, A_χ by (4.2) and (4.3). Put $K_a = aK$. For every $z \in C^{1,\gamma}(\mathcal{D})$ and $a \in [0,1]$, the Dirichlet problem*

$$\operatorname{tr}_{A_\chi}\left(K_a + \frac{l}{\sqrt{D}}\operatorname{Hess} f\right) = \eta f, \qquad f|_{\partial\mathcal{D}} = 0, \qquad t + \frac{1}{2k}\log(1 + l(t)^2|df|^2) = z$$

has a unique solution $f = f_a[z] \in C^{3,\gamma}$. The solution map is continuous from $[0,1] \times C^{1,\gamma}(\mathcal{D})$ to $C^{3,\gamma}(\mathcal{D})$ and maps bounded sets to bounded sets. All constants may depend on $N, \eta, \mathcal{D}, g, K$ and the prescribed bound for z. There is no restriction on the minimum of the implicitly determined t.

Proof. 1. Coefficient growth for arbitrary trials. Let $b = \sqrt{D}/l$. On a bounded trial range the coefficients satisfy, for every $\sigma = |df|$,

$$b \leq C(1 + \sigma), \qquad \frac{c}{1+\sigma} \leq \chi \leq 1. \tag{5.49}$$

We verify the bounds also at large slope, allowing t to decrease without a prescribed lower bound. The implicit relation implies $t < 0$ when σ is sufficiently large. There $p = N$, $l = e^{Nt}$, and

$$e^{2kz} = e^{2kt} + \sigma^2 e^{2(k+N)t}.$$

Uniformly on a bounded z range, as $\sigma \to \infty$,

$$t = \frac{kz - \log\sigma}{k+N} + o(1), \qquad d \asymp \sigma^{-2k/(k+N)}, \qquad \chi \asymp \frac{k}{k+N}\sigma^{-2k/(k+N)}.$$

The exponent is less than one because $N > k$. Moreover $b = \sigma/\sqrt{v} \asymp \sigma$ there. The remaining bounded slope range is compact in (z, σ), and the coefficients are smooth and positive on it. This proves (5.49).

2. A global gradient bound. Continue from $\Delta_g f = f$ by solving, for $0 \leq q \leq 1$,

$$A_q : \operatorname{Hess} f - c_q f = -qb\operatorname{tr}_{A_\chi} K_a, \qquad A_q = (1-q)I + qA_\chi, \qquad c_q = 1 - q + q\eta b. \tag{5.50}$$

The scalar c_q is positive. Extrema give a common height bound $\|f\|_\infty \le H$, since

$$\frac{qb|\operatorname{tr}_{A_\chi} K_a|}{c_q} \le \frac{C}{\eta}.$$

On this height range the right side of the equation written as $A_q : \operatorname{Hess} f = G(x, f, df)$ satisfies, for a covector ξ,

$$|G(x, f, \xi)| \le C(1 + |\xi|), \tag{5.51}$$

uniformly in q, a. The eigenvalues of A_q are one in the transverse directions and $\chi_q = 1 - q + q\chi \ge c/(1 + |\xi|)$ along ξ.

Here is a gradient estimate using precisely this lower axial bound. Choose K_0 sufficiently large, then a positive distance d_0 smaller than the boundary tubular radius, the injectivity radius of a smooth extension of the compact domain, and $(2K_0)^{-1}$. For $M > 0$ set

$$\psi(d) = K_0^{-1} \log(1 + K_0 M d), \qquad \psi'' = -K_0(\psi')^2. \tag{5.52}$$

By increasing M we ensure both $\psi' \ge 1$ on $[0, d_0]$ and $\psi(d_0) > 2H$. Indeed $\psi'(d_0)$ tends to $(K_0 d_0)^{-1} > 2$, while $\psi(d_0)$ tends to infinity.

First compare with $\pm\psi(d(x, \partial\mathcal{D}))$ in the boundary collar. At a contact, the gradient is normal and has length $\sigma = \psi'$. The tangential Hessian contraction is bounded by $C\sigma$, whereas

$$\chi_q \psi'' \le -\frac{cK_0\sigma^2}{1 + \sigma} \le -\frac{cK_0}{2}\sigma.$$

Choose K_0 large enough to dominate the collar curvature and (5.51). The positive barrier is a strict supersolution; the negative one is a strict subsolution. On the inner edge of this collar they dominate $\pm f$ by the height bound, and both vanish at the boundary. Therefore

$$|f(x)| \le \psi(d(x, \partial\mathcal{D})) \quad \text{when } d(x, \partial\mathcal{D}) \le d_0. \tag{5.53}$$

Consider now $f(x) - f(y) - \psi(d(x, y))$ for pairs with $0 \le d(x, y) \le d_0$, using the distance of the smooth ambient extension. It is nonpositive on the diagonal, for $d = d_0$, and when either endpoint lies on the boundary, by (5.53) and the monotonicity of ψ. Suppose it had a positive interior maximum. The endpoint gradients are parallel along their unique short joining geodesic and have the same length $\sigma = \psi' \ge 1$. Write e for their common parallel direction. Varying one endpoint in the axial direction gives

$$\operatorname{Hess} f_x(e, e) \le \psi'', \qquad \operatorname{Hess} f_y(e, e) \ge -\psi''.$$

Varying both endpoints in parallel transverse directions and summing the second-variation inequalities gives

$$\operatorname{tr}_{e^\perp} \operatorname{Hess} f_x - \operatorname{tr}_{e^\perp} \operatorname{Hess} f_y \le Cd(x, y)\sigma.$$

For completeness, the simultaneous endpoint variation is generated by a parallel vector field along the geodesic; its index form is bounded above by $Cd(x, y)$ since the ambient curvature is bounded. This is the displayed estimate after multiplication by ψ'. The transverse eigenvalues are exactly one at both endpoints, so

$$\begin{aligned} A_{q,x} : \operatorname{Hess} f_x - A_{q,y} : \operatorname{Hess} f_y &\le Cd(x, y)\sigma + (\chi_{q,x} + \chi_{q,y})\psi'' \\ &\le Cd_0\sigma - cK_0\sigma. \end{aligned}$$

The right sides of (5.50) differ by a quantity at least $-2C(1+\sigma)$. A sufficiently large K_0 contradicts this. Both signed axial Hessian bounds have the favorable sign, so the comparison holds independently of $\chi_{q,x} - \chi_{q,y}$. We conclude $|f(x) - f(y)| \le \psi(d(x,y))$ and hence $|df| \le \psi'(0) = M$.

3. Continuation and regularity. The coefficient bounds in (5.49) now give uniform ellipticity throughout the continuation. Its normalized forcing is bounded. Lemma 5.4 gives a common positive Hölder exponent for df, including the boundary. Since the coefficients are smooth functions of (x, z, df), the Dirichlet Schauder estimate first gives $f \in C^{2,\gamma_0}$ for a possibly smaller exponent. The resulting Lipschitz bound for df then makes those coefficients $C^{0,\gamma}$, so the estimate improves to $C^{2,\gamma}$. Differentiating the equation once gives $C^{3,\gamma}$ estimates: the trial is $C^{1,\gamma}$ and all differentiated coefficient and source terms are controlled. These scalar estimates are uniform along the continuation; they are the interior and smooth Dirichlet estimates of [21] on a finite coordinate covering.

The linearization in f is a uniformly elliptic scalar operator with bounded Hölder first-order coefficients and zero-order coefficient $-c_q < 0$. Its homogeneous Dirichlet kernel is zero by the maximum principle. The scalar linear Dirichlet continuity theorem, starting from $\Delta_g - 1$ and using the Schauder estimate, makes it invertible from $C_0^{2,\gamma}$ to $C^{0,\gamma}$. The implicit function theorem gives openness of the nonlinear continuation set; the bounds and compactness give closedness. The initial solution is $f = 0$, hence existence follows at $q = 1$.

4. Uniqueness and continuity. For uniqueness compare two solutions at a positive maximum of their difference. Their gradients agree there, so their coefficient matrices and their positive factors b agree; Hessian ordering and the strictly negative zero-order term give a contradiction. For continuity in the stated norm, regard the normalized trace operator as a map

$$[0,1] \times C^{1,\gamma} \times C_0^{3,\gamma} \longrightarrow C^{1,\gamma}.$$

It is continuously differentiable in the trial and solution variables, since its coefficient functions are smooth on every bounded jet range. At a solution its linearized principal, first-order, and zero-order coefficients are $C^{1,\gamma}$. The same maximum principle and the Dirichlet Schauder estimate at one additional derivative therefore make that linearization an isomorphism from $C_0^{3,\gamma}$ onto $C^{1,\gamma}$. The implicit function theorem supplies local continuous dependence into $C^{3,\gamma}$; uniqueness joins these local maps. The a priori estimates already proved give the bounded-set assertion. □

5.7 A compact map and the two homotopies

We now return to $\mathcal{D} = \mathcal{M}_{N,R}$ and $\eta = \ell^{3/2}$. Fix N, R with $R \ge R_{\min}(N)$, and let

$$X = \{z \in C^{1,\gamma}(\mathcal{M}_{N,R}) : z|_{S_R} = 0\}, \qquad 0 < \gamma < 1.$$

For a trial $z \in X$, solve $f = f_a[z]$ by Lemma 5.7. Evaluate t, D, u, A_χ, w and $\mathcal{T}$ using (f, z), including the derivatives of t obtained by differentiating its implicit definition. Define $T_a(z) = Z_{\text{out}}$ to be the solution of the *linear* Dirichlet problem

$$\operatorname{div}(kuA_\chi \nabla Z_{\text{out}}) = u\Xi_a - \operatorname{div}\big(uA_\chi K_a(w,\cdot)\big), \qquad Z_{\text{out}}|_{S_R} = 0. \tag{5.54}$$

All coefficients and the source Ξ_a are evaluated at the input (f, z); the displayed gradient of Z_{out} is the sole unknown in this linear problem.

On bounded subsets of X, the trace lemma gives bounded $C^{3,\gamma}$ norms for f. Consequently the principal coefficient kuA_χ is $C^{1,\gamma}$ and uniformly positive definite, the drift flux is $C^{1,\gamma}$, and

$u\Xi_a$ is $C^{0,\gamma}$. The linear scalar Dirichlet theorem and Schauder estimate therefore give a unique $C^{2,\gamma}$ output. The map $T_a : X \to X$ is continuous and compact, uniformly on bounded sets and $a \in [0,1]$. The same construction, with $a = 0$ and source $su\Xi_0$, defines the second compact homotopy $T_{0,s}$. Its trace solution is zero for every trial, since $K = 0$.

Every fixed point bootstraps to smoothness. Indeed the first Schauder step just described gives $z \in C^{2,\gamma}$; the trace equation then gives $f \in C^{4,\gamma}$, and repeated application to the second equation and then the first gives arbitrary derivatives. Thus the a priori estimates proved above apply to all fixed points.

For either homotopy parameter λ, set

$$\mathcal{U}_\lambda = \{z \in X : \min_{\mathcal{M}_{N,R}} t[z, df_\lambda[z]] > -\epsilon,\ \|z\|_{C^{1,\gamma}} < M\}. \tag{5.55}$$

The associated set of parameter–trial pairs is

$$\mathcal{U} = \{(\lambda, z) \in [0,1] \times X : z \in \mathcal{U}_\lambda\}.$$

It is relatively open in $[0,1] \times X$ because the trace solution map and the implicit function defining t are continuous. Choose M larger than the common $C^{1,\gamma}$ bound for all fixed points with $t \geq -\epsilon$ supplied by Proposition 5.6. There is no fixed point on the norm boundary, and Lemmas 5.2 and 4.3 exclude one where $\min t = -\epsilon$.

Degree is constant along these moving domains by compactness and excision. The fixed points in the closure of $\mathcal{U}$ form a compact subset of $[0,1] \times X$: the parameter interval is compact, the trial norms are bounded, and compactness of the homotopy gives a convergent subsequence of every sequence of such fixed points. Its limit cannot lie on either excluded boundary. Near any one parameter, take a fixed open neighborhood of that compact fiber whose closure remains inside $\mathcal{U}_\lambda$ for all sufficiently nearby parameters; shrink the parameter interval to exclude other fixed points outside this neighborhood, using compactness. Excision and ordinary fixed-domain homotopy invariance identify the degrees at those nearby parameters. A finite chain of such intervals identifies the degree at the two ends. These are the usual excision and homotopy properties of Leray–Schauder degree [16, Chapter 2].

At the endpoint $a = 0, s = 0$, (5.54) has zero source and zero boundary value, so $T_{0,0}(z) = 0$ for every trial. Its sole fixed point is zero, with $t = 0 > -\epsilon$, and its degree is one. First follow the second homotopy to $s = 1$, then the first to $a = 1$. The degree remains one, yielding a smooth solution of the desired filled system on every sufficiently large truncation.

5.8 Exhaustion, end decay, and the energy flux

The degree argument gives solutions on every sufficiently large truncation. Uniform local estimates let the outer boundary escape while N remains fixed. The positive zeroth-order term in the trace equation then gives exponential decay of the graph; a scalar change of variable converts the end equation for Z to a Poisson estimate. These are the precise asymptotics needed to evaluate the energy.

Completion of the proof of Theorem 5.1. Exhaustion at fixed N. Keep N fixed and let $R \to \infty$. The estimates of Proposition 5.6 on compact sets are independent of R. A diagonal subsequence therefore converges smoothly on every compact subset of $\mathcal{M}_N$ to a solution (f, Z). The polynomial bounds and height comparisons survive. They give $t \geq -\epsilon$; equality would be an interior global minimum and contradict Lemma 4.3.

We prove the remaining end estimates on the truncation solutions, uniformly in R at this fixed N; the estimates then pass to the smooth limit just obtained.

Exponential decay of the graph. On the end $K = 0$ and the trace equation reads

$$A_\chi : \operatorname{Hess} f - \frac{\eta\sqrt{D}}{l} f = 0, \qquad \frac{\eta\sqrt{D}}{l} \geq \frac{\eta}{e} > 0. \tag{5.56}$$

For small $a_N > 0$ and a sufficiently large fixed $r_2(N)$, $v_N(r) = e^{-a_N(r-r_2)}$ obeys

$$A_\chi : \operatorname{Hess} v_N - \frac{\eta\sqrt{D}}{l} v_N < 0 \quad (r \geq r_2).$$

In fact $0 < A_\chi \leq I$ and the end metric bounds give $A_\chi : \operatorname{Hess} v_N \leq C(a_N^2 + a_N/r)v_N$; choose $Ca_N^2 < \eta/(4e)$ and then r_2 large. Multiples of $\pm v_N$ dominate f at S_{r_2} and at the outer zero sphere. The maximum principle for the operator in (5.56), with its coefficients evaluated at the solution, proves exponential decay uniformly on all truncations. The already established scalar local estimates, on balls and boundary half-balls of a fixed size, then give the same decay for every derivative of f. In particular $t - Z$ and its derivatives are exponentially small.

Decay of the conformal variable. The second equation can consequently be written, with an error exponentially small through every fixed number of derivatives, as

$$\begin{gathered} k\Delta_g Z + kB_N(Z)|dZ|_g^2 = \rho - \vartheta(N(Z+\epsilon))C_N\rho_0 + O_j(e^{-a_N r}), \\ B_N(z) = (k-1) + p(z) - \delta_0 s_{p(z)}. \end{gathered} \tag{5.57}$$

To verify this formula, set $f = 0$ in the coefficients. Then $D = 1$, $t = Z$, $w = 0$, $A_\chi = I$, $\mathcal{T} = ks_{p(Z)}|dZ|^2$, and $V = ku(Z)\nabla Z$ with $(\log u)' = k - 1 + p$. Every difference from these expressions has an exponentially small factor involving a derivative of f; its other factors are bounded at fixed N by local regularity. This also proves the differentiated error assertion. The same fixed-size local estimates hold at the outer boundary uniformly in R.

Define a smooth increasing function Q_N on the bounded range of Z by

$$Q_N(0) = 0, \qquad Q_N'(z) = \exp\left(\int_0^z B_N(\tau)\, d\tau\right).$$

Its derivative has positive upper and lower bounds at fixed N. The end equation (5.57) and the chain rule give

$$|\Delta_g Q_N(Z)| \leq C_N' r^{-n-\beta}, \qquad Q_N(Z)|_{S_R} = 0. \tag{5.58}$$

Here C_N' denotes a constant at fixed N, not necessarily the penalty polynomial. The function

$$b_*(r) = r^{2-n}(1 - r^{-\beta/2})$$

is positive for $r > 1$ and satisfies

$$\Delta_g b_* = -\frac{\beta}{2}\left(n - 2 + \frac{\beta}{2}\right) r^{-n-\beta/2} + O(r^{2-2n}) < 0$$

sufficiently far out. The correction dominates the metric error because $\beta/2 < n - 2$. Taking a sufficiently large multiple of b_* dominates both the source in (5.58) and the inner boundary values. Comparison with its two signs, also valid on the outer zero sphere, gives $|Q_N(Z)| \leq C_N' r^{2-n}$ on every truncation. Monotonicity of Q_N gives $Z = O(r^{2-n})$, and hence the same bound for t.

Rescale successive end annuli to unit size. The metric coefficients have uniformly controlled derivatives and converge to the Euclidean ones. The right side of the equation for $Q_N(Z)$ is a smooth function of Z times $r^{-n-\beta}$ plus exponentially small terms. Interior Poisson $W^{2,p}$ estimates, then Schauder estimates and differentiation, give $Z, t = O_j(r^{2-n})$ for each j; the constants may depend on N, j. Finally $|dZ|^2 = O(r^{2-2n}) = O(r^{-n-\beta})$, since $\beta < n-2$. The end equation (5.57) now proves the stated $\Delta_g t$ bound and finishes the theorem. □

For the resulting solution, Proposition 4.4 now applies with all of its end hypotheses verified. In particular, writing $\omega = \omega_{n-1}$ and $\mathfrak{F}_N = \lim_{r\to\infty} \int_{S_r} g(V, \nu_g)\, dA_g$,

$$\widehat{E} - E_0 = -\frac{\mathfrak{F}_N}{k\omega}, \qquad \mathfrak{F}_N = \int_{\mathcal{M}_N} u\Xi\, dV_g \geq -\mathcal{P}_N(\ell + \ell^{1/2}). \tag{5.59}$$

Here E_0 is the energy of the fixed prepared data. The metric also satisfies $\widehat{g} \geq e^{-2\epsilon} g$ pointwise. The flux estimate uses the polynomial penalty-band calculation of that proposition; the arbitrarily N-dependent constants in the end regularity estimates are used only to justify the flux and its normalization.

Remark 5.8. The construction first fixes N and then exhausts the radii. The bounds in (5.1) and (5.2) are the estimates used with $N \to \infty$ in the next section. No bound for a high derivative that depends arbitrarily on N is substituted for a polynomial bound. In particular, this proof takes no elliptic limit as $N \to \infty$ and imposes no equation on a limiting singular hypersurface.

6 Height separation and a weak minimizing enclosure

Throughout this section the data satisfy Definition 2.1. In particular, K has compact support, the dominant energy inequality is strict, and every component of S has strictly negative future expansion. Write a for the full-cut infimum of these prepared data. Fix $0 < \epsilon < 1$. We use the smooth solution on the filled manifold $\mathcal{M}_N$ supplied by Theorem 5.1, with the notation of Proposition 4.1. Thus

$$\ell = e^{-N\epsilon}, \qquad \eta = \ell^{3/2}, \qquad h = \eta f, \qquad \widehat{g} = e^{2t}(g + l^2 df \otimes df). \tag{6.1}$$

On the filling, g denotes its smooth extension. All comparisons with the original cut infimum will take place in the original exterior.

Recall the estimates from the analytic construction. After increasing a polynomial $P_N = P(N)$, they give

$$-\epsilon < t \leq Z \leq P_N, \qquad \ell \leq l \leq e, \qquad |f| + |df|_g \leq e^{P_N}, \qquad |h| \leq C_0. \tag{6.2}$$

Moreover, $|h| \leq C r^{-b_1}$ on the original end, with C independent of N. Local geometry of the background filling is uniformly bounded; its added product lengths and volume are $O(1+N)$. In particular it has uniform coordinate balls of some radius $r_0 > 0$, with uniform local isoperimetric constants. For each fixed N, the end estimates are

$$t = O_j(r^{2-n}), \qquad \Delta_g t = O(r^{-n-\beta}), \qquad f \text{ and its derivatives decay exponentially}, \tag{6.3}$$

where $\beta > 0$. Constants in (6.3) may depend arbitrarily on fixed N. Only (6.2) and the uniform height barrier will be used in estimates required to be polynomial in N.

The height separates the filling from infinity: it remains uniformly negative on the filling and tends uniformly to zero on the original end. We use this separation to place a large constant conformal factor around the filling. The cutoff vanishes on a distant tail, so it leaves the ADM energy unchanged. Its transition derivatives are small by weighted coercivity. We then minimize perimeter subject to a distance obstacle inside the plateau. Contact with that obstacle would give positive volume densities on both sides of the frontier; the plateau would turn the resulting perimeter into an area larger than a distant competitor. Detachment places the exterior and a neighborhood of its frontier in the region of controlled scalar curvature. The construction and its estimates are at fixed prepared data and fixed ϵ.

6.1 A negative height on the whole filling

Lemma 6.1 (The cap and collar barrier)**.** *Fix prepared data satisfying Definition 2.1 and $0 < \epsilon < 1$. For the final solution of Theorem 5.1 with the original prepared K, let $h = \eta f$ as in (6.1). The product lengths in the filling can be chosen $O(1+N)$ so that, for some $b_2 > 0$ independent of N, every sufficiently large N satisfies*

$$h < -b_2 \quad \text{on the whole added filling and a neighborhood of } S. \tag{6.4}$$

Proof. We construct a supersolution that is constant on the cap, rises along each neck, and crosses the trapped collar with positive slope. Use the signed transverse coordinate s increasing from each product neck toward S and the original end. The fixed transition from product geometry to an original collar has been chosen so that its leaves have

$$H_s + \operatorname{tr}_{TS_s} K \le -4\delta \tag{6.5}$$

for some fixed $\delta > 0$. Shrink δ when necessary. On the cap and the product part $K = -L_0 g$, where L_0 is a fixed large constant.

It is useful to write the trace equation directly in terms of h. At a point where a test function has the same gradient as h, put $a_0 = \eta/l$ and let χ have its value for the solution there. The operator applied to the test function v_* is

$$\mathcal{Q}(v_*) = \operatorname{tr}_{A_\chi} K + \frac{\operatorname{tr}_{A_\chi} \operatorname{Hess}_g v_*}{\sqrt{a_0^2 + |dv_*|_g^2}}. \tag{6.6}$$

The equation is $\mathcal{Q}(h) = h$. The coefficients in (6.6) are evaluated at the solution; we do not replace its value of t by one computed from the barrier. Nevertheless,

$$0 < \chi \le d = \frac{a_0^2}{a_0^2 + |dh|_g^2}, \qquad \frac{\eta}{e} \le a_0 \le \frac{\eta}{\ell} = \ell^{1/2}. \tag{6.7}$$

These inequalities suffice for a comparison at a positive maximum of $h - v_*$.

Set $v_* = -\delta$ on the cap. On a product neck let $q_N = v_*' \ge 0$ start from zero, increase approximately exponentially, and end at a small fixed number $q_* > 0$. One explicit construction is to take an exponential $q_* e^{s - L_N}$ where its size lies between a constant multiple of η and q_*, smooth its end to the constant q_* on an interval of fixed length, and multiply its initial part by a fixed smooth cutoff on an interval where it is $O(\eta)$. Choose

$$L_N = \log(q_*/\eta) + O(1) = O(1+N).$$

The cutoff can be flat at its initial endpoint. This construction gives

$$|q_N'| \le C(q_N + \eta), \qquad \int q_N \, ds \le C q_* + C\eta, \tag{6.8}$$

with C independent of N. Thus v_* glues smoothly to the constant on the cap. The same construction is made separately on every neck. The cap value is common to all components.

The product Hessian has only an axial component. The bounds (6.7)–(6.8) give, even where the slope vanishes,

$$\frac{\chi |v_*''|}{\sqrt{a_0^2 + |v_*'|^2}} \le C \frac{q_N + \eta}{\sqrt{(\eta/e)^2 + q_N^2}} \le C'.$$

Consequently $\mathcal{Q}(v_*) \le -kL_0 + C'$ on this part. Choose L_0 large enough that this is less than -2δ. Since $v_* \ge -\delta$, this is a strict supersolution inequality $\mathcal{Q}(v_*) < v_*$. It also holds on the constant cap, where the trace of K is $-nL_0$.

Continue with the fixed slope q_* across the fixed transition and up to S. Choose q_* so small that all the increase up to S, including (6.8), is less than $\delta/4$. Beyond S, still inside the fixed strictly trapped collar, increase the slope smoothly until $v_* > C_0 + 1$ at the outer edge of that collar. This requires only a fixed, possibly large, slope and fixed derivatives; they do not depend on N. Keep $v_*' \ge q_*$ on the entire transition and collar. For such a function of signed distance,

$$\mathcal{Q}(v_*) = \operatorname{tr}_{TS_s} K + \chi K(\partial_s, \partial_s) + \frac{v_*'}{\sqrt{a_0^2 + (v_*')^2}} H_s + \frac{\chi v_*''}{\sqrt{a_0^2 + (v_*')^2}}.$$

The bound (6.7) shows that this converges uniformly to the left side of (6.5). For example, the absolute error is bounded by $C\ell/q_*^2 + C\ell \sup |v_*''|/q_*^3$, with fixed C. For large N it is therefore less than δ. As $v_* \ge -\delta$, again $\mathcal{Q}(v_*) < v_*$.

Apply the maximum test on the compact region consisting of the cap, necks, transitions, and these outer collars. On its outer boundary $h - v_* < 0$. At a positive interior maximum, the gradients agree and $\operatorname{Hess} h \le \operatorname{Hess} v_*$. Positive definiteness of A_χ would give

$$h = \mathcal{Q}(h) \le \mathcal{Q}(v_*) < v_* < h,$$

a contradiction. Thus $h \le v_*$. Up to S the barrier is less than $-3\delta/4$. Continuity of the fixed collar barrier gives the same strict negative bound on a small neighborhood of S. Taking, for example, $b_2 = \delta/2$ proves the assertion. □

6.2 Conformal area increase and exterior scalar curvature

Choose $0 < b_3 < b_2/4$ so small that

$$\mu - |J|_g - \rho - b_3|\tau| \ge \tfrac{1}{2}\rho > 0 \quad \text{on the original exterior.} \tag{6.9}$$

This follows from the strict prepared inequality and the choice of ρ in the scalar construction: outside the compact support of τ no smallness condition on b_3 is needed, and on that compact set choose $b_3|\tau| \le \rho/2$. Let $\zeta \in C^\infty(\mathbb{R})$ be nonnegative, equal to zero for $|z| \le b_3/4$, equal to one for $|z| \ge b_3/2$, and with values in $[0,1]$. Define

$$A_N = (\ell/\eta)^{1/2} = \ell^{-1/4}, \qquad \psi_N = A_N \zeta(h), \qquad \widetilde{g}_N = e^{2\psi_N} \widehat{g}. \tag{6.10}$$

The height barrier on the end puts $\{|h| \ge b_3/4\}$ in a fixed original coordinate radius, together with the added filling. Hence ψ_N vanishes identically near infinity. It changes neither the ADM energy nor the asymptotic estimates of $\widehat{g}$.

Lemma 6.2 (Curvature on the small-height exterior)**.** *Fix prepared data satisfying Definition 2.1 and $0 < \epsilon < 1$, and use the final solution of Theorem 5.1. Choose b_3 and ζ as in Section 6.2, and let $\widetilde{g}_N$ be the metric (6.10). For all sufficiently large N,*

$$R_{\widetilde{g}_N} \geq 0 \quad \text{on } \Omega \cap \{|h| \leq b_3\}.$$

On the original exterior the metric satisfies $\widetilde{g}_N \geq e^{-2\epsilon} g$. Its ADM energy obeys

$$E(\widetilde{g}_N) \leq E + P_N(\ell + \ell^{1/2}). \tag{6.11}$$

Proof. The solution supplied by Theorem 5.1 has $t > -\epsilon$ and the parameters of the scalar construction, so its curvature is given by (4.24). On $|h| \leq b_3$, the constraint and trace terms have lower bound $\rho/2$ by (6.9); the remaining terms are nonnegative and retain the positive contribution $(1-\delta_0)\mathcal{T}$.

For the new conformal factor we use (4.26), with $\eta = \ell^{3/2}$. Its constants are polynomial in N for these fixed prepared data. The factor $\sqrt{d}$ in the height Laplacian cancels the axial loss in weighted coercivity, as shown immediately after that equation. The bound thus has polynomial dependence on N from weighted coercivity, independently of the unweighted second-derivative estimates at fixed N.

The conformal scalar-curvature law in dimension $n = k+1$ reads

$$\tfrac{1}{2} e^{2(t+\psi_N)} R_{\widetilde{g}_N} = \tfrac{1}{2} e^{2t} R_{\widehat{g}} - k e^{2t} \Delta_{\widehat{g}} \psi_N - \tfrac{k(k-1)}{2} e^{2t} |d\psi_N|^2_{\widehat{g}}. \tag{6.12}$$

The chain rule, (4.26), and the fixed derivative bounds of ζ imply

$$|e^{2t} \Delta_{\widehat{g}} \psi_N| \leq P_N \ell^{1/4}(1 + \sqrt{\mathcal{T}}) + C\ell^{3/4},$$
$$e^{2t} |d\psi_N|^2_{\widehat{g}} \leq C\ell^{1/2}.$$

On the support of these errors $b_3/4 \leq |h| \leq b_3/2$. This region is disjoint from the filling by Lemma 6.1, and is contained in a fixed compact portion of the original exterior. There $\rho \geq c_* > 0$. Young's inequality bounds the error containing $\sqrt{\mathcal{T}}$ by $(1-\delta_0)\mathcal{T}/2 + P_N \ell^{1/2}$. All remaining errors tend to zero. The curvature identity (4.24) therefore proves nonnegative scalar curvature there for large N. Where ζ is constant, (6.12) has no error terms and the same conclusion follows directly.

The lower bound $t > -\epsilon$, the nonnegative graph term, and $\psi_N \geq 0$ give the metric comparison. Finally $\psi_N = 0$ near infinity, so (6.11) follows from Proposition 4.4. In particular its sign is the upper energy estimate required for the Riemannian inequality. □

6.3 Perimeter minimization and detachment from a distance obstacle

We use full perimeter in the filled manifold. For a measurable set F and an open set U, write $P_\gamma(F;U) = |D\mathbf{1}_F|_\gamma(U)$ for perimeter in a smooth metric γ, and $P_\gamma(F) = P_\gamma(F;\mathcal{M}_N)$. The frontier is the support of this perimeter measure. For a set of finite perimeter its integration boundary is the reduced boundary $\partial^* F$, and $P_\gamma(F;U) = \mathcal{H}^k_\gamma(\partial^* F \cap U)$. These conventions distinguish full perimeter from a relative frontier inside Ω.

The final exterior and an ambient neighborhood of its frontier must lie in the original small-height region where Lemma 6.2 applies. We therefore enclose the whole set where $|h| \geq b_3$, which contains the filling, and use a thin distance neighborhood of that set as the obstacle. Its distance collar will supply the filled-side density if a minimizing frontier touches it. Put

$$C^h_N = \{|h| \geq b_3\}, \qquad O_N = \{x : \mathrm{dist}_g(x, C^h_N) \leq s_N\}, \qquad s_N = e^{-Q_N}, \tag{6.13}$$

where Q_N is a sufficiently large polynomial. It can be chosen so that $s_N < r_0/10$ and

$$3s_N \sup |dh|_g < b_3/2. \tag{6.14}$$

Indeed $|dh| \le \eta e^{P_N}$ by (6.2). The entire $2s_N$-neighborhood of O_N then lies in $\{|h| > b_3/2\}$, where $\widetilde{g}_N = e^{2A_N}\widehat{g}$. The compact obstacle O_N contains the whole added filling and a neighborhood of S. The distance obstacle may have a nonsmooth boundary. Outside O_N we are in $\operatorname{int}\Omega \cap \{|h| < b_3\}$; detachment from O_N will provide the required ambient neighborhood as well.

Lemma 6.3 (Existence with a compact support bound)**.** *Fix prepared data satisfying Definition 2.1 and $0 < \epsilon < 1$, and use the final solution of Theorem 5.1. Let $\widetilde{g}_N$ be the conformal metric constructed in Section 6.2 and defined by (6.10). Let O_N be the distance obstacle in (6.13), with its scale satisfying (6.14). There is a bounded finite-perimeter set $F_N \supset O_N$, modulo null sets, minimizing $P_{\widetilde{g}_N}$ among all bounded sets containing O_N. One may choose it to contain every other minimizer modulo null sets. Its perimeter satisfies*

$$P_{\widetilde{g}_N}(F_N) \le e^{P_N}. \tag{6.15}$$

Proof. Clipping and compactness. At fixed N the end is smooth and asymptotically Euclidean with the differentiated falloff in (6.3). Consequently sufficiently large coordinate spheres have positive mean curvature for $\widetilde{g}_N$ toward infinity. The required radius may depend arbitrarily on N.

Here is the clipping argument that gives the compact support needed for the direct method. All objects in this paragraph use $\widetilde{g}_N$. On the large end let $Y = \nabla r/|\nabla r|$, so $|Y| = 1$ and $\operatorname{div} Y = H_r > 0$. For a bounded finite-perimeter competitor F and almost every large R, Gauss–Green on $F \cap \{r > R\}$ gives

$$\int_{F\cap\{r>R\}} H_r \, dV = \int_{\partial^* F\cap\{r>R\}} \langle Y, \nu_F\rangle \, dA - \mathcal{H}^k(F^{(1)} \cap S_R).$$

The spherical normal here is $-Y$. Boundedness of F eliminates an outer flux; alternatively first cut off Y beyond its support. The first boundary integral is at most $P(F; \{r > R\})$. The BV truncation formula therefore yields the stronger clipping inequality

$$P_{\widetilde{g}_N}(F \cap \{r < R\}) \le P_{\widetilde{g}_N}(F) - \int_{F\cap\{r>R\}} H_r \, dV_{\widetilde{g}_N}. \tag{6.16}$$

The Gauss–Green and truncation formulas hold at almost every radius by the trace and slicing theorems for BV functions [3]. The notation $\{r < R\}$ here includes the entire compact core. In particular it contains O_N once R is sufficiently large.

Choose a fixed interval of such radii beyond the obstacle. Applying (6.16) to each term of a minimizing sequence, with an almost-everywhere admissible radius in that interval, confines the sequence to one compact set. BV compactness gives L^1 convergence along a subsequence, and lower semicontinuity gives a minimizer. Containment of O_N is preserved almost everywhere. The minimum is the unrestricted bounded-competitor minimum, so local variations are not subject to an artificial outer constraint.

A distant competitor. Choose a fixed original coordinate sphere S_{R_*} so far out that $|h| < b_3/4$ on and beyond it for every N. It encloses O_N when N is large. On it $\psi_N = 0$, and (6.2) implies $\widehat{g} \le e^{P_N} g$. Its filled side is a competitor; increasing P_N gives (6.15). The fixed sphere need not lie in the region used for clipping: the latter was needed only for compactness at fixed N.

A maximal minimizer. All minimizers have a common compact support bound: if a minimizer had positive volume beyond an almost-everywhere admissible clipping radius, the strictly positive last integral in (6.16) would contradict minimality. Among minimizers, maximize the enclosed volume. Compactness, lower semicontinuity, and convergence of volume show that this maximum is attained. Perimeter submodularity implies

$$P(F \cup G) + P(F \cap G) \le P(F) + P(G).$$

For two minimizers both sets on the left contain O_N, so equality holds and their union is a minimizer. Maximal volume therefore implies $G \subset F_N$ modulo null sets for every minimizer G. □

Lemma 6.4 (Two densities and detachment)**.** *With the prepared data, metric $\widetilde{g}_N$, and distance obstacle O_N of Lemma 6.3, let F_N be the maximal minimizer selected there. For all sufficiently large N,*

$$\operatorname{supp} |D\mathbf{1}_{F_N}| \cap O_N = \varnothing. \tag{6.17}$$

Proof. At a contact point, the distance obstacle supplies a density bound for the filled set, and minimization supplies one for its complement. These two arguments are illustrated in Figure 1.

Density of the filled set. Suppose p belongs to the intersection in (6.17). It cannot lie in the interior of O_N, because F_N contains O_N almost everywhere. Thus $\operatorname{dist}_g(p, C_N^h) = s_N$. The complete background metric has a minimizing geodesic from some $z \in C_N^h$ to p. In particular $B^g_{s_N}(z) \subset O_N$. The midpoint of this geodesic is the center of a ball of radius $s_N/4$ contained in both O_N and $B^g_{s_N}(p)$. Uniform background ball-volume bounds give

$$\operatorname{Vol}_g(F_N \cap B^g_{s_N}(p)) \ge c s_N^n. \tag{6.18}$$

This is the filled-side density, obtained from the distance obstacle.

Density of the complement. We derive the other density from minimization. On $B^g_{s_N}(p)$ the high conformal multiplier is constant. Remove it and write $\gamma = \widehat{g}$. The bounds in (6.2) give, on this ball and every k-plane,

$$e^{-P_N} g \le \gamma \le e^{P_N} g, \qquad e^{-P_N} dA_g \le dA_\gamma \le e^{P_N} dA_g, \tag{6.19}$$

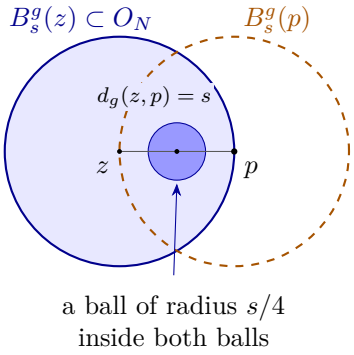
after changing the polynomial. This comparison is pointwise and independent of derivatives of γ.

Put $U = \mathcal{M}_N \setminus F_N$ and $V(r) = \operatorname{Vol}_g(U \cap B^g_r(p))$. Since p is in the support of perimeter, $V(r) > 0$ for every $r > 0$: otherwise $\mathbf{1}_{F_N}$ would be constant almost everywhere on some neighborhood of p. For almost every $0 < r < s_N$, filling the ball is an admissible competitor. The unchanged exterior perimeter and the common constant factor e^{kA_N} cancel in the comparison. The perimeter truncation formula therefore gives

$$P_\gamma(U; B^g_r(p)) \le \mathcal{H}^k_\gamma(U^{(1)} \cap \partial B^g_r(p)).$$

Here $U^{(1)}$ is the measure-theoretic interior; the displayed trace agrees with it for almost every radius. The coarea formula in the smooth background metric and (6.19) then yield

$$P_g(U; B^g_r(p)) \le e^{P_N} V'(r) \quad \text{for almost every } r \in (0, s_N). \tag{6.20}$$



Obstacle inclusion gives
$\mathrm{Vol}_g(F_N \cap B_s^g(p)) \geq cs^n$.

Filling-ball comparison gives
$\mathrm{Vol}_g(F_N^c \cap B_s^g(p)) \geq e^{-P_N} s^n$.

Figure 1: The two densities at a hypothetical contact point $p \in \mathrm{supp}\,|D\mathbf{1}_{F_N}| \cap \partial O_N$, with $z \in C_N^h$ chosen so that $d_g(z,p) = s = s_N$. The circles depict background metric balls schematically; no regularity of the obstacle or of the contact is assumed. Their overlap provides the first density. Minimality provides the second. Relative isoperimetry then forces perimeter inside the large constant conformal plateau, contradicting the competitor bound.

The absolute isoperimetric inequality on a uniformly controlled background ball, applied to the zero extension of $\mathbf{1}_{U \cap B_r^g(p)}$, includes its spherical trace:

$$V(r)^{(n-1)/n} \leq C\big(P_g(U; B_r^g(p)) + V'(r)\big) \leq e^{P_N} V'(r).$$

Local Euclidean isoperimetry in a uniform coordinate ball proves this form; the smooth background metric changes its constant by a fixed factor. Since V is absolutely continuous and positive for $r > 0$, its nth root satisfies

$$(V^{1/n})'(r) \geq n^{-1} e^{-P_N} \quad \text{almost everywhere on } (0, s_N).$$

Integrate from $a > 0$ to r and let $a \downarrow 0$. The result is

$$V(r) \geq e^{-P_N} r^n \qquad (0 < r \leq s_N), \tag{6.21}$$

with another polynomial. This proves the unfilled-side density from a one-sided comparison, without assuming that the obstacle contact is a regular point or that the deformed metric has controlled derivatives.

The perimeter contradiction. Relative isoperimetry in $B_{s_N}^g(p)$, together with (6.18) and (6.21), now gives

$$P_g(F_N; B_{s_N}^g(p)) \geq e^{-P_N} s_N^{n-1}.$$

The factor $s_N^{n-1} = e^{-(n-1)Q_N}$ can be absorbed into the polynomial exponential. Applying (6.19) and restoring the constant multiplier gives

$$P_{\widetilde{g}_N}(F_N) \geq \exp(kA_N - P_N) = \exp(ke^{N\epsilon/4} - P_N). \tag{6.22}$$

For fixed $\epsilon > 0$, $e^{N\epsilon/4}$ dominates every polynomial in N. The lower bound (6.22) contradicts (6.15) for large N. □

6.4 The exterior, its full boundary, and its area

Proof of Theorem 2.2. The connected exterior and its curvature region. Choose F_N as in Lemma 6.3. Its frontier is compact and, by Lemma 6.4, has positive distance from O_N. Every sufficiently small variation of the set near a frontier point remains admissible. Thus its boundary is locally perimeter minimizing in the smooth metric $\widetilde{g}_N$.

We use the regularity theorem for codimension-one minimizing boundaries: the support of the perimeter measure is a smooth embedded minimal hypersurface outside a relatively closed set of Hausdorff dimension at most $n-8$. A directly applicable smooth-ambient formulation is [42, Chapter 7, Section 5, Theorem 5.8]; its hypersurface dimension is $k=n-1$, so the singular bound is $k-7=n-8$. Multiplicity one for the boundary of a set is recorded in [42, Chapter 7, Section 5, Remark 5.2]. The compactness and tangent-cone conclusions used in the minimizing-boundary regularity argument are Theorems 5.3 and 5.5 of that section.

We also specify the representative of the filled set. Since the frontier is detached from the obstacle, both insertion and deletion of a sufficiently small ball are admissible. The coarea and isoperimetric argument used in the preceding lemma, now applied to each side, gives positive lower volume densities for both F_N and its complement at every point in the support of perimeter. On a ball not meeting that support the indicator has zero distributional derivative, hence is constant almost everywhere. Let U_N be the union of those balls on which that constant is zero, and let the closed filled representative be $\mathcal{M}_N \setminus U_N$. The two density bounds identify its topological frontier with the perimeter support: every neighborhood of a support point contains both filled and unfilled points. Conversely a point outside the support has a constant-indicator neighborhood. This representative agrees with F_N up to a null set; the regularity theorem makes its frontier n-dimensional-volume null. Its perimeter and all the preceding minimizing comparisons are therefore unchanged.

Put $\Sigma_N=\partial U_N$. There is only one component of U_N. Indeed there is exactly one unbounded component, since F_N is bounded and the ambient manifold has just one end. Filling all bounded complementary components cannot increase perimeter. This statement can be seen directly on regular boundary charts, where it deletes the separating sheets and adds no sheet; the singular set has zero $\mathcal{H}^k$ measure. Equivalently it is the component decomposition and perimeter additivity for finite-perimeter sets [3]. The resulting set is still an admissible bounded competitor. If any bounded open component existed, its positive volume would strictly increase the filled volume; minimality would either be contradicted by a perimeter decrease or preserved with a volume increase. Both alternatives contradict the choice of F_N. Thus U_N is connected. We take $X_N=\overline{U_N}$, with its ambient boundary included.

The exterior lies in $\Omega\cap\{|h|<b_3\}$, because the filled set contains $O_N\supset C_N^h$. In fact it has a smooth ambient neighborhood in that region: detachment gives positive separation between Σ_N and O_N, and compactness permits a smooth compact domain containing O_N whose boundary is still strictly inside the filled set. Remove that domain. On the remaining smooth ambient exterior, Lemma 6.2 gives nonnegative scalar curvature, also on a neighborhood of Σ_N.

All local variations of Σ_N have zero first variation. More precisely, if Y is a compactly supported smooth ambient vector field near the frontier and Φ_s is its flow, then $\Phi_s(F_N)$ remains admissible for both signs of small s. Differentiation of the perimeter area formula gives

$$\int_{\partial^* F_N} \operatorname{div}_{T_x\partial^* F_N} Y \, d\mathcal{H}^k_{\widetilde{g}_N} = 0. \tag{6.23}$$

This is stationarity of the full boundary varifold, including the assertion that there is no additional first-variation term at the singular set. Global outer minimization follows because every bounded

superset of F_N also contains O_N. The maximal choice of F_N also makes it the outermost member among enclosures with this same minimum perimeter.

Comparison with full original cuts. We verify the comparison with original full cuts in some detail. Detachment and Lemma 6.1 put Σ_N in $\operatorname{int}\Omega$, separated from a neighborhood of the entire original filling, including every component of S. There are bounded smooth open sets F_j in the filled manifold such that

$$\mathbf{1}_{F_j} \longrightarrow \mathbf{1}_{F_N} \text{ in } L^1, \qquad P_g(F_j) \longrightarrow P_g(F_N), \tag{6.24}$$

and each F_j still contains a fixed neighborhood of the whole filling. Here is a way to impose the last condition in strict BV approximation. Choose a smaller compact neighborhood on which $\mathbf{1}_{F_N} = 1$ almost everywhere, separated from Σ_N, and a compact exterior region where it is zero. Smooth the characteristic function in finitely many coordinate charts only in the remaining region, using a partition of unity and leaving those two constant regions fixed. The resulting smooth functions $u_j \in [0,1]$ converge in L^1 with total variation converging to $P_g(F_N)$. Choose $\delta_j \downarrow 0$ so slowly that $\|u_j - \mathbf{1}_{F_N}\|_{L^1}/\delta_j \to 0$. By coarea there is a regular value $t_j \in (\delta_j, 1-\delta_j)$ for which

$$P_g(\{u_j > t_j\}) \le \frac{\int |du_j|_g \, dV_g}{1-2\delta_j} + o(1) = P_g(F_N) + o(1).$$

The L^1 difference of this level set from F_N is at most $\|u_j - \mathbf{1}_{F_N}\|_{L^1}/\delta_j$, and hence tends to zero. Lower semicontinuity gives the reverse perimeter liminf. Thus $F_j = \{u_j > t_j\}$ satisfies (6.24). This is the strict approximation/coarea construction of [3], localized away from the two constant regions.

For each j retain the component of $\mathcal{M}_N \setminus \overline{F_j}$ that contains the distant end, and let D_j be its closure. A compact smooth boundary has finitely many components; retaining this exterior component merely selects some of those entire boundary components. It preserves smoothness and can only decrease the full boundary area. Because F_j contains a neighborhood of the whole filling, $D_j \subset \Omega$, its interior lies in $\operatorname{int}\Omega$, and D_j is closed and connected with a smooth compact intrinsic boundary Γ_j. It is an admissible full cut in Definition 1.2. Thus

$$a \le \operatorname{Area}_g(\Gamma_j) \le P_g(F_j).$$

Taking the limit gives $P_g(F_N) \ge a$. The passage to D_j chooses an admissible exterior domain, whose entire intrinsic boundary is counted in Γ_j. The fixed neighborhood keeps these approximating cuts disjoint from S; the original definition still allows coincidence with S for its other competitors.

Since $\widetilde{g}_N \ge e^{-2\epsilon} g$ on Ω, comparison on every approximate tangent k-plane now yields

$$\mathcal{H}^k_{\widetilde{g}_N}(\Sigma_N) = P_{\widetilde{g}_N}(F_N) \ge e^{-k\epsilon} P_g(F_N) \ge e^{-k\epsilon} a. \tag{6.25}$$

The first equality uses the regularity theorem and the zero $\mathcal{H}^k$ measure of the singular set.

Completeness. The filled metric is complete, since $\widetilde{g}_N \ge e^{-2\epsilon} g$ and the background filled metric is complete. Its restriction to X_N is complete with the frontier included. Here completeness is understood in the enclosing-set convention of the Riemannian input: the intrinsic length distance on X_N may be extended-valued, and its restriction to each finite-distance equivalence class is complete. For the intrinsic-length formulation, a Cauchy sequence has a subsequence whose successive intrinsic distances are less than 2^{-j}. Join successive terms by curves of length less than 2^{1-j} in X_N. Their concatenation has an ambient endpoint, by ambient completeness, and that

endpoint belongs to X_N because X_N is closed. Appending this endpoint gives a curve in X_N whose tail length tends to zero, proving intrinsic convergence of the subsequence and then of the original Cauchy sequence. The frontier is included in the sense of a closed enclosing set used in the Riemannian input in Section 2.

Finally the end satisfies $\widetilde{g}_N - I = O_j(r^{2-n})$. The conformal law applied to (6.3) shows

$$R_{\widetilde{g}_N} = O(r^{-n-\beta}) + O(r^{-2n+2}).$$

If needed decrease β so that $0 < \beta < n-2$; then the displayed bound is $O(r^{-n-\beta})$. All graph errors decay exponentially, and ψ_N is zero on the end. Scalar curvature is integrable there and, being smooth and bounded on the compact ambient neighborhood of the frontier, integrable on the remainder of X_N. The energy estimate is Lemma 6.2. This proves all the stated properties. □

Remark 6.5 (Flux and weak boundary conventions). The scalar flux in Proposition 4.4 is computed on the complete smooth filled manifold before an enclosure is chosen:

$$\mathfrak{F}_N = \lim_{R\to\infty} \int_{S_R} g(V, \nu_g)\, dA_g = \int_{\mathcal{M}_N} u\Xi\, dV_g, \qquad E(\widehat{g}) - E = -\frac{\mathfrak{F}_N}{k\omega}.$$

Thus no elliptic boundary condition or unaccounted singular-boundary flux is involved. If one restricts a smooth vector field to X_N, the Gauss–Green theorem for finite-perimeter sets gives its ordinary reduced-boundary flux. For instance at a regular truncation radius,

$$\int_{U_N \cap B_R} \operatorname{div}_{\widetilde{g}_N} Y\, dV_{\widetilde{g}_N} = \int_{S_R} \widetilde{g}_N(Y, \nu_R)\, dA_{\widetilde{g}_N} - \int_{\partial^* F_N} \widetilde{g}_N(Y, \nu_{F_N})\, dA_{\widetilde{g}_N},$$

when Y is supported in the indicated truncation or has its stated outer trace. The normal ν_{F_N} points from the filled set toward the end. There is no further measure on the singular set. This flux convention and (6.23) are the weak identities attached to the constructed boundary.

7 Weakly asymptotically flat data in three dimensions

In three dimensions, we extend the inequality of Theorem 2.3 to data with two differentiated metric bounds and one differentiated tensor bound. The approximation in Proposition 7.13 replaces the distant end by a Schwarzschild reference slice with the original ADM vector, bends the slice to rest, and repairs the small constraint error. Its energy tends to the original rest mass, while compactness of minimizing enclosures gives convergence of the full enclosing infimum. We first deduce the exterior inequality from this approximation, then construct it. A positive conformal family removes timelikeness as an assumption, and truncation of the other ends gives the complete-data statement.

7.1 Initial data, normals, and enclosing area

We work in three spatial dimensions, with zero cosmological constant and $G = c = 1$. In this section μ, J denote the physical densities: the geometric densities of (1.1) are $8\pi\mu, 8\pi J$, while the ADM energy and momentum retain their previous normalization. For a Riemannian metric g and a symmetric covariant two-tensor K, define the constraint densities by

$$16\pi\mu = R_g + (\operatorname{tr}_g K)^2 - |K|_g^2, \qquad 8\pi J = \operatorname{div}_g(K - (\operatorname{tr}_g K)g). \tag{7.1}$$

Here J is a covector field. The dominant energy condition is

$$\mu \geq |J|_g. \tag{7.2}$$

All initial data considered here are smooth, including at a compact boundary when one is present.

On an asymptotically flat end with Euclidean coordinates x and $r = |x|$, our basic assumptions are

$$g_{ij} - \delta_{ij} = O_2(r^{-q}), \qquad K_{ij} = O_1(r^{-1-q}), \qquad q > \tfrac{1}{2}. \tag{7.3}$$

Recall that $O_j(r^{-a})$ includes the corresponding coordinate derivative bounds through order j. We assume that μ and $|J|_g$ are integrable and that the following ADM limits exist and are finite:

$$E = \frac{1}{16\pi} \lim_{r\to\infty} \int_{S_r} (\partial_j g_{ij} - \partial_i g_{jj}) n^i \, \mathrm{d}A_\delta, \tag{7.4}$$

$$P_i = \frac{1}{8\pi} \lim_{r\to\infty} \int_{S_r} (K_{ij} - (\mathrm{tr}_g K) g_{ij}) n^j \, \mathrm{d}A_\delta. \tag{7.5}$$

The normal and area form in these definitions are Euclidean. Whenever $E > |P|_\delta$, write

$$m = \sqrt{E^2 - |P|_\delta^2}. \tag{7.6}$$

For a two-sided surface Σ with specified unit normal ν, we use

$$H_\Sigma = \mathrm{div}_\Sigma \, \nu, \qquad \theta_+(\Sigma) = H_\Sigma + \mathrm{tr}_\Sigma \, K.$$

Thus Euclidean spheres have positive mean curvature for the normal toward infinity. A future marginally outer trapped surface, abbreviated MOTS, satisfies $\theta_+ = 0$. A weakly future outer trapped surface satisfies $\theta_+ \leq 0$. Our spacetime sign convention is

$$K(Y, Z) = \mathbf{g}(\nabla_Y n, Z), \tag{7.7}$$

where n is the future unit timelike normal. Equivalently, in Gaussian normal coordinates the spatial metric has normal derivative $2K$.

Definition 7.1 (Exterior and enclosing area)**.** An exterior is a connected orientable smooth three-manifold Ω with nonempty compact smooth boundary, complete as a metric space with its boundary included, and with exactly one end, which is asymptotically flat. An enclosing cut is the entire compact smooth embedded two-sided intrinsic boundary of a connected closed outer domain containing the sufficiently distant end, with its manifold interior in the open exterior. Equivalently, it separates the entire inner boundary from infinity, with bounded complementary pockets filled on the inner side. Every frontier component is counted, including portions coinciding with the inner boundary. A cut may have several components. Its normal points toward the retained end. We put

$$a_g(\partial\Omega) = \inf\{|\Gamma|_g : \Gamma \text{ is an enclosing cut}\}.$$

When using perimeter compactness, we take the closure of this class among filled inner sets and include the area of any frontier coinciding with the obstacle. Smooth outward approximation gives the same infimum. Auxiliary perimeter arguments may use competitors containing the inner obstacle before filling.

Remark. Filling a bounded complementary pocket deletes its frontier, cannot increase perimeter, and leaves the enclosing infimum unchanged. The definition permits coincidence because first variations of the minimum need not be realized by a cut lying strictly outside the obstacle. It permits disconnected cuts because neither the minimizing hull nor an intermediate trapped boundary need be connected. The perimeter formulation and its compactness properties are established in Lemma 7.6.

7.2 The one-ended exterior inequality

Theorem 7.2 (Minimum-enclosing-area inequality)**.** *Let (Ω, g, K) be an exterior as in Definition 7.1. Assume (7.1)–(7.3), integrability of μ and $|J|_g$, and the finite ADM limits (7.4)–(7.5). Orient $\partial\Omega$ by the normal into Ω, and suppose $\theta_+(\partial\Omega) \le 0$. If $E > |P|_\delta$, then*

$$\sqrt{E^2 - |P|_\delta^2} \ge \sqrt{\frac{a_g(\partial\Omega)}{16\pi}}. \tag{7.8}$$

Proof of Theorem 7.2. Let $B = \partial\Omega$ and $m = \sqrt{E^2 - |P|^2} > 0$. We use the approximating sequence of Proposition 7.13, whose construction follows below. Each datum is a smooth complete one-ended exterior on Ω with the same nonempty compact boundary B. The proposition gives a compactly supported tensor, $O_d(r^{-1})$ metric decay for every fixed derivative order d, scalar curvature $O(r^{-3-\delta})$ with $0 < \delta < 1$, integrable constraints, strict future trapping, and a global weighted DEC margin. These are the geometric and asymptotic conditions in Definition 2.1.

The enclosing class also agrees with its full-cut class: both retain a connected end-containing outer domain and count its entire intrinsic frontier, including coincidence with the obstacle. Filling pockets removes only surplus frontier and leaves either infimum unchanged. Thus the prepared area is $A_j = a_{g_j}(B) > 0$, where positivity follows from Lemma 7.3.

Recall that the densities used here are physical densities. Those of the prepared theorem satisfy $2\mu_{\mathrm{A}} = R + (\operatorname{tr} K)^2 - |K|^2$ and $J_{\mathrm{A}} = \operatorname{div}(K - (\operatorname{tr} K)g)$. Hence

$$\mu_{\mathrm{A}} = 8\pi\mu, \qquad J_{\mathrm{A}} = 8\pi J.$$

The global weighted margin becomes $\mu_{\mathrm{A},j} - |J_{\mathrm{A},j}|_{g_j} \ge 8\pi c_j \rho^{-3-\delta}$, and all sign and integrability conditions are preserved. At $n = 3$, $k = n - 1 = 2$ and $\omega_2 = 4\pi$, so the energy and momentum prefactors in that theorem are respectively

$$\frac{1}{2k\omega_2} = \frac{1}{16\pi}, \qquad \frac{1}{k\omega_2} = \frac{1}{8\pi}.$$

Thus neither ADM energy nor momentum nor invariant mass is rescaled.

Theorem 2.3 therefore applies to each fixed prepared datum and gives

$$E_j \ge \frac{1}{2}\left(\frac{a_{g_j}(B)}{4\pi}\right)^{1/2} = \sqrt{\frac{a_{g_j}(B)}{16\pi}}.$$

For each datum, the elliptic exhaustion and the limits in N and ϵ are completed before taking $j \to \infty$. Thus the passage to the limit uses only $E_j \to m$ and the full convergence $a_{g_j}(B) \to a_g(B)$, and gives

$$\sqrt{E^2 - |P|^2} \ge \sqrt{\frac{a_g(B)}{16\pi}},$$

as asserted. □

7.3 A positive lower bound for every enclosing cut

Before preparing the end, we record a geometric fact about the full enclosing infimum. A thin tube joining the inner boundary to infinity carries a fixed flux through every enclosing cut. A bound for the flux density then gives a uniform positive area bound. This argument uses neither the constraint equations nor a sign for the boundary expansion.

Lemma 7.3 (Positive enclosing area)**.** *Let (Ω, g) be a connected orientable smooth three-dimensional exterior, complete with its nonempty compact smooth boundary B included, with one asymptotically flat end and compact complement. Let $a_g(B)$ be the infimum of the full areas of the enclosing cuts in Definition 7.1, including all components and all portions coincident with B. Then*

$$0 < a_g(B) < \infty.$$

Proof. A sufficiently large coordinate sphere encloses B, so the infimum is finite. To obtain a lower bound, choose a point of B and a smooth proper embedded ray starting there, initially normal to B and pointing into Ω. Choose the ray to be straight in the asymptotic coordinates outside a compact set. A tubular neighborhood has product coordinates $(t, y) \in [0, \infty) \times D^2$, with its initial disk contained in B. On the straight tail the transverse coordinate disk can have a fixed positive width.

Choose a smooth two-form η supported in the interior of D^2 with $\int_{D^2} \eta = 1$. Pull it back by $(t, y) \mapsto y$ and extend it by zero across the lateral boundary of the tube. This gives a smooth closed two-form ω on Ω, smooth also at B. Its pointwise comass is bounded: this follows from compactness on the initial portion of the tube and from asymptotic flatness and fixed coordinate width on the tail. Write

$$C_\omega = \sup_{x \in \Omega} \sup_{v,w \text{ orthonormal in } (T_x\Omega, g)} |\omega_x(v, w)| < \infty.$$

Orient the transverse disk so that the flux through a sufficiently large outward-oriented coordinate sphere S_R is one.

Let Γ be a smooth enclosing cut lying in the interior of Ω, oriented toward the end, and choose S_R outside it. The region between Γ and S_R is compact and disjoint from B; all of its inner boundary components are counted in Γ. Stokes' theorem gives

$$\int_\Gamma \omega = \int_{S_R} \omega = 1, \qquad 1 \le C_\omega \operatorname{Area}_g(\Gamma).$$

This calculation also applies when the cut is disconnected. If a smooth cut touches or coincides with B, push it into the interior by a smooth collar flow pointing from B into Ω, filling the collar on the obstacle side. The resulting surfaces are enclosing cuts and their areas converge to the full area of the original cut. Thus the same lower bound holds for boundary-coincident cuts. Taking the infimum yields $a_g(B) \ge C_\omega^{-1} > 0$. The smooth/perimeter equivalence in the definition of the full enclosing infimum transfers this bound to its filled-perimeter formulation as well. □

For the construction, retain the exterior Ω of Definition 7.1 and write $B = \partial\Omega$, with normal pointing into Ω. Recall the differentiated asymptotic assumptions:

$$g - \delta = O_2(r^{-q}), \qquad K = O_1(r^{-1-q}), \qquad q > \tfrac{1}{2}. \tag{7.9}$$

The density, finite-charge and weak-trapping hypotheses are those stated in Theorem 7.2. The preparation assumes $E > |P|$; the conformal construction immediately below does not.

7.4 A strict conformal direction

Strict conformal deformation by a spacetime Poisson equation was developed by Jaracz [28]. Here a regularized drift equation with nonzero inner Neumann data gives strict trapping, an energy slope, unchanged momentum, and comparison of all enclosing cuts.

Lemma 7.4 (Conformal constraint formulas)**.** *Let g be a smooth Riemannian metric and K a smooth symmetric covariant two-tensor on a smooth three-manifold Ω, with physical constraint densities defined by (7.1). For a smooth function f on Ω, set*

$$g_f = e^{4f} g, \qquad K_f = e^{2f} K.$$

The corresponding constraint densities satisfy

$$16\pi e^{4f} \mu_f = 16\pi\mu - 8\Delta_g f - 8|df|_g^2, \tag{7.10}$$

$$8\pi J_f = e^{-2f}\big(8\pi J + 4K(\nabla f, \cdot)\big). \tag{7.11}$$

Here the second identity is an identity of covectors. Consequently,

$$\begin{aligned} 16\pi e^{4f}(\mu_f - |J_f|_{g_f}) \geq\ & 16\pi(\mu - |J|_g) \\ & + 8\big(-\Delta_g f - |df|_g^2 - |K|_g |df|_g\big). \end{aligned} \tag{7.12}$$

On a fixed smooth boundary $B = \partial\Omega$, when present, use the normal ν pointing into Ω and the expansion $\theta_+ = \operatorname{div}_B \nu + \operatorname{tr}_B K$. Then

$$\theta_{+,f} = e^{-2f}(\theta_+ + 4\partial_\nu f). \tag{7.13}$$

Equivalently, if $u = e^f > 0$, the last summand in (7.12) is $8u^{-1}(-\Delta_g u - |K|_g |du|_g)$.

Proof. Apply the pointwise identities in (3.2) with $n = 3$, $k = 2$, $p = 1$, and $s = 2f$. The densities in that identity are $\mu_{\mathrm{A}} = 8\pi\mu$ and $J_{\mathrm{A}} = 8\pi J$ in this section's physical normalization. Setting the vector argument to zero gives $8\pi e^{4f}\mu_f = 8\pi\mu - 4\Delta f - 4|df|^2$; multiplication by two is (7.10). The covector formula proved there gives $8\pi J_f = e^{-2f}(8\pi J + 4K(\nabla f, \cdot))$. Its new-metric norm introduces a further factor e^{-2f}, so the triangle inequality gives (7.12) with its factor 8. The boundary formula has $k\partial_\nu s = 4\partial_\nu f$ and proves (7.13). Finally, the identity (3.3) uses $U = e^f = u$ here and gives the stated positive-factor form. These are pointwise identities for smooth data, independently of their end decay. □

Lemma 7.5 (Strict direction under the original decay)**.** *Let (Ω, g, K) be an exterior as in Definition 7.1, satisfying (7.2) and (7.9), with integrable μ and $|J|_g$ and finite ADM limits (7.4)–(7.5). Write $B = \partial\Omega$. Choose*

$$0 < \delta < \beta < \min(q, 1).$$

There are smooth positive functions w and ϕ on Ω, with $w = r^{-3-\delta}$ on the end, such that

$$\partial_\nu \phi = -1 \quad \text{on } B, \qquad \phi = O_2(r^{-1}), \tag{7.14}$$

$$-\Delta_g \phi - |K|_g |d\phi|_g \geq w, \qquad \frac{-\Delta_g \phi}{w} \longrightarrow 1. \tag{7.15}$$

The Euclidean normal flux

$$L_\phi = \lim_{r\to\infty} \int_{S_r} \partial_r \phi \, \mathrm{d}A_\delta$$

is finite and negative. For every finite $s \geq 0$, set

$$u_s = 1 + s\phi, \qquad g_s = u_s^4 g, \qquad K_s = u_s^2 K. \tag{7.16}$$

These data are complete up to B and satisfy DEC, strictly for $s > 0$. If $\theta_+ \le 0$ on B, their boundary expansion remains nonpositive and is strictly negative for $s > 0$. Their decay exponent can be taken to be $\min(q, 1) > 1/2$, their constraint densities are integrable, and

$$E_s = E + sA_\phi, \qquad P_s = P, \qquad A_\phi = -\frac{L_\phi}{2\pi} > 0. \tag{7.17}$$

Their full enclosing infima satisfy

$$a_g(B) \le a_{g_s}(B) \le (1 + s\|\phi\|_\infty)^4 a_g(B). \tag{7.18}$$

In particular $E_s \to E$ and $a_{g_s}(B) \to a_g(B)$ as $s \downarrow 0$.

If K is compactly supported and the metric has symbol estimates of all orders with $g - \delta = O(r^{-1})$, the function can instead be chosen with $\phi = O_k(r^{-1})$ for every k, while retaining (7.14) and (7.15).

Proof. We solve a regularized drift equation on bounded exteriors and use a radial barrier to pass to the limit. Its flux determines the energy change, and the conformal identities give the constraint and area bounds.

The drift equation on bounded exteriors. Choose smooth $b \ge |K|_g$, positive everywhere and equal to $b_0 r^{-1-\beta}$ sufficiently far out, where $b_0 > 0$ is chosen large enough. This is possible because $\beta < q$. Choose a smooth positive regularizer ζ, equal to r^{-2} on the end, and extend w smoothly and positively to Ω. We solve

$$-\Delta_g \phi = b\sqrt{|d\phi|_g^2 + \zeta^2} + w, \qquad \partial_\nu \phi = -1, \qquad \phi \longrightarrow 0. \tag{7.19}$$

The regularizer makes this equation smooth even at critical points.

First truncate at a large coordinate sphere S_R and prescribe $\phi = 0$ there. The Neumann and Dirichlet conditions occur on disjoint boundary components. For $0 \le t \le 1$, replace b in (7.19) by tb. At $t = 0$, the mixed Laplacian is invertible: its Dirichlet face gives the Poincare inequality, and the weak solution obtained from its coercive quadratic form is smooth by boundary regularity. At any solution, the linearization is a Laplacian with a smooth drift of norm at most b and with the homogeneous mixed boundary conditions. The strong maximum principle and the Hopf boundary principle make its kernel zero. It is a Fredholm perturbation of the mixed Laplacian of index zero, and is therefore invertible. The boundary estimates used here are the ordinary elliptic estimates for Dirichlet and normal Neumann conditions: positive definiteness of g verifies their principal complementing condition, and the disjoint boundary components can be covered separately [1]. The maximum principles and local elliptic estimates below are used in their usual uniformly elliptic forms [21].

To close continuation at $t = 1$, first work on a fixed truncation. The elliptic $W^{2,p}$ estimate and gradient interpolation give

$$\|\phi\|_{W^{2,p}} \le C_p(1 + \|\phi\|_{L^p} + \|d\phi\|_{L^p}) \le \tfrac{1}{2}\|\phi\|_{W^{2,p}} + C_p'(1 + \|\phi\|_{L^p}).$$

The constants include the fixed boundary data and are uniform in t. If the supremum norms of solutions on this truncation were unbounded, divide them by their supremum norms. The preceding estimate with $p > 3$, compactness, and the equation give a nonzero limit v with homogeneous mixed data satisfying

$$-\Delta_g v = t_* b|dv|_g.$$

This can be written $\Delta_g v + A \cdot dv = 0$ with a bounded measurable drift: take $A = t_* b \nabla v / |dv|$ off the critical set and zero on it. The strong and Hopf maximum principles exclude a nonzero solution with the homogeneous mixed conditions. Thus the supremum norms are bounded. The $W^{2,p}$ estimate, Schauder estimates, and the smooth equation now give the bounds needed for closedness of continuation. Existence on each truncation follows.

Every such solution is nonnegative. A negative minimum cannot be interior because the right side of (7.19) is positive. At an inner-boundary minimum, the outward-domain derivative would be negative by the Hopf principle, whereas the prescribed derivative in that direction is 1. The outer value is zero.

Bounds independent of the outer radius. To pass to the complete exterior, we need bounds independent of R. For fixed sufficiently large A and then sufficiently large r_0, the positive radial function

$$v_0(r) = r^{-1}(1 - Ar^{-\delta})$$

satisfies, on $r \geq r_0$,

$$-\Delta_g v_0 - b|dv_0|_g = A\delta(1+\delta)r^{-3-\delta} + O(r^{-3-q}) + O(r^{-3-\beta}) \geq c_0 w.$$

The constants are fixed independently of the truncation. Since $b\zeta = o(w)$, a sufficiently large multiple of v_0 is a supersolution of the regularized equation. The comparison principle gives

$$0 \leq \phi(x) \leq C(1 + M_R)v_0(r), \qquad M_R = \sup_{\Omega \cap \{r \leq r_0\}} \phi, \quad r_0 \leq r \leq R. \tag{7.20}$$

The notation for the compact core includes all points outside the end chart. Were M_R unbounded along expanding truncations, the normalized solutions ϕ / M_R would have locally uniform $W^{2,p}$ bounds, including at B, by the preceding local estimates and (7.20). A subsequence would converge locally in C^1 to a nonnegative function with maximum one on the core, homogeneous inner Neumann data, and a bounded-drift homogeneous equation. The bound (7.20) makes its value tend to zero at infinity. It consequently attains a positive maximum, contradicting the strong or Hopf principle for that homogeneous equation. This proves the uniform core bound.

Decay and flux. Local compactness and diagonal extraction give a smooth solution of (7.19), with $\phi = O(r^{-1})$. The claimed derivative count can be checked directly on annuli. Put

$$\phi_\rho(y) = \rho\phi(\rho y), \qquad g_\rho(y) = g(\rho y), \qquad 1 < |y| < 4.$$

On smaller fixed annuli the equation is

$$-\Delta_{g_\rho}\phi_\rho = \rho b(\rho y)\sqrt{|d\phi_\rho|^2_{g_\rho} + (\rho^2\zeta(\rho y))^2} + \rho^3 w(\rho y).$$

Its coefficients have uniform $C^{1,1}$ bounds from the two derivatives in (7.9); the scaled drift is $O(\rho^{-\beta})$ and the scaled source is $O(\rho^{-\delta})$. The bounded zeroth-order norm, interior $W^{2,p}$ estimates, and gradient interpolation first give uniform $C^{1,\alpha}$ bounds for some $\alpha > 0$. The right side is then uniformly C^α. Schauder estimates give uniform $C^{2,\alpha}$ bounds on smaller annuli. This proves $\phi = O_2(r^{-1})$ without using a third derivative of g or a second derivative of K.

The right side of (7.19) is $w + O(r^{-3-\beta})$. It is integrable and proves (7.15). The divergence theorem, with outward-domain normal $-\nu$ on B, gives

$$\lim_{r\to\infty} \int_{S_r} \partial_{n_g}\phi \, \mathrm{d}A_g = -|B|_g - \int_\Omega \left(b\sqrt{|d\phi|^2_g + \zeta^2} + w\right) \mathrm{d}V_g. \tag{7.21}$$

The Euclidean flux differs by $O(r^{-q})$ and has the same finite limit.

The conformal family. For every finite $s \geq 0$, the factor u_s in (7.16) is bounded and at least one, so the transformed data are complete up to B. The u-form of Lemma 7.4 gives

$$16\pi u_s^4(\mu_s - |J_s|_{g_s}) \geq 16\pi(\mu - |J|_g) + \frac{8s}{u_s}(-\Delta_g\phi - |K|_g|d\phi|_g) \geq \frac{8s}{u_s}w,$$
$$\theta_{+,s} = u_s^{-2}\left(\theta_+ - \frac{4s}{u_s}\right).$$

This proves the asserted signs for all finite s, without a smallness restriction.

Since $u_s - 1 = O_2(r^{-1})$, the transformed decay exponent is $\min(q, 1) > 1/2$. The exact energy transformation is

$$16\pi u_s^4\mu_s = 16\pi\mu - 8su_s^{-1}\Delta_g\phi.$$

Here $\Delta_g\phi = O(r^{-3-\delta})$, and the additional current term is bounded by a fixed-s constant times $|K|_g|d\phi|_g = O(r^{-3-q})$. Both are integrable in dimension three. The bounded positive conformal factor also preserves integrability in the transformed volume measure.

The leading metric change is $4s\phi\delta$, whose ADM energy flux is $-8s\int_{S_r}\partial_r\phi\,\mathrm{d}A_\delta$ before division by 16π. All remaining metric flux terms are $O_s(r^{-q}) + O_s(r^{-1})$. Writing $\pi = K - (\mathrm{tr}_g\,K)g$, one has $\pi_s = u_s^2\pi$, so the change in its Euclidean momentum flux is $O_s(r^2r^{-1}r^{-1-q}) = O_s(r^{-q})$. This proves (7.17); (7.21) gives $A_\phi > 0$. The flux argument applies without a $1/r$ coefficient expansion for ϕ. Finally, the cut class is unchanged and every cut has area $\int_\Gamma u_s^4\,\mathrm{d}A_g$. Bounding u_s between 1 and $1 + s\|\phi\|_\infty$ and taking infima proves (7.18), including coincident and disconnected cuts.

For the final assertion, take b nonnegative, compactly supported, and at least $|K|$; the proof is unchanged. The end equation is then simply $-\Delta_g\phi = w$. Once the two-derivative estimate has been established, successive annular elliptic estimates use the assumed symbol bounds for g and w to give $\phi = O_k(r^{-1})$ for every k. □

7.5 Compactness of enclosing minimizers

The end replacement will change the metric on receding annuli. The next lemma keeps area-minimizing enclosures in a fixed compact set, where local metric convergence controls their areas.

Lemma 7.6 (Enclosing areas under receding changes)**.** *Let (Ω, g) be an exterior as in Definition 7.1, and set $B = \partial\Omega$. Let g_j be smooth metrics on Ω, converging to g locally uniformly up to B. Suppose that in a fixed chart outside a compact set they satisfy, with constants independent of j,*

$$c\delta \leq g_j \leq C\delta, \qquad r|\partial g_j|_\delta \leq C. \tag{7.22}$$

Suppose also that each g_j has an asymptotically flat end, possibly in a different chart, and admits a fixed compact enclosing competitor with uniformly bounded area. Then minimizing filled enclosures exist, their boundaries lie in one fixed compact subset of Ω, and

$$a_{g_j}(B) \longrightarrow a_g(B).$$

The perimeter and smooth enclosing infima agree. Each minimizing boundary is $C^{1,1}$, and is smooth minimal away from contact with B.

Proof. We first obtain a minimizing enclosure for each metric. A local area lower bound then confines all minimizers to a common compact set, where uniform metric convergence gives convergence of their areas.

Existence. For purposes of perimeter minimization only, extend the metrics through a collar of B and attach a fixed compact filling. The filling need not satisfy any constraint or curvature condition. Extend g_j so that the extensions converge uniformly on this fixed collar and filling. A filled enclosure is then a bounded set containing the prescribed inner obstacle.

For each fixed metric, sufficiently large spheres in its asymptotically flat chart have strictly positive outward mean curvature. Clipping a bounded enclosure at any such sphere cannot increase perimeter. To see this, integrate the divergence of the outward unit normal field of the sphere foliation over the part of the enclosure outside the sphere. Its divergence is positive. Its flux through the exterior boundary is at most the area of that boundary, whereas its flux on the cutting sphere is the negative of the new cutting area. Thus the discarded boundary area is at least the inserted area. Approximation gives the same statement for finite-perimeter sets. The direct method on the resulting bounded domain now gives a minimizer. The clipping observation makes this a minimizer against every bounded competitor, including competitors that extend beyond the chosen truncation.

Confinement to a compact set. Off the obstacle the boundary is locally perimeter minimizing. If it contains a point of radius r sufficiently large, the coordinate ball of radius $c_1 r$ about that point avoids the obstacle. After scaling this ball to unit size, (7.22) gives uniform ellipticity and uniform first-derivative bounds for its area integrand. The local perimeter density estimate gives

$$\mathrm{Area}_{g_j}(\partial F_j \cap B_{c_1 r}) \geq c_2 r^2.$$

Here $c_1, c_2 > 0$ are independent of j and r. To justify the uniform lower bound, work in a rescaled coordinate ball avoiding the obstacle, and let $v(s) = |F_j \cap B_s|_\delta$. Removing $F_j \cap B_s$ is an admissible local comparison. For almost every s, minimality and uniform metric comparability give

$$P_\delta(F_j; B_s) \leq C v'(s), \qquad P_\delta(F_j \cap B_s) \leq C' v'(s).$$

The Euclidean isoperimetric inequality therefore yields $v(s)^{2/3} \leq C'' v'(s)$. At a point in the support of the minimizing frontier, both phases meet every ball. Integrating this differential inequality gives $v(s) \geq c s^3$. Filling the complement in B_s gives the same estimate for $|B_s \setminus F_j|_\delta$. The relative isoperimetric inequality now gives $P_\delta(F_j; B_s) \geq c' s^2$ on a smaller fixed ball. Returning to the original scale proves the displayed area bound. The argument also applies at all support points by taking limits of reduced-boundary points. The estimate depends on metric comparability and the first-derivative bound in (7.22). The fixed competitor bounds the total area independently of j. Consequently the minimizing boundaries lie in a fixed compact set.

Regularity and convergence of the infima. The local obstacle regularity theorem for pure area minimization with a C^2 obstacle gives $C^{1,1}$ regularity, and smoothness off contact, in dimension three; its local estimates require only the appropriate bounded geometry of the smooth metric [24, Regularity Theorem 1.3]. This applies to the fixed smooth obstacle and each of the smooth metrics g_j and g after the collar extension. Push a minimizing boundary slightly away from its contact set by a smooth vector field agreeing there with the outward obstacle normal. In a small obstacle collar its flow increases the signed distance to the obstacle. The resulting C^1 embedded boundary is a positive distance from the obstacle. Smooth local graph approximation, within that distance, preserves enclosure and changes area by an arbitrarily small amount. This proves equality of the smooth and perimeter infima for each metric.

Enlarge the fixed compact region so that its complement is a connected coordinate tail. Each minimizing inner set is either empty or the whole tail there, since its frontier is absent. Boundedness excludes the latter, so the sets themselves also lie in a common compact set. Choose a fixed smooth auxiliary metric on this compact set. Uniform metric comparability gives uniformly bounded auxiliary perimeters, and BV compactness gives a limiting filled enclosure. The obstacle is still contained in the limit. Fill any newly bounded complementary pockets; this only removes frontier. Uniform metric convergence makes the area integrands converge uniformly relative to the auxiliary metric, so lower semicontinuity implies

$$a_g(B) \le \liminf_j a_{g_j}(B).$$

Conversely, using a fixed enclosure with g-area arbitrarily close to $a_g(B)$ gives the reverse upper-limit inequality. □

7.6 Schwarzschild reference charges

The area infimum is now stable under uniformly controlled changes on receding annuli. We next arrange those changes. The vacuum reference end must first have the original ADM charges. This matching makes the later constraint repair possible with a mass change tending to zero.

Lemma 7.7 (A reference end with prescribed timelike charges)**.** *Let $E > |P|$ and $m = (E^2 - |P|^2)^{1/2}$. There is a spacelike plane near spatial infinity of Schwarzschild spacetime of mass m whose induced data, in asymptotically Euclidean coordinates on the plane, have ADM charges (E, P). The reference data satisfy*

$$g_* - \delta = O_k(r^{-1}), \qquad K_* = O_k(r^{-2})$$

for every derivative order k.

Proof. Apply Lemma 3.2 with $n = 3$, $k = 2$, $p = 1$, and $M = m$. In static isotropic coordinates (T, y) its spacetime metric is

$$-\left(\frac{1 - m/(2r)}{1 + m/(2r)}\right)^2 \mathrm{d}T^2 + \left(1 + \frac{m}{2r}\right)^4 \delta_{ij}\,\mathrm{d}y^i\,\mathrm{d}y^j, \qquad r = |y|. \tag{7.23}$$

For $P \neq 0$, put $v = P/E$ and rotate the boost axis into its direction; for $P = 0$, take $v = 0$. Then $\gamma = (1 - |v|^2)^{-1/2} = E/m$, and the reference plane has charges $(m\gamma, m\gamma v) = (E, P)$. The common lemma uses the same future-normal convention for K, and its ADM prefactors become

$$\frac{1}{2k\omega_2} = \frac{1}{16\pi}, \qquad \frac{1}{k\omega_2} = \frac{1}{8\pi}.$$

Thus its charges have precisely the present signs and normalization; the use of physical constraint densities here does not rescale them.

For the later bend, write the plane as $T = v \cdot y$ and set $y = Ax$, where $A = (I - v \otimes v)^{-1/2}$. The induced Minkowski metric in the x coordinates is δ. The perturbation of the metric above from Minkowski space has symbol decay of order r^{-1} on this fixed spacelike plane, and Lemma 3.2 gives the induced metric and tensor orders r^{-1} and r^{-2} through every fixed derivative order. These estimates concern the exact reference end and impose no additional derivative assumption on the original data. □

7.7 Moments and compact linear corrections

The reference end has the required charges, but interpolation can violate DEC. Compact linear corrections reduce this error before the conformal repair. This is related to localized constraint deformation and its adjoint obstructions [14, 13]. The construction below uses compact inverses and direct moment estimates to control the constraint error.

All operators in the next two lemmas are Euclidean. For symmetric covariant tensors define

$$\mathcal{L}h = \partial_i\partial_j h_{ij} - \Delta \operatorname{tr} h, \qquad (\mathcal{D}k)_i = \partial_j(k_{ij} - (\operatorname{tr} k)\delta_{ij}). \tag{7.24}$$

The formal adjoint kernel of $\mathcal{L}$ consists of affine functions; the formal adjoint kernel of $\mathcal{D}$ consists of Euclidean Killing fields. Orthogonality to these kernels is the compatibility condition for the compact inverses below.

Lemma 7.8 (Compact scalar and symmetric divergence inverses)**.** *Use the operators $\mathcal{L}$ and $\mathcal{D}$ in (7.24). Let $\mathcal{A} \subset \mathbb{R}^3$ be a fixed connected open annulus and let $Q \Subset \mathcal{A}$. Smooth sources supported in Q admit the following compact corrections, supported in a fixed larger compact subset of $\mathcal{A}$.*

(i) *If $\int f = 0$ and $\int x^i f = 0$ for $i = 1, 2, 3$, there is a smooth symmetric tensor h with $\mathcal{L}h = f$ and $\|h\|_{W^{j+2,p}} \le C_{j,p}\|f\|_{W^{j,p}}$.*

(ii) *If $\int V \cdot Y = 0$ for every Euclidean Killing field Y, there is a smooth symmetric tensor k with $\mathcal{D}k = V$ and $\|k\|_{W^{k_0+1,p}} \le C_{k_0,p}\|V\|_{W^{k_0,p}}$.*

The estimates hold for integers $j, k_0 \ge 0$ and $1 < p < \infty$.

Proof. Apply Lemma 3.3 in dimension three, with the prescribed source support Q and the same connected annulus. The scalar operator there is exactly $\mathcal{L}$ here, and its affine moments are the four conditions in (i). The vector operator is exactly $\mathcal{D}$; its translational and rotational moments are (ii). The inverse of trace reversal in that proof is $S \mapsto S - (\operatorname{tr} S)\delta/(n-1)$, which becomes $S \mapsto S - (\operatorname{tr} S)\delta/2$. In particular its symmetric correction is obtained from a skew tensor $D_{ijk} = -D_{jik}$ by

$$C_{ijk} = D_{ijk} - D_{ikj} - D_{jki}. \tag{7.25}$$

The identities $C_{ijk} = -C_{ikj}$ and $(C_{ijk} - C_{jik})/2 = D_{ijk}$ cancel the skew part without changing divergence. The two scalar inversions gain two derivatives, and the row-divergence inversion followed by that correction gains one. All supports and bounds are those of the fixed annulus. Thus the full conclusions, including linear dependence and every stated Sobolev order, are the specializations of that lemma. □

Lemma 7.9 (Integrable sources give sublinear first moments)**.** *Use the linear operators in (7.24). For data satisfying (7.9) and integrable $\mu, |J|$, put $h = g - \delta$ and $\pi = K - (\operatorname{tr}_\delta K)\delta$. Then*

$$\begin{aligned} \mathcal{L}h &= 16\pi\mu + O(r^{-2-2q}), \\ \partial_j \pi_{ij} &= 8\pi J_i + O(r^{-2-2q}). \end{aligned} \tag{7.26}$$

In particular these Euclidean sources are integrable. Their scalar boundary fluxes paired with affine functions having zero constant term, and their vector boundary fluxes paired with rotational Killing fields, are $o(R)$ on coordinate spheres of radius R.

Proof. The scalar-curvature expansion has remainder bounded by

$$C(|h||\partial^2 h| + |\partial h|^2 + |K|^2),$$

and the replacement of the momentum constraint by Euclidean divergence has remainder bounded by $C(|h||\partial K| + |\partial h||K|)$. These are $O(r^{-2-2q})$, using exactly the derivatives in (7.9). They are integrable because $2 + 2q > 3$.

For any integrable function F on the end and fixed $c > 1$,

$$\frac{1}{R}\int_{r_0<|x|<cR} |x|\,|F(x)|\,\mathrm{d}x \longrightarrow 0. \tag{7.27}$$

Indeed the contribution inside any fixed radius M tends to zero after division by R; the remaining contribution is at most $c\int_{|x|>M}|F|$. Let $R \to \infty$ and then $M \to \infty$.

For an affine function f, the vector field

$$\mathcal{Q}_f(h)^i = f(\partial_j h_{ij} - \partial_i \operatorname{tr} h) - (\partial_j f)h_{ij} + (\partial_i f)\operatorname{tr} h$$

satisfies $\partial_i \mathcal{Q}_f(h)^i = f\mathcal{L}h$. For a Euclidean Killing field Y,

$$\partial_j(\pi_{ij}Y^i) = Y^i\partial_j\pi_{ij},$$

because π is symmetric. Integrating from a fixed inner sphere and using (7.27) proves the asserted $o(R)$ estimates. □

Proposition 7.10 (Annular replacement with a vanishing deficit)**.** *Let (Ω, g, K) satisfy the hypotheses of Theorem 7.2, including $E > |P|$. Let (g_*, K_*) be the reference end from Lemma 7.7, identified with the original end in the common asymptotically Euclidean coordinates. For all sufficiently large R there are smooth data $(\widetilde{g}_R, \widetilde{K}_R)$ equal to (g, K) on $r \le R$ and to (g_*, K_*) on $r \ge 4R$ such that*

$$16\pi(\widetilde{\mu}_R - |\widetilde{J}_R|_{\widetilde{g}_R}) \ge -\eta_R R^{-3}, \qquad \eta_R \longrightarrow 0. \tag{7.28}$$

The possible negative part is supported in $R < r < 4R$. The metrics there are uniformly Euclidean comparable and have uniformly bounded scaled first derivatives. The construction uses no derivatives of the original data beyond those in (7.9) for its quantitative estimates.

Proof. After rescaling to a fixed annulus, interpolation produces compact commutator errors. Charge matching and integrability make their moments small; compact linear corrections remove the remaining errors up to a quadratic remainder.

Interpolation and commutators. Fix $\frac{1}{2} < \beta < \min(q, 1)$ and work on the annulus $\mathcal{A} = \{1 < |y| < 4\}$ with $x = Ry$. The scaled fields are

$$g_R(y) = g(Ry), \qquad k_R(y) = RK(Ry),$$

and likewise for the reference data. Their constraint densities are the original densities multiplied by R^2. If $h_R = g_R - \delta$, then

$$\|h_R\|_{C^2(\mathcal{A})} + \|k_R\|_{C^1(\mathcal{A})} + \|h_{*,R}\|_{C^2(\mathcal{A})} + \|k_{*,R}\|_{C^1(\mathcal{A})} \le CR^{-\beta}. \tag{7.29}$$

Choose $\chi \in C^\infty(\mathcal{A})$ with $0 \le \chi \le 1$, equal to one near $|y| = 1$ and zero near $|y| = 4$, with all its derivatives supported in a fixed compact subannulus. Blend both fields by this cutoff.

Write the constraints in the unnormalized form

$$\mathcal{C}(g,k) = (\mathrm{Scal}_g + (\mathrm{tr}_g\, k)^2 - |k|_g^2, \mathrm{div}_g(k - (\mathrm{tr}_g\, k)g)).$$

Their Euclidean linear part is $(\mathcal{L}h, \mathcal{D}k)$. On fields satisfying (7.29), the nonlinear remainder is bounded in supremum norm by $CR^{-2\beta}$. Thus the blended constraints equal the blended original constraint sources, plus a quadratic error and the compact commutators

$$f_R = \mathcal{L}(\chi(h_R - h_{*,R})) - \chi\mathcal{L}(h_R - h_{*,R}),$$
$$V_R = \mathcal{D}(\chi(k_R - k_{*,R})) - \chi\mathcal{D}(k_R - k_{*,R}).$$

They have $C^{0,1}$ norms at most $CR^{-\beta}$. In detail the scalar commutator applied to a symmetric tensor h is

$$2\partial_i\chi(\partial_j h_{ij} - \partial_i \,\mathrm{tr}\, h) + (\partial_i\partial_j\chi)h_{ij} - (\Delta\chi)\,\mathrm{tr}\, h;$$

one derivative of this expression uses at most two derivatives of h. The vector commutator is $(\partial_j\chi)(k_{ij} - (\mathrm{tr}\, k)\delta_{ij})$; one derivative uses at most one derivative of k.

Moment estimates and corrections. Every affine moment of f_R and every Killing-field moment of V_R is $o(R^{-1})$. For the constant moments, integration by parts on $\mathcal{A}$ gives R^{-1} times the difference between the physical translation fluxes on its two boundary spheres, less the integrals of the blended linear sources. The flux difference tends to zero because the reference and original ADM vectors agree. Replacing the physical momentum integrand by its Euclidean trace reversal introduces an error $O(R^{1-2q})$, which tends to zero. By Lemma 7.9, the physical linear-source integrals on this receding annulus tend to zero as well. For a linear scalar test or a rotational vector test the corresponding factor is R^{-2}. Its physical boundary fluxes are $o(R)$ by the same lemma, while the weighted source integrals are $o(R)$ by (7.27). This proves the claim.

These small moments can be removed by smooth bumps with fixed support. For a basis $p_1, \dots, p_4$ of affine functions and a fixed nonnegative bump ω positive on a ball inside the annulus, the Gram matrix

$$G_{ab} = \int \omega p_a p_b$$

is positive definite. Subtract $\omega \sum_{a,b} p_a (G^{-1})_{ab} \int p_b f_R$ from f_R. For the vector moments use the same construction with the Gram matrix $\int \omega Y_a \cdot Y_b$ of a basis of the six Euclidean Killing fields. It too is positive definite, since a nonzero Killing field cannot vanish on an open ball. Both bump corrections have every fixed smooth norm $o(R^{-1})$.

Lemma 7.8, applied to the negative of the moment-free commutators, gives compact corrections. Since the commutators have $W^{1,p}$ norms $O(R^{-\beta})$, choosing $p > 3$ gives metric and tensor corrections of size $O(R^{-\beta})$ in $W^{3,p}$ and $W^{2,p}$, respectively. Sobolev embedding gives the needed $C^{2,\alpha}$ and $C^{1,\alpha}$ bounds. Their supports stay in a fixed compact subset of $\mathcal{A}$. Extend them by zero and undo the scaling. The metric remains positive definite for large R.

The remaining constraint deficit. Denote the original scaled sources by (A_R, B_R), so $A_R \geq 2|B_R|_{g_R}$. The reference sources vanish. The corrected sources therefore have the form

$$(\chi A_R, \chi B_R) + o(R^{-1}) + O(R^{-2\beta})$$

in supremum norm. The norm of B_R changes by at most $CR^{-\beta}|B_R| \leq CR^{-2\beta}$ when the metric is changed, since the pointwise constraint formula gives $|B_R| \leq CR^{-\beta}$. All other nonlinear correction terms have the same quadratic bound. Because $2\beta > 1$, the scaled DEC deficit is

$o(R^{-1})$. Multiplication by R^{-2} gives (7.28). Outside the correction annulus the data are either the original DEC data or the exact vacuum reference data. Smoothness follows from smoothness of the original fields and of the compact inverse constructions, irrespective of the sizes of their unneeded higher derivatives. □

7.8 Bending to a rest slice and repairing the deficit

The annular correction has reduced the possible DEC failure to $o(R^{-3})$ on a region of radius comparable to R. We now work inside the exact vacuum reference part. Bending makes the distant tensor vanish; a radial conformal factor then repairs the earlier deficit with mass change $o(1)$. Both operations must retain the uniform geometry needed by Lemma 7.6.

Lemma 7.11 (A vacuum bend with uniform geometry)**.** *The data in Proposition 7.10 can be modified farther out, inside their exact Schwarzschild part, so that they are induced by a constant-static-time slice outside a compact set. This modification adds no constraint deficit. Write (g_R, K_R) for the resulting unscaled data. In the common end coordinates the metrics are uniformly Euclidean comparable, and throughout all transition annuli*

$$|\partial g_R|_\delta + |K_R|_\delta \le C/r. \tag{7.30}$$

There is a smooth proper radius ρ_R on the end which equals the original coordinate radius on the gluing annulus and equals the rest isotropic radius beyond the transitions, with

$$cr \le \rho_R \le Cr, \qquad |d\rho_R|^2_{g_R} \ge c, \qquad |\Delta_{g_R}\rho_R| \le C/r. \tag{7.31}$$

All transition annuli lie between fixed multiples of R; the constants can depend on the prescribed timelike ADM vector, but not on R.

Proof. A logarithmic cutoff bends the reference plane to a static slice while keeping it uniformly spacelike. A second interpolation then joins the original radius to the static isotropic radius.

The spacelike graph. Use the description $T = v\cdot y$, $y = Ax$, from Lemma 7.7. Choose $R_b = c_b R$ so large, with c_b fixed, that the region $|y| \ge R_b$ is entirely inside the exact reference part. Let ψ be a smooth function equal to one on $(-\infty, 0]$, zero on $[1, \infty)$, and taking values in $[0, 1]$. Replace the plane by the graph

$$T_R(y) = (v\cdot y)\psi\left(\frac{\log(|y|/R_b)}{L}\right). \tag{7.32}$$

Its Euclidean gradient satisfies

$$|dT_R|_\delta \le |v|(1 + \|\psi'\|_\infty/L).$$

Choose a fixed L making the right side strictly smaller than one. The lapse and spatial coefficients in (7.23) converge to their Minkowski values, so for all sufficiently large R this graph is uniformly spacelike. In fact for positive Schwarzschild mass the lapse is smaller than one and the spatial conformal factor exceeds one, which only improves this estimate on the static exterior.

The graph agrees with the original plane before R_b and with a constant-time slice after $e^L R_b$. Its first derivative is bounded, and its second and third derivatives are $O(r^{-1})$ and $O(r^{-2})$. The induced metric is uniformly comparable with δ; differentiating its pullback formula gives the first bound in (7.30). The second fundamental form uses the graph's second derivatives and ambient first derivatives, divided by a uniformly positive spacelikeness factor, and gives the other bound.

These estimates remain true in the x coordinates because A is a fixed invertible linear map. The constraints vanish on the entire bent part by the Gauss–Codazzi equations in exact vacuum Schwarzschild spacetime.

The interpolated radius. On the bent region keep $\rho_R = |x| = |A^{-1}y|$. Beyond the bend, where the slice is already static, write $y = r\omega$ and put $a(\omega) = |A^{-1}\omega|$. Choose R_c a sufficiently large fixed multiple of $R_b e^L$, and a cutoff χ_0 equal to one before zero and zero after one. In that static region set

$$\rho_R(r,\omega) = r \exp\left[\chi_0\left(\frac{\log(r/R_c)}{L_0}\right)\log a(\omega)\right].$$

For fixed sufficiently large L_0 its radial derivative is bounded below by a positive constant: differentiating in r gives the positive factor ρ_R/r times $1 + L_0^{-1}\chi_0' \log a$, which is at least $1/2$. The function and its first two Euclidean derivatives have the bounds $\rho_R \asymp r$, $|d\rho_R| \le C$, and $|\partial^2 \rho_R| \le C/r$. Before this interpolation, the same bounds hold for $|A^{-1}y|$, and uniform metric comparability gives the gradient lower bound. Combining these facts with (7.30) proves (7.31). After the interpolation, $\rho_R = r$ is precisely the static isotropic radius. All radius ratios used in the construction are fixed, independently of R. □

Lemma 7.12 (A radial DEC repair with vanishing mass change)**.** *Let (Ω, g, K) satisfy the hypotheses of Theorem 7.2, including DEC, integrable constraint densities, weak future trapping of $B = \partial\Omega$, and $E > |P|$. Set $m = (E^2 - |P|^2)^{1/2}$. Let (g_R, K_R) be obtained by applying Proposition 7.10 followed by Lemma 7.11. There is a smooth positive conformal factor u_R, constant on the unchanged inner region and tending to one at infinity, such that $(u_R^4 g_R, u_R^2 K_R)$ satisfy DEC everywhere. Moreover*

$$\|u_R - 1\|_\infty = o(R^{-1}), \qquad |du_R| = o(R^{-2}) \quad \text{on the transition region.} \tag{7.33}$$

The final end is exactly Schwarzschild with $K = 0$ and mass $m + o(1)$. The boundary expansion retains its original weak sign.

Proof. We construct a radial factor whose negative Laplacian controls both the constraint deficit and the first-order term on the transition annuli. On the static end, its radial flux is continued to a constant, giving a Schwarzschild end with a mass change that tends to zero.

The correction on the transition annuli. All geometry estimates below are uniform in R. Enlarge the fixed transition region slightly so that, in terms of $t = \rho_R/R$, it lies inside an interval (a, b) with $0 < a < b < \infty$. Arrange that $t \ge b$ is in the static spherical region, and that the support of the possible deficit lies in a fixed smaller interval $I \Subset (a, b)$. Increasing b if needed makes these assertions follow from Lemma 7.11. We take $a = 1$: the compact supports in the annular correction leave a fixed neighborhood of $t = 1$ free of deficit. The factor constructed below will consequently be constant throughout the original region $r \le R$, and is extended by that constant over the compact core of Ω.

Let $\eta_R \to 0$ be as in (7.28), replacing it by a positive majorant if necessary. We will choose $\varepsilon_R = C_0\eta_R$ and construct

$$u_R = 1 + \frac{\varepsilon_R}{R} F_R(\rho_R/R),$$

where F_R and the relevant derivatives are bounded independently of R. Write $z_R = -F_R'$. On $[a, b+1]$, initially define z_R by

$$z_R' = C_1 z_R + \omega_1, \qquad z_R(a) = 0, \tag{7.34}$$

where ω_1 is smooth, nonnegative, zero near a, and at least one on I. Take C_1 sufficiently large, depending only on the constants in (7.30) and (7.31). The solution is nonnegative and has uniform bounds on this fixed interval. For a function with $F_R' = -z_R$ the chain rule gives

$$\begin{aligned} -\Delta_{g_R} u_R - |K_R|_{g_R} |du_R|_{g_R} &= \frac{\varepsilon_R}{R^3}\left[z_R' |d\rho_R|_{g_R}^2 + R z_R \Delta_{g_R}\rho_R - R z_R |K_R|_{g_R} |d\rho_R|_{g_R}\right] \\ &\geq \frac{\varepsilon_R}{R^3}(c_1 z_R' - C_2 z_R). \end{aligned}$$

Choose C_1 so that this is nonnegative and is at least $c_1 \varepsilon_R R^{-3}$ on I.

Continuation along the spherical end. On the spherical tail, $K_R = 0$ and the Schwarzschild radial Laplace flux coefficient is $(r + m/2)^2$. Superharmonicity there is controlled by this radial flux. Put $c_R = m/(2R)$ and, on the still spherical interval near b, define

$$A_R(t) = (t + c_R)^2 z_R(t).$$

It has positive derivative wherever $z_R > 0$. Continue it to a positive constant on $[b+1, \infty)$ while keeping $A_R' \geq 0$. For an explicit smooth continuation, continue the solution of (7.34) to $b+1$, multiply the derivative of $(t+c_R)^2 z_R(t)$ by a nonnegative cutoff equal to one near b and zero near $b+1$, and integrate with the initial value $A_R(b)$. Define $z_R = A_R(t)/(t+c_R)^2$ on this tail, and set it to zero for $t \leq a$. All joins are smooth, the functions are nonnegative, and their bounds on finite intervals are uniform in R. The constant tail value, denoted A_R^∞, is bounded independently of R.

Now define

$$F_R(t) = \int_t^\infty z_R(s)\, \mathrm{d}s.$$

It is nonnegative, uniformly bounded, and constant before a. The preceding construction gives a superharmonic u_R on the spherical tail, because its signed radial flux is $-\varepsilon_R A_R(t)$ and is nonincreasing. It also gives (7.33). For large R, $1 \leq u_R \leq 2$. Increasing C_0 ensures

$$-\Delta_{g_R} u_R - |K_R||du_R| \geq \frac{u_R}{8}\left[16\pi(|J_R|_{g_R} - \mu_R)\right]_+.$$

On the original inner region the conformal factor is constant and the original data satisfy DEC. Elsewhere this inequality and Lemma 7.4 prove DEC for the repaired data. The u_R-form of the conformal inequality includes the gradient-square term in the formula for $\log u_R$.

The final mass and boundary sign. On the final end,

$$u_R = 1 + \frac{\varepsilon_R A_R^\infty}{r + m/2}.$$

Writing $a_R = \varepsilon_R A_R^\infty = o(1)$, one obtains the identity

$$u_R^4 \left(1 + \frac{m}{2r}\right)^4 \delta = \left(1 + \frac{m/2 + a_R}{r}\right)^4 \delta. \tag{7.35}$$

Thus the final mass is $m + 2a_R = m + o(1)$ and the momentum is zero. Multiplication of K_R by u_R^2 preserves its compact support. On the inner boundary, where u_R is constant, the expansion is simply multiplied by u_R^{-2}, so its weak sign is preserved. □

7.9 Approximation by strict data

The preceding construction has replaced the end while preserving DEC and the boundary expansion sign. A final strict conformal change puts the data in the class used for the elliptic deformation.

Proposition 7.13 (Reduction to strict data with a controlled rest end)**.** *Let (Ω, g, K) satisfy the exterior hypotheses of Theorem 7.2, including $\theta_+ \le 0$ on its smooth compact boundary, and let $m = (E^2 - |P|^2)^{1/2} > 0$. There is a sequence of smooth exterior data (g_j, K_j) on Ω with the following properties:*

(i) *$\mu_j > |J_j|_{g_j}$ everywhere and $\theta_{+,j} < 0$ on B. The constraint densities are integrable. On the end their DEC margin is bounded below by $c_j r^{-3-\delta}$ for some $c_j > 0$ and one fixed $0 < \delta < 1$. For a smooth positive radius extension ρ, the constant can be decreased so that $\mu_j - |J_j|_{g_j} \ge c_j \rho^{-3-\delta}$ on all of Ω.*

(ii) *K_j is compactly supported, and in a rest chart $g_j - \delta = O_k(r^{-1})$ and $R_{g_j} = O_k(r^{-3-\delta})$ for every k. In particular $\mathrm{Ric}_{g_j} = O(r^{-3})$ there. Their ADM momentum is zero.*

(iii) *$(g_j, K_j) \to (g, K)$ smoothly on every fixed compact subset up to the boundary, their ADM energies satisfy $E_j \to m$, and $a_{g_j}(B) \to a_g(B)$.*

Proof. Proposition 7.10, Lemma 7.11, and Lemma 7.12, applied along any sequence $R_j \to \infty$, give DEC exteriors with compactly supported second fundamental form and an exact rest Schwarzschild end with energy tending to m. They agree with the original data on expanding compact subsets except for a constant conformal factor tending to one. Hence they converge smoothly on each fixed compact subset. Their boundary expansion is weakly negative.

The last part of Lemma 7.5 gives a strict conformal deformation of each of these exteriors, using the same chosen $0 < \delta < 1$ and a compactly supported drift coefficient. Choose its positive parameter s in (7.16) small enough that the energy change, $\|\log(1 + s\phi)\|_\infty$, and the supremum of its first derivative multiplied by the fixed original coordinate radius are at most $1/j$. Also require the change in the first j smooth norms on the first j members of a compact exhaustion to be at most $1/j$. All these quantities are finite for the fixed j-th exterior, so these choices are possible. The parameter is chosen separately for each exterior after its bending radius has been fixed.

The resulting data have pointwise strict DEC and strict trapping. The quantitative end gap is at least $c_j r^{-3-\delta}$. On its compact complement the continuous positive function $(\mu_j - |J_j|_{g_j})\rho^{3+\delta}$ has a positive minimum. Replacing c_j by the smaller of this minimum and the end constant proves the global weighted assertion in (i). The strict conformal factor has symbol decay of all orders and does not enlarge the support of K. On the far Schwarzschild end the drift vanishes, so $-\Delta\phi = w = r^{-3-\delta}$. Since the rest Schwarzschild metric has zero scalar curvature and $K = 0$ there, the u-formula gives

$$R_{g_j} = 8s(1 + s\phi)^{-5} r^{-3-\delta}.$$

This proves the scalar-curvature bound in (ii). The Ricci bound follows from the two-derivative metric decay. The energy change tends to zero and the momentum change is zero. The prescribed smallness gives smooth local convergence, proving (ii) and the local convergence and charge assertions in (iii).

The metrics before the strict conformal deformation satisfy the uniform comparisons and scaled first-derivative bounds of Lemma 7.6, in the fixed original end chart: the reference plane,

the bend, and the radius interpolation involve only fixed linear maps and fixed radius ratios. The radial repair preserves these bounds. The parameters of the final conformal deformation were chosen to preserve them as well. That lemma therefore gives the convergence of enclosing infima. □

7.10 Future timelikeness from the exterior inequality

The conformal direction already constructed allows us to move the ADM energy upward while fixing the momentum and increasing every enclosing area. If the original ADM vector were not future timelike, we could approach the future null cone through data to which the timelike inequality applies. The positive area bound would persist along this family, giving a contradiction.

Theorem 7.14 (Future timelikeness and the exterior inequality)**.** *Let (Ω, g, K) satisfy all the hypotheses of Theorem 7.2 except $E > |P|_\delta$. Then*

$$E > |P|_\delta, \qquad \sqrt{E^2 - |P|_\delta^2} \geq \sqrt{\frac{a_g(B)}{16\pi}}, \qquad B = \partial\Omega.$$

Proof. Lemma 7.5 supplies a positive function ϕ, a constant $A_\phi > 0$, and, for every finite $s \geq 0$, the data

$$g_s = (1 + s\phi)^4 g, \qquad K_s = (1 + s\phi)^2 K.$$

These data retain completeness, integrable constraint densities, DEC, weak future trapping of B, and the required differentiated decay. The family is defined for the original ADM vector, and its charge and area identities, (7.17) and (7.18), give

$$E_s = E + sA_\phi, \qquad P_s = P, \qquad a_{g_s}(B) \geq a_g(B) > 0,$$

where the last strict inequality is Lemma 7.3.

Suppose $E \leq |P|_\delta$, and set $s_0 = (|P|_\delta - E)/A_\phi$. This is a finite nonnegative number. For every $s > s_0$ the deformed vector is future timelike, so Theorem 7.2, applied to these fixed data, implies

$$\sqrt{(E + sA_\phi)^2 - |P|_\delta^2} \geq \sqrt{\frac{a_{g_s}(B)}{16\pi}} \geq \sqrt{\frac{a_g(B)}{16\pi}} > 0.$$

As $s \downarrow s_0$ the left side tends to zero, while the final lower bound is fixed and positive. This contradiction proves $E > |P|_\delta$. The timelike exterior inequality now applies directly to (Ω, g, K) and gives the stated bound. □

7.11 Complete data with finitely many ends

We now pass from a one-ended exterior to a complete boundaryless initial-data manifold. The ambient enclosing class allows other ends to remain on the chosen side. We first isolate the chosen end to use the exterior theorem, and then compare the two area infima in the direction required for the conclusion.

Definition 7.15 (Admissible boundary and ambient enclosing area)**.** Let (M, g, K) be a smooth three-dimensional initial data set without boundary, with M connected, and fix one asymptotically flat end e. An admissible boundary for e is a nonempty compact smooth embedded two-sided surface $S = \partial D^+$, where D^+ is a smooth open region containing the entire sufficiently distant

part of e. The surface and the region may be disconnected, and D^+ may contain other ends. For the normal ν pointing into D^+, require

$$H_S + \operatorname{tr}_S K \le 0.$$

Its minimum enclosing area is

$$A_{\min}(S) = \inf \left\{ \operatorname{Area}_g(\partial D'^+) : \begin{array}{l} D'^+ \subseteq D^+ \text{ is a smooth open region containing} \\ \text{the entire sufficiently distant part of } e, \\ \partial D'^+ \text{ is compact, smooth, and embedded} \end{array} \right\}. \tag{7.36}$$

The full boundary area is counted, including portions coincident with S. Competitor regions D'^+ may be disconnected and may retain other ends. Their boundaries need be neither trapped nor minimal. The definition imposes no sign condition on the inward null expansion.

Theorem 7.16 (Spacetime Penrose inequality for complete initial data)**.** *Let (M, g, K) be a connected orientable smooth three-dimensional initial data set without boundary, with g complete. Suppose that outside a compact set M has finitely many asymptotically flat ends. On each end, in its asymptotic coordinates, assume*

$$g - \delta = O_2(r^{-q}), \qquad K = O_1(r^{-1-q}), \qquad q > \tfrac{1}{2},$$

and finite ADM energy and momentum limits with the normalizations (7.4)–(7.5). The decay exponents may differ between ends. Assume that the physical constraint densities (7.1) obey $\mu \ge |J|_g$ and that both μ and $|J|_g$ are integrable on M.

For every chosen end and every admissible boundary S in Definition 7.15, its ADM vector (E, P) satisfies

$$E > |P|_\delta, \qquad \sqrt{E^2 - |P|_\delta^2} \ge \sqrt{\frac{A_{\min}(S)}{16\pi}}. \tag{7.37}$$

Proof. We isolate the chosen end by truncating the other ends. The resulting exterior satisfies the boundary sign condition. Its inequality transfers through a one-sided comparison of the enclosing areas.

Restriction to one end. Let D_0 be the connected component of D^+ containing the distant chosen end. Its boundary is a union of components of S. It is nonempty: otherwise D_0 would be both open and closed in the connected manifold M, and hence equal to M, contradicting $S \neq \varnothing$. The closure $X_0 = \overline{D_0}$ is therefore a smooth manifold with nonempty compact boundary. Since S is compact, each sufficiently distant end of M lies either entirely in D_0 or entirely outside it. Thus X_0 has finitely many ends, precisely those original end tails contained in D_0.

For every end of X_0 other than the chosen one, remove a sufficiently distant open coordinate tail. The tails can be chosen disjoint and disjoint from S. Denote the remainder by Ω. It is connected: a path segment entering one of the removed tails can be replaced by a path on the connected coordinate sphere bounding that tail. Consequently Ω is a one-ended smooth exterior with compact boundary

$$B = \partial D_0 \cup \bigcup_{\text{other ends of } X_0} S_{R_i}.$$

This boundary is nonempty because ∂D_0 is nonempty.

The set Ω is closed in the complete manifold (M, g), and it is complete for its intrinsic distance with B included. Indeed, an intrinsic Cauchy sequence is ambient Cauchy and has its ambient

limit in Ω. At an interior limit point the two distances are locally comparable. At a boundary limit point, a smooth boundary chart identifies Ω locally with a half-ball and gives the same comparison, so the sequence converges intrinsically as well. Restricting the data preserves DEC and integrability. The chosen end is unchanged, so its decay and ADM vector remain the original ones.

On ∂D_0 the normal into Ω is the normal into D^+, and the future expansion is nonpositive. On an added sphere S_{R_i}, the normal into Ω points toward decreasing radius in the removed end. The differentiated asymptotic assumptions give

$$H_{S_{R_i}} = -\frac{2}{R_i} + O(R_i^{-1-q_i}), \qquad \operatorname{tr}_{S_{R_i}} K = O(R_i^{-1-q_i}).$$

Choose each R_i large enough that their sum is negative. Every component of B is then weakly future outer trapped for the normal into Ω. Theorem 7.14 applies and yields

$$E > |P|_\delta, \qquad \sqrt{E^2 - |P|_\delta^2} \geq \sqrt{\frac{a_g(B)}{16\pi}}. \tag{7.38}$$

Comparison of the enclosing areas. We claim

$$a_g(B) \geq A_{\min}(S). \tag{7.39}$$

Take first a smooth enclosing cut Γ contained in the interior of Ω. The connected component on its distant chosen-end side is a smooth open region $D'^+ \subset D_0 \subset D^+$. Its boundary consists of components of Γ, so it is compact and smooth and has area at most $\operatorname{Area}_g(\Gamma)$. It is therefore an admissible competitor in (7.36). For a cut that meets or coincides with B, a smooth collar push into the interior, with the obstacle side filled, gives enclosing cuts whose areas converge to its full area. This applies also to contact with the added coordinate spheres. Hence every full enclosing cut of Ω has area at least $A_{\min}(S)$, proving (7.39) after taking the infimum. The ambient definition permits the chosen-end component alone as a competitor, so the comparison also holds when other components of D^+ are discarded. Combining the area comparison with (7.38) proves the theorem. □

The auxiliary exterior excludes the other distant ends, whereas the ambient competitor class allows them. The proof uses only the one-sided comparison (7.39). Rigidity concerns the original data and requires a separate argument beyond these conformal changes and end truncations.

8 Weak asymptotics in four dimensions

We prove a lower bound for the ADM rest mass in terms of the least three-volume needed to separate a chosen asymptotically flat end from the compact weakly trapped boundary and the other ends. The original end has only two controlled metric derivatives and one controlled derivative of the second fundamental form. To apply Theorem 2.3, we replace it by a static end with zero momentum, while controlling the area of every enclosing cut. A small conformal change then makes the dominant energy and boundary inequalities strict.

8.1 Initial data and enclosing cuts

Four is the spatial dimension; the associated spacetime dimension is five, and a hypersurface area is a three-dimensional Riemannian volume. Here $\omega_3 = |S^3| = 2\pi^2$, and we write $\tau = \operatorname{tr}_g K$. Specializing the constraint convention already fixed in the paper gives

$$2\mu = R_g + (\operatorname{tr}_g K)^2 - |K|_g^2, \qquad J_i = \nabla^j (K_{ij} - (\operatorname{tr}_g K) g_{ij}).$$

Recall that R_g denotes scalar curvature and ∇ is the Levi–Civita connection of g. The cosmological constant is zero. We use $\Delta_g = \operatorname{div}_g \operatorname{grad}_g$. For a hypersurface with specified unit normal ν, write

$$H = \operatorname{div}_{\tan} \nu, \qquad \operatorname{tr}_{\tan} K = (g^{ij} - \nu^i \nu^j) K_{ij}, \qquad \Theta_K = H + \operatorname{tr}_{\tan} K.$$

At an inner boundary the normal points into the exterior. In an auxiliary trapped-region construction it points out of the bounded region.

Recall that $O_j(r^{-a})$ includes coordinate derivatives of order $k \le j$, bounded by $O(r^{-a-k})$. On an asymptotically flat end the ADM normalization is

$$E = \frac{1}{6\omega_3} \lim_{R\to\infty} \int_{|x|=R} (\partial_j g_{ij} - \partial_i g_{jj}) n_\delta^i \, \mathrm{d}A_\delta, \tag{8.1}$$

$$P_i = \frac{1}{3\omega_3} \lim_{R\to\infty} \int_{|x|=R} (K_{ij} - \tau g_{ij}) n_\delta^j \, \mathrm{d}A_\delta. \tag{8.2}$$

Here n_δ and $\mathrm{d}A_\delta$ are Euclidean. Whenever a charge is used below, its stated limit is required to exist.

Definition 8.1 (Enclosing cuts)**.** Let M have nonempty compact boundary S and finitely many asymptotically flat ends, and fix an end e. An admissible outer domain D_e is a connected smooth codimension-zero submanifold with boundary, closed as a subset of M, such that its manifold interior lies in $\operatorname{int} M$, it contains the entire sufficiently distant part of e, and it contains no sufficiently distant part of any other end. Its entire intrinsic manifold boundary $\Gamma = \partial_{\mathrm{man}} D_e$ is required to be compact. The enclosing volume is

$$A_e(S) = \inf_{D_e} |\partial_{\mathrm{man}} D_e|_g.$$

All frontier coincident with S is counted. The cut may be disconnected. In the one-ended case, e denotes the unique end, and $D_e = M$ is allowed and has intrinsic boundary S. When the metric or exterior varies, we display that dependence in the arguments of A_e.

The infimum over smooth cuts need not be attained. The other ends and the entire original boundary are obstacles to the selected outer domain. This end-isolating convention differs from the three-dimensional complete-data class, whose chosen-end region may be disconnected and may retain other ends.

Theorem 8.2. *Let (M^4, g, K) be smooth and connected, with M orientable, K an arbitrary smooth symmetric covariant two-tensor, and $S = \partial M$ nonempty, compact, and smooth. Suppose g is complete as a metric space with S included. Outside a compact set let M have finitely many, and at least one, ends diffeomorphic to the complement of a closed ball in $\mathbb{R}^4$. Assume*

$$\mu \ge |J|_g, \qquad \mu, |J|_g \in L^1(M, \mathrm{d}V_g),$$

and, on every end for a common $q > 1$,

$$g - \delta = O_2(r^{-q}), \qquad K = O_1(r^{-1-q}),$$

with finite energy and momentum limits (8.1)–(8.2). On every component of S assume $\Theta_K \le 0$, with normal into M.

Then for each end e,

$$E_e \ge \sqrt{|P_e|_\delta^2 + \frac{1}{4}\left(\frac{A_e(S)}{\omega_3}\right)^{4/3}}. \tag{8.3}$$

In particular $E_e > |P_e|_\delta$, and the invariant mass satisfies

$$\sqrt{E_e^2 - |P_e|_\delta^2} \ge \frac{1}{2}\left(\frac{A_e(S)}{\omega_3}\right)^{2/3}. \tag{8.4}$$

The proof first treats one end with timelike ADM vector. Proposition 8.8 replaces that end by a static end whose energy approaches the original rest mass and whose enclosing area has the required lower bound. Lemma 8.9 then produces prepared data. The prepared inequality is applied before taking either limit: first the strictification tends to zero at a fixed replacement radius, then that radius tends to infinity. Lemma 8.5 excludes nontimelike ADM vectors, and Lemma 8.3 handles the other ends and establishes $A_e(S) > 0$. Proposition 8.10 gives this deduction. The maximal-DEC consequence is Corollary 8.12. The tensor K is arbitrary subject to the stated hypotheses; its trace $\tau = \operatorname{tr}_g K$ requires no additional decay assumption.

The coefficient is sharp. On the time-symmetric Schwarzschild–Tangherlini exterior [43],

$$g_m = \left(1 + \frac{m}{2r^2}\right)^2 \delta, \qquad K = 0, \qquad r \ge \sqrt{m/2},$$

the ADM energy is m, the momentum is zero, and the boundary has three-volume $\omega_3(2m)^{3/2}$. The areal radius $r + m/(2r)$ is nondecreasing on this exterior. Radial projection to its boundary is therefore nonexpanding on three-dimensional tangent volumes; its degree on an enclosing cut is one. The minimum enclosing volume is consequently the boundary volume, and (8.3) is an equality for this family. The theorem concerns the numerical bound. Classification at equality requires a separate argument on the original data.

8.2 Positivity and removal of other ends

Lemma 8.3 (Cut reduction and positivity)**.** *Under the hypotheses of Theorem 8.2, the quantity $A_e(S)$ is finite and strictly positive for every end e. Truncating all other ends at sufficiently large coordinate spheres gives a connected, complete one-ended exterior M_T with compact smooth boundary S_T. All its boundary components have future nonpositive expansion with the normal into M_T. Its selected-end charges are the original charges, and*

$$A_e(M_T, g) \ge A_e(S), \qquad \lim_{T\to\infty} A_e(M_T, g) = A_e(S). \tag{8.5}$$

The common parameter T in the limit can be replaced by any exhaustion in which every unwanted truncation radius tends to infinity.

Proof. Positivity. A sufficiently distant sphere in the chosen end bounds an admissible outer domain, so the infimum is finite. To obtain a positive lower bound, choose an embedded arc from an open disk in S to the chosen end, and continue it along a straight coordinate ray. The compact part of the arc can be chosen embedded and disjoint from the boundary except at its initial point. A boundary collar and the tubular neighborhood theorem give a proper tube with coordinates

$$[0, \infty) \times B^3 \longrightarrow M.$$

Its initial section lies in S; its far part is a Euclidean tube of fixed transverse coordinate size. Choose a smooth nonnegative function η compactly supported in B^3, with integral one, and define in this tube

$$\alpha = \eta(z)\, \mathrm{d}z^1 \wedge \mathrm{d}z^2 \wedge \mathrm{d}z^3.$$

Extend α by zero across the lateral tube boundary. It is a smooth closed three-form on M. Its comass, meaning the supremum of its absolute value on unit simple three-vectors, has a finite bound C_α: the initial part is compact, and the far metric is uniformly comparable with the Euclidean metric. A sufficiently distant selected-end sphere has α-integral one, after fixing orientation.

Let D_e be any admissible outer domain with cut Γ. Cut off its selected-end tail at a sphere beyond Γ. The resulting domain is compact, since D_e contains no distant part of another end. Its boundary is the distant sphere together with the *entire* intrinsic boundary Γ, with the boundary orientations. Stokes' theorem gives

$$\left| \int_\Gamma \alpha \right| = 1, \qquad |\Gamma|_g \geq C_\alpha^{-1}.$$

This proof applies to disconnected cuts and to frontier on S. In particular, it proves positivity before any end is truncated.

Truncation of the other ends. On an unwanted end, the normal of the truncation sphere that points into the retained manifold is the inward radial normal. The given decay therefore gives

$$H = -3T^{-1} + O(T^{-1-q}), \qquad \mathrm{tr}_{S_T} K = O(T^{-1-q}).$$

The new boundary has strictly negative future expansion for large T. The other boundary components are the original ones. Removing the open tails leaves a connected manifold: a path entering a removed tail can have that portion replaced by a path along its connected spherical cross-section. The retained manifold is closed in M. It is complete for its intrinsic metric as well. Indeed, an intrinsic Cauchy sequence is ambient Cauchy and has a limit in the retained set; near a smooth boundary or an interior point, the intrinsic and ambient local distances are comparable in a half-ball or ball chart.

Every admissible outer domain in M_T, viewed as a submanifold of M, is an admissible outer domain for e. A part of its frontier on a truncation sphere becomes a hypersurface in the interior of M, and remains part of its intrinsic boundary. This proves the first inequality in (8.5). Conversely, a fixed original admissible domain has compact frontier and contains no tail of any unwanted end. It is consequently contained in M_T for all sufficiently large truncation radii and is then an admissible domain there. Applying this observation to cuts with area within any prescribed positive error of $A_e(S)$ proves the limit. The selected-end data have not been changed, so neither selected-end charge changes. □

It therefore suffices to prove the inequality for a one-ended exterior, with its boundary included and with the smooth-cut infimum of Definition 8.1.

8.3 Conformal changes and an energy-raising shell

Lemma 8.4 (Conformal constraint and boundary formulas)**.** *Let g be a smooth Riemannian metric and K a smooth symmetric covariant two-tensor on a smooth four-dimensional manifold M. Let μ, J be defined by (1.1), with the hypersurface expansion Θ_K of Section 8.1. Let $u > 0$ be smooth, and set $g_u = u^2 g$, $K_u = uK$. Then, as an identity of scalar functions and an identity of covectors, respectively,*

$$u^2 \mu_u = \mu - 3u^{-1}\Delta_g u, \tag{8.6}$$

$$uJ_u = J + 3K(\operatorname{grad}_g \log u, \cdot). \tag{8.7}$$

Consequently,

$$u^2(\mu_u - |J_u|_{g_u}) \geq \mu - |J|_g + 3u^{-1}(-\Delta_g u - |K|_g |\,\mathrm{d}u|_g). \tag{8.8}$$

For any consistently oriented hypersurface,

$$\Theta_{K_u} = u^{-1}(\Theta_K + 3\partial_\nu \log u). \tag{8.9}$$

Proof. With $n = 4$, $k = 3$, $p = 2$, and $s = \log u$, (3.2) has the same constraint normalization. At zero vector argument its first identity gives $u^2\mu_u = \mu - 3\Delta \log u - 3|d \log u|^2 = \mu - 3u^{-1}\Delta u$. The covector formula proved with that identity gives $uJ_u = J + 3K(\nabla \log u, \cdot)$; taking its g_u norm adds another factor u^{-1}. The triangle inequality then gives (8.8), equivalently the $U = u$ specialization of (3.3). The boundary formula reads $\Theta_{K_u} = u^{-1}(\Theta_K + 3\partial_\nu \log u)$, with the same chosen normal. These identities hold for smooth fields and $u > 0$, and hence apply under the original O_2/O_1 decay hypotheses. □

Lemma 8.5 (Prescribed energy increase in four dimensions)**.** *For any one-ended data in Theorem 8.2 and every $\lambda > 0$, there is a conformal change preserving all its hypotheses, with*

$$E_\lambda = E + 2\lambda, \qquad P_\lambda = P, \qquad A_e(g_\lambda) \geq A_e(g).$$

The conformal factor is constant near the compact boundary and is at least one everywhere.

Proof. Choose $0 < \xi < \min(q, 1)$. For $T(r) = r^{-2} - r^{-2-\xi}$, at sufficiently large radius,

$$T' < 0, \qquad -\Delta_g T - |K|_g |\,\mathrm{d}T|_g = \xi(2 + \xi)r^{-4-\xi} + O(r^{-4-q}) > 0.$$

Take a nondecreasing smooth switch β from zero to one in this region, constant on neighborhoods of its endpoints, and set

$$G(r) = -\int_r^\infty \beta(s)T'(s)\,\mathrm{d}s.$$

Extend G constantly throughout the interior. Then $G \geq 0$, it equals T sufficiently far out, and

$$-\Delta_g G - |K|_g |\,\mathrm{d}G|_g = \beta(-\Delta_g T - |K|_g |\,\mathrm{d}T|_g) - \beta' T' |\,\mathrm{d}r|_g^2 \geq 0.$$

For $u = 1 + \lambda G$, Lemma 8.4 and the preceding inequality preserve the dominant energy condition and the boundary expansion sign. The new densities are integrable: outside a compact set the additional terms in (8.6)–(8.7) are bounded by constants, depending on λ, times $r^{-4-\xi} + r^{-4-q}$. The new falloff exponent can be taken to be $\min(q, 2) > 1$.

The leading change in the metric flux numerator is $-6\partial_i(\lambda G)$. Products involving $g-\delta$, $u-1$, or their first derivatives have vanishing limiting flux. Hence (8.1) gives

$$E_\lambda - E = -\frac{\lambda}{\omega_3}\lim_{r\to\infty}\int_{S_r}\partial_r G\,\mathrm{d}A_\delta = 2\lambda.$$

The transformed momentum tensor is $u(K-\tau g)$. Its additional flux is $O(r^{-q})$, so $P_\lambda = P$ by (8.2). Since $u \geq 1$, every cut has at least its original area, proving the cut comparison. □

Once the inequality has been proved for $E > |P|$, the shell excludes the remaining ADM vectors, as shown in Proposition 8.10.

8.4 Matching the charges before making a rest end

The energy of the replacement end must approach $\sqrt{E^2-|P|^2}$. We first attach a vacuum reference plane carrying the original charges (E,P), then bend it to a static slice. Compact linear corrections reduce the interpolation deficit; a positive conformal change restores DEC while changing the mass by a quantity tending to zero.

8.4.1 The reference plane

Lemma 8.6 (Charges of the reference plane)**.** *For any $E > |P|$, put $m = (E^2-|P|^2)^{1/2}$. A spacelike plane in the Schwarzschild–Tangherlini spacetime of mass m, sufficiently far from its central region, has induced vacuum data (g_*, K_*) with charges exactly (E,P), and*

$$g_* - \delta = O_k(r^{-2}), \qquad K_* = O_k(r^{-3}) \quad \textit{for every integer } k \geq 0.$$

Here K_* is one half the metric rate in the future normal direction.

Proof. Lemma 3.2 applies with $n=4$, $k=3$, $p=2$, and $M=m$. Its isotropic spacetime metric is

$$\mathbf{g} = -\left(\frac{1-m/(2|y|^2)}{1+m/(2|y|^2)}\right)^2 \mathrm{d}b^2 + \left(1+\frac{m}{2|y|^2}\right)^2 \sum_{i=1}^{4}(\mathrm{d}y^i)^2. \tag{8.10}$$

Choose the first axis in the direction of P and put $v = |P|/E$, $\gamma = (1-v^2)^{-1/2}$. The Lorentz plane in that lemma has $E_* = m\gamma = E$ and $P_{*1} = m\gamma v = |P|$, with all transverse momenta zero. These are the present factors $1/(6\omega_3)$ and $1/(3\omega_3)$, with the same mass parameter. The leading energy flux is $6m\gamma^2 x_i \rho^{-4}$, the leading momentum tensor is the $k=3$ instance of (3.10), and the angular integral in (3.11) is ω_3/γ. Thus the sign and normalization are the same ones used here. The second form uses the same future-normal convention. Finally the metric and tensor remainders in that lemma are $O_d(r^{-4})$ and $O_d(r^{-5})$, with leading orders r^{-2} and r^{-3}, through every fixed derivative order. Gauss–Codazzi in the vacuum metric (8.10) proves that the reference data are vacuum. □

8.4.2 Compact inverses for the linear constraints

Localized constraint deformation and finite-dimensional adjoint obstructions have a substantial history [14, 13]. The compact operators below reduce the interpolation defect to a size that the subsequent conformal change can correct. The comparison of enclosing cuts follows from a separate metric estimate.

On Euclidean four-space write

$$\mathcal{L}b = \partial_i\partial_j(b_{ij} - (\operatorname{tr}_\delta b)\delta_{ij}), \qquad (\mathcal{D}k)_i = \partial_j(k_{ij} - (\operatorname{tr}_\delta k)\delta_{ij}). \tag{8.11}$$

These are the linearizations at $(\delta, 0)$ of 2μ and J. The next lemma identifies every compatibility condition needed for a compact correction.

Lemma 8.7 (Compact linear constraint corrections)**.** *Let $\mathcal{L}$ and $\mathcal{D}$ be the operators in (8.11). Fix a compact subannulus U_0 of a bounded connected Euclidean annulus $U \subset \mathbb{R}^4$. If f is a smooth scalar function and F a smooth covector field, both supported in U_0, and*

$$\begin{aligned} \int f\phi \, \mathrm{d}z = 0 \quad &\text{for every affine } \phi, \\ \int F_i Y^i \, \mathrm{d}z = 0 \quad &\text{for every Euclidean Killing field } Y, \end{aligned} \tag{8.12}$$

there are symmetric smooth tensors b, k, supported in a fixed compact subset of U, with $\mathcal{L}b = f$, $\mathcal{D}k = F$. For every integer $j \geq 0$ and every $1 < p_1 < \infty$, they can be chosen linearly in the sources so that

$$\|b\|_{W^{j+2,p_1}} \leq C\|f\|_{W^{j,p_1}}, \qquad \|k\|_{W^{j+1,p_1}} \leq C\|F\|_{W^{j,p_1}}. \tag{8.13}$$

The constant depends only on U_0, U, j, p_1 and fixed localization choices. The moment conditions are also necessary for compactly supported solutions.

Proof. Lemma 3.3 applies with $n = 4$ on the given connected annulus. Its scalar operator S is $\mathcal{L}$, and its vector operator D is $\mathcal{D}$. Its affine and Euclidean Killing moments are those in (8.12). The inverse trace reversal specializes to

$$b = S - \tfrac{1}{3}(\operatorname{tr}_\delta S)\delta, \tag{8.14}$$

so $b - (\operatorname{tr}_\delta b)\delta = S$. The scalar construction gains two Sobolev derivatives. The symmetric divergence construction first gains one derivative, uses a further divergence inverse to correct the skew part, and differentiates that correction once; its net gain is one. This proves (8.13), with support in a fixed larger compact subset and linear dependence on the sources. The commutator calculation in Proposition 8.8 will establish the needed source norms from the O_2/O_1 bounds of the initial data.

For necessity, integration by parts pairs $\mathcal{L}b$ with $\operatorname{Hess}\phi - (\Delta\phi)\delta$, which vanishes for affine ϕ, and pairs $\mathcal{D}k$ with the symmetric gradient of a Killing field after trace reversal. The latter also vanishes. Thus all and only the stated moments obstruct these compact inverses. □

8.5 Rest-end replacement and comparison of all cuts

Proposition 8.8 (Rest-end replacement)**.** *Let (M, g, K) satisfy the one-ended hypotheses of Theorem 8.2, and suppose $E > |P|$. Put $m = (E^2 - |P|^2)^{1/2}$. There is a sequence of smooth data (g_R, K_R) on the same manifold, complete with the same compact boundary, satisfying the dominant energy condition and the same future boundary inequality, such that:*

(i) *K_R is compactly supported, and the ultimate end is an exact static Schwarzschild–Tangherlini spatial end;*

(ii) *its energy is $m_R = m + o(1)$, its momentum is zero, and both constraint densities are integrable;*

(iii) $A_e(g_R) \geq (1 - o(1))A_e(g)$.

Every error here tends to zero as $R \to \infty$ for the fixed original data and fixed timelike vector.

Remark. The constants may depend on the timelike vector as $E - |P|$ tends to zero.

Proof. The reference end in Lemma 8.6 has the original charges. We join the data to this plane on $R < r < 4R$, then bend it to a static slice. Matching charges makes the compatibility moments small enough for the corrected interpolation to have a vanishing constraint deficit on the physical scale. A conformal change repairs that deficit, and a proper radial map gives the comparison for every enclosing cut. All constants below may depend on the original data and its timelike vector.

Scaled commutators. Fix $1 < p < \min(2, q)$, write $x = Rz$, and on $1 < |z| < 4$ use the scaled component fields

$$g(Rz), \qquad k(z) = RK(Rz),$$

and their reference counterparts. This corresponds to rescaling lengths by R^{-1}, so both constraint densities scale by R^2. The metric deviations and tensors have size $O(R^{-p})$ in C^2 and C^1, respectively. Expanding the constraint map about $(\delta, 0)$ gives its linear part $(\mathcal{L}, \mathcal{D})$ from (8.11), with remainder bounded in C^0 by CR^{-2p}. This follows directly from the scalar curvature formula: the nonlinear metric terms are bounded by $C(|b||D^2b|+|Db|^2)$; the other scalar terms are quadratic in k. The momentum remainder is bounded by $C(|b||Dk| + |Db||k|)$.

Choose a fixed smooth radial ψ, equal to one near $|z| = 1$ and zero near $|z| = 4$, with $0 \leq \psi \leq 1$. Blend the metric components and tensors by ψ. If b is the difference of the two scaled metric perturbations and k the difference of the two scaled tensors, the errors of the linear constraints relative to the blended sources are $f_R = [\mathcal{L}, \psi]b$ and $F_R = [\mathcal{D}, \psi]k$. For $T_{ij} = b_{ij} - (\operatorname{tr}_\delta b)\delta_{ij}$,

$$[\mathcal{L}, \psi]b = (\partial_i\partial_j\psi)T_{ij} + 2(\partial_i\psi)\partial_j T_{ij}, \tag{8.15}$$

$$([\mathcal{D}, \psi]k)_i = (\partial_j\psi)(k_{ij} - (\operatorname{tr}_\delta k)\delta_{ij}). \tag{8.16}$$

These expressions are supported in a fixed compact subannulus, and their C^1 norms are $O(R^{-p})$. In particular, the first bound uses only two metric derivatives, and the second only one tensor derivative.

Compatibility moments. We prove the sharper bound $o(R^{-2})$ for every moment in (8.12). Work momentarily in the physical coordinates, with

$$h = g - g_*, \quad T = K - (\operatorname{tr}_\delta K)\delta - K_* + (\operatorname{tr}_\delta K_*)\delta, \quad f = \mathcal{L}h, \quad F_i = \partial_j T_{ij}.$$

Both f and F belong to L^1 on the end. Indeed, the difference between the physical constraints and their Euclidean linear parts is $O(r^{-2-2\min(q,2)})$, which is integrable in dimension four. The physical densities are integrable by hypothesis and by reference vacuum, and Euclidean and physical measures and norms are uniformly comparable there.

An elementary consequence of integrability is

$$\int_{r_0<r<4R} r(|f| + |F|)\, \mathrm{d}x = o(R). \tag{8.17}$$

To see this, split at a fixed radius L. The integral up to L, divided by R, tends to zero. The remaining integral divided by R is at most four times the L^1 tail beyond L, which can be made arbitrarily small. Thus integrability alone gives the stated sublinear first-moment bound.

For an affine function ϕ, define the flux primitive

$$Q_{\phi,i} = \phi(\partial_j h_{ij} - \partial_i h_{jj}) - (\partial_j \phi) h_{ij} + (\partial_i \phi) h_{jj}.$$

Its divergence is ϕf. For a Killing field Y, symmetry of T gives $\partial_j(Y^i T_{ij}) = Y^i F_i$. Let $\mathfrak{F}_\phi(r)$ and $\mathfrak{G}_Y(r)$ be the corresponding outward Euclidean sphere fluxes. For constant tests, matching the ADM charges gives

$$\mathfrak{F}_1(r) = o(1), \qquad \mathfrak{G}_{e_i}(r) = o(1).$$

In the momentum flux the replacement of $\operatorname{tr}_g K\, g$ by $(\operatorname{tr}_\delta K)\delta$ changes the flux by at most $O(r^{2-2\min(q,2)}) = o(1)$. For homogeneous linear scalar tests and rotational fields, integration from a fixed sphere and (8.17) instead give

$$\mathfrak{F}_\phi(r) = o(r), \qquad \mathfrak{G}_Y(r) = o(r).$$

Put $\psi_R(x) = \psi(x/R)$, and let $A_R = \{R < r < 4R\}$. Integration by parts, with $\psi_R = 1$ near the inner sphere and zero near the outer sphere, gives

$$\int_{A_R} \phi[\mathcal{L}, \psi_R] h \, \mathrm{d}x = -\mathfrak{F}_\phi(R) - \int_{A_R} \psi_R \phi f \, \mathrm{d}x, \tag{8.18}$$

$$\int_{A_R} Y^i([\mathcal{D}, \psi_R](K - K_*))_i \, \mathrm{d}x = -\mathfrak{G}_Y(R) - \int_{A_R} \psi_R Y^i F_i \, \mathrm{d}x. \tag{8.19}$$

In scaled coordinates, constant moments have the factor R^{-2}: the source scales by R^2 and volume by R^{-4}. A linear test in z contributes the additional factor R^{-1}. Thus (8.18)–(8.19) and the preceding estimates show

$$\int f_R \phi \, \mathrm{d}z = o(R^{-2}), \qquad \int (F_R)_i Y^i \, \mathrm{d}z = o(R^{-2}) \tag{8.20}$$

for each element of a fixed basis of the affine functions or Killing fields.

Correction and the physical deficit. Choose a nonnegative smooth bump ζ, supported in a fixed ball within the annulus and positive on a smaller ball. For a basis of the affine tests, its Gram matrix $\int \zeta \phi_a \phi_b$ is positive definite. For a basis of Killing fields, the matrix $\int \zeta\, Y_a \cdot Y_b$ is likewise positive definite: a nonzero affine Killing field cannot vanish on an open ball. Subtract the corresponding bump combinations from f_R, F_R to erase their moments. By (8.20), the coefficients, and every fixed smooth norm of these bump corrections, are $o(R^{-2})$.

Lemma 8.7, applied to the negatives of the moment-free commutators, gives tensors whose addition to the blend cancels those commutators. With $j = 1$ and $p_1 > 4$, Sobolev embedding and (8.13) bound the metric correction in C^2 and the tensor correction in C^1 by CR^{-p}. Their supports stay in a fixed enlarged compact subannulus. The new metric is positive for sufficiently large R and agrees exactly with the original data inside R and the plane outside $4R$.

Let $(\widetilde{g}_R, \widetilde{K}_R)$ denote these physical data. Their scaled constraints equal ψ times the original scaled constraints, plus an error $o(R^{-2}) + O(R^{-2p})$; the reference constraints vanish. The change in the momentum norm contributes another $O(R^{-2p})$, because the metric change is $O(R^{-p})$ and the scaled momentum density is $O(R^{-p})$. Since $\psi \geq 0$, the original dominant energy condition implies, after undoing the scaling,

$$\widetilde{\mu}_R - |\widetilde{J}_R|_{\widetilde{g}_R} \geq -\eta_R R^{-4}, \qquad \eta_R \longrightarrow 0. \tag{8.21}$$

Indeed the remaining physical error is $R^{-2}[o(R^{-2}) + O(R^{-2p})] = o(R^{-4})$, since $p > 1$. The possible deficit is confined to a fixed closed subannulus on the scale R. Replace η_R by a positive majorant tending to zero if necessary.

Bending to a static slice. The bending takes place in the reference region. In its spatial coordinates y, write the plane as $b = v \cdot y$. Choose a smooth switch ψ_1 from one to zero on $[0, 1]$, constant outside that interval, and replace the plane farther out by the graph

$$b = (v \cdot y)\psi_1 \left(L^{-1} \log(|y|/R_b) \right). \tag{8.22}$$

Here R_b is a sufficiently large fixed multiple of R, so the graph initially lies wholly beyond the gluing annulus. For fixed $|v| < 1$, choose the fixed L so large that

$$|\operatorname{grad}_y b| \le |v|(1 + \|\psi_1'\|_\infty/L) < 1.$$

The Minkowski graph is uniformly spacelike. Since the perturbation (8.10) is $O(r^{-2})$, it remains uniformly spacelike in that metric for large R. It agrees with the plane on an open inner region and with $b = 0$ on an open outer region. These induced data are vacuum everywhere in the bend, so it adds no deficit.

In the fixed x chart, use $y = Ax$ where $A = (I - v \otimes v)^{-1/2}$, with the chosen rotation. The derivatives of (8.22) give $D^2 b = O(r^{-1})$, and the induced data, now denoted (g_R^0, K_R^0), satisfy on all transition regions

$$c\delta \le g_R^0 \le C\delta, \qquad |\partial g_R^0| + |K_R^0| \le C/r. \tag{8.23}$$

Here $c, C > 0$ may depend on the fixed boost and switches, but not on large R. Outside a fixed multiple of R, the data are static in the y chart and have tensor zero.

For the repair choose a smooth radius $\mathfrak{r}$, equal to $|x|$ through the gluing annulus and through the bend, and equal to $|y|$ farther out in the static region. It can be chosen with

$$\mathfrak{r} \asymp r, \qquad c \le |\,\mathrm{d}\mathfrak{r}|_{g_R^0} \le C, \qquad |\Delta_{g_R^0}\mathfrak{r}| \le C/r. \tag{8.24}$$

Here is an explicit way to make the interpolation. In the static region put $a(n) = |A^{-1}n|$, $n = y/|y|$, and interpolate

$$\log \mathfrak{r} = \log|y| + (1 - \chi(\log(|y|/R_c))) \log a(n).$$

Take R_c beyond the bend and make χ change from zero to one over a sufficiently long fixed logarithmic interval. Its slope can be made so small that the derivative of $\log \mathfrak{r}$ with respect to $\log|y|$ lies between $1/2$ and $3/2$. Angular derivatives are bounded, second spatial derivatives are $O(r^{-1})$, and (8.23) proves (8.24). All transition intervals are contained between fixed multiples of R.

Conformal repair. Write $s = \mathfrak{r}/R$. We construct a nonnegative function $F_R(s)$, constant on the inside, and use

$$u_R = 1 + \varepsilon_R R^{-2} F_R(s), \qquad \varepsilon_R = C_0 \eta_R.$$

Choose a fixed interval $[a, b_1]$ in the s variable that starts before the possible deficit, contains all transition regions, and ends where the radius is $|y|$ and the data are static. This interval lies in the end for all large R. Put $z_R = -F_R'$ and prescribe

$$z_R(a) = 0, \qquad z_R' = C_1 z_R + \omega_1, \tag{8.25}$$

where $\omega_1 \geq 0$ is smooth, vanishes on a neighborhood of a, and is at least one on a neighborhood of the possible deficit interval. Choose it to vanish near b_1. The solution is nonnegative and its norms on this fixed interval are bounded independently of large R.

Differentiating u_R and using (8.23) and (8.24) gives fixed constants $c', C' > 0$ such that

$$-\Delta_{g_R^0} u_R - |K_R^0||\,\mathrm{d}u_R|_{g_R^0} \geq \varepsilon_R R^{-4}(c' z_R' - C' z_R) \quad \text{on } a \leq s \leq b_1. \tag{8.26}$$

In fact the first positive term is $\varepsilon_R R^{-4} z_R' |\,\mathrm{d}\mathfrak{r}|^2$; the other terms are $\varepsilon_R R^{-3} z_R \Delta\mathfrak{r}$ and the negative tensor term, both bounded below by $-C\varepsilon_R R^{-4} z_R$ here. Choose $C_1 > C'/c'$. The ODE (8.25) then makes (8.26) nonnegative everywhere on this interval and at least $c'\varepsilon_R R^{-4}$ on the deficit region.

On the static region put $c_m = m/2$ and continue z_R so that

$$q_R(s) = s^3 \left(1 + \frac{c_m}{R^2 s^2}\right)^2 z_R(s)$$

is nondecreasing and becomes a positive constant $q_{R,\infty}$. This can be done smoothly with a uniform bound on $q_{R,\infty}$. Indeed, immediately past b_1, continue (8.25); both its derivative and that of the positive prefactor make q_R' nonnegative for large R. Multiply this nonnegative derivative by a smooth switch that equals one initially and zero after one more fixed interval, and integrate. This preserves all matching derivatives and keeps q_R nondecreasing. The radial Laplacian of the static metric then shows $-\Delta_{g_R^0} u_R \geq 0$ there, since its radial divergence is proportional to $-q_R'$; also $K_R^0 = 0$.

Set $F_R(s) = \int_s^\infty z_R(t)\,\mathrm{d}t$, extending it constantly before a. Its norm is bounded uniformly: the finite intervals have bounded length and data, while on the tail $z_R = O(s^{-3})$. Thus $1 \leq u_R \leq 2$ for large R. Choose the fixed C_0 large enough that $3u_R^{-1} c' \varepsilon_R \geq \eta_R$. In the conformal inequality of Lemma 8.4, the lower bound (8.26) then offsets the entire negative term in (8.21). Outside that region the correction is nonnegative. Thus the dominant energy condition holds everywhere for

$$g_R = u_R^2 g_R^0, \qquad K_R = u_R K_R^0.$$

The repair is constant in a neighborhood of the original boundary, and therefore preserves its expansion sign.

On the ultimate tail the integral for F_R is explicit:

$$u_R = 1 + \frac{a_R}{|y|^2 + c_m}, \qquad a_R = \tfrac{1}{2}\varepsilon_R q_{R,\infty} = O(\varepsilon_R) = o(1).$$

Consequently

$$g_R = \left(1 + \frac{c_m + a_R}{|y|^2}\right)^2 \delta, \qquad K_R = 0 \quad \text{on that tail.} \tag{8.27}$$

Its energy is $m + 2a_R = m + o(1)$, with zero momentum, by the direct static instance of Lemma 8.6. The densities vanish on this tail and are smooth on the remaining compact part, so they are integrable. The data are complete with boundary: a fixed compact core is smooth, and the end has a uniform positive metric lower bound for each R.

Comparison of enclosing cuts. The final metric dominates the original g for $r \leq R$, because the intermediate data there are the original ones and $u_R \geq 1$. For $r \geq R$, (8.23) and the static tail give $g_R \geq c\delta$ in the original x chart with a fixed $c > 0$. Choose $0 < \kappa < \min(1, \sqrt{c}/2)$. Let $L_R = R^{1/2}$,

and choose a smooth strictly increasing radial map ρ_R equal to r for $r \le L_R$, with derivative decreasing from one to κ on $[L_R, 2L_R]$, and derivative κ afterward. It defines a diffeomorphism

$$\Phi_R(x) = \rho_R(|x|)\,\frac{x}{|x|}$$

on the end, extended by the identity on the core. It is proper because $\rho_R(r) \to \infty$, preserves the selected end, and fixes the boundary. Its radial and tangential Euclidean dilations are ρ_R' and ρ_R/r. They are at most one everywhere, and for $r \ge R$ are at most $\kappa + O(R^{-1/2})$.

On the nonidentity region with $r \le R$, both r and $\rho_R(r)$ tend uniformly to infinity; the original metric is therefore $(1 + o(1))\delta$ at both points. The dilation bound and $g_R \ge g$ show $\Phi_R^* g \le (1 + o(1))g_R$ there. For $r \ge R$, the choice of κ, the same original-end decay, and $g_R \ge c\delta$ give this inequality as well. It is immediate on the identity core. We have thus proved the global tensor comparison

$$\Phi_R^* g \le (1 + \epsilon_R)g_R, \qquad \epsilon_R \longrightarrow 0. \tag{8.28}$$

Let Γ be the full boundary of any admissible outer domain for g_R. Its image under this proper diffeomorphism is the full boundary of an admissible original outer domain, with all coincident frontier retained. By (8.28),

$$|\Phi_R(\Gamma)|_g \le (1 + \epsilon_R)^{3/2}|\Gamma|_{g_R}.$$

Taking infima gives

$$A_e(g) \le (1 + \epsilon_R)^{3/2} A_e(g_R),$$

which proves the last assertion by a comparison valid for each admissible cut, whether or not the infimum is attained. □

8.6 Strict trapping and a positive energy margin

The rest-end replacement preserves weak trapping and the dominant energy condition. The prepared construction requires strict inequalities, which we obtain by a conformal change tending uniformly to the identity. Its ADM energy change is determined by a flux integral. The rest-end metric is fixed throughout this construction; in particular, its replacement radius R stays fixed.

The drift-Poisson equation used below is related to Jaracz's conformal strictification [28]. The proof here supplies the four-dimensional equation, its nonzero inner Neumann condition, and the charge and full-cut estimates needed for the numerical argument.

Lemma 8.9 (Strictification of a fixed rest end)**.** *Let (M, g, K) be a smooth connected four-dimensional exterior, complete with its nonempty compact smooth boundary included, with one Euclidean coordinate end and compact complement. Suppose $\mu \ge |J|_g$ and $\Theta_K \le 0$ on every boundary component, where the boundary normal ν points into M. Assume that K has compact support and that, outside a compact set,*

$$g = \left(1 + \frac{c}{r^2}\right)^2 \delta$$

in the end coordinates, for a constant c. Fix $0 < \delta_1 < 1$, and extend the coordinate radius to a positive smooth function r on M.

There is a nonnegative smooth function ϕ, with $\phi = O_k(r^{-2})$ for every fixed derivative order k, such that, for all sufficiently small $s > 0$, the data

$$(g_s, K_s) = (e^{2s\phi} g, e^{s\phi} K)$$

satisfy

$$\Theta_{K_s} < 0, \qquad \mu_s - |J_s|_{g_s} \geq c_s r^{-4-\delta_1} \quad \text{on } M, \qquad c_s > 0. \tag{8.29}$$

They are complete, have compactly supported second fundamental form, integrable constraint densities, zero ADM momentum and finite ADM energy. Their end satisfies

$$g_s - \delta = O_k(r^{-2}) \quad \text{for every fixed } k, \qquad R_{g_s} = O(r^{-4-\delta_1}). \tag{8.30}$$

For the full enclosing-cut infimum on this fixed exterior,

$$E_{g_s} \longrightarrow E_g, \qquad A_e(g_s) \longrightarrow A_e(g) \quad \text{as } s \downarrow 0. \tag{8.31}$$

Proof. We use the solution of Lemma 3.5 to make both geometric inequalities strict, then estimate the changes in energy and enclosing area from its flux and boundedness.

The conformal factor. The static tail has $g - \delta = O_d(r^{-2})$ for every d, and K is compactly supported. Choose any $1 < q_* < 2$ to express its O_6/O_5 bounds, set $n = 4$, $k = 3$, and take $\beta = \delta_1$. The condition $0 < \delta_1 < \min(1, q_*)$ is the one required by that lemma. Its smooth nonnegative majorant can be chosen compactly supported: write it as $b_0 \geq |K|_g$. The mixed-boundary existence and decay conclusions of that lemma therefore give a positive smooth solution of

$$-\Delta_g \phi = b_0 \sqrt{|\mathrm{d}\phi|_g^2 + r^{-6}} + r^{-4-\delta_1}, \qquad \partial_\nu \phi = -1, \qquad \phi \longrightarrow 0. \tag{8.32}$$

The exponent -6 is $-2k$. The all-order clause of the same lemma gives

$$|\partial^j \phi| \leq C_j r^{-2-j} \qquad (j \geq 0). \tag{8.33}$$

The higher differentiated bounds used here belong to the static tail produced by rest replacement; the original end retains its stated weak asymptotics.

Expansion and the energy condition. Put

$$F = b_0 \sqrt{|\mathrm{d}\phi|^2 + r^{-6}} + r^{-4-\delta_1}.$$

The inner outward normal is $-\nu$, and the finite-flux conclusion of the lemma is

$$L_\phi = \lim_{T \to \infty} \int_{S_T} \partial_r \phi \, \mathrm{d}A_\delta = -|\partial M|_g - \int_M F \, \mathrm{d}V_g < 0. \tag{8.34}$$

The geometric and Euclidean fluxes differ by $O(T^{-2})$ on the static tail, by (8.33).

With $u = e^{s\phi}$, Lemma 8.4 gives

$$e^{2s\phi}(\mu_s - |J_s|_{g_s}) \geq 3s\big(r^{-4-\delta_1} - s|\mathrm{d}\phi|^2\big).$$

The number $B = \sup_M r^{4+\delta_1}|\mathrm{d}\phi|^2$ is finite: smoothness controls the compact core and the tail is $O(r^{-2+\delta_1})$. Choose $sB \leq 1/2$. This proves the global margin with $c_s = (3s/2)e^{-2s\|\phi\|_\infty} > 0$. On each boundary component,

$$\Theta_{K_s} = e^{-s\phi}(\Theta_K + 3s\partial_\nu \phi) = e^{-s\phi}(\Theta_K - 3s) < 0.$$

Furthermore

$$g \leq g_s \leq e^{2s\|\phi\|_\infty} g. \tag{8.35}$$

Thus completeness with the boundary included is preserved, and the tensor remains compactly supported. On the tail, $R_g = 0$, $K_s = 0$ and

$$R_{g_s} = 6e^{-2s\phi}(sr^{-4-\delta_1} - s^2|\mathrm{d}\phi|^2) = O(r^{-4-\delta_1}).$$

This proves the scalar decay and integrability there; the rest is smooth and compact. All differentiated metric bounds follow from (8.33), and momentum is zero because the tensor vanishes on a distant tail.

Energy and enclosing area. The leading metric change is $2s\phi\delta + O_1(r^{-4})$, whose energy flux numerator is $-6s\partial_i\phi + O(r^{-5})$. The latter remainder has vanishing sphere flux. Division by $6\omega_3$ therefore gives

$$E_{g_s} - E_g = -\frac{s}{\omega_3}L_\phi \longrightarrow 0. \tag{8.36}$$

Every full cut has its three-dimensional area element multiplied by $e^{3s\phi}$; the cut class itself is unchanged. Taking infima yields

$$A_e(g) \le A_e(g_s) \le e^{3s\|\phi\|_\infty}A_e(g). \tag{8.37}$$

A distant coordinate sphere makes the infimum finite, so these bounds prove its convergence as $s \downarrow 0$, including when the infimum is unattained. If the datum is (g_R, K_R), every constant above may depend on R. □

8.7 Applying the prepared theorem and returning to the original end

The proper radial comparison and the conformal metric bounds allow the prepared inequality to pass through the two limits while retaining the original enclosing-area lower bound.

Proposition 8.10 (Transfer from prepared data)**.** *Theorem 2.3, applied in dimension four, implies Theorem 8.2.*

Proof. We first establish the bound for a timelike ADM vector on one end. The positive shell then excludes the other ADM vectors, and truncation reduces the general case to this one-ended result.

The timelike one-ended case. Let the original exterior have one end and $E > |P|$. Set $m = (E^2 - |P|^2)^{1/2} > 0$. Proposition 8.8 gives data (g_R, K_R) with zero momentum and energy $m_R \to m$. Its full-cut comparison is

$$A_e(g_R) \ge (1+\epsilon_R)^{-3/2}A_e(g), \qquad \epsilon_R \longrightarrow 0. \tag{8.38}$$

Fix one sufficiently large replacement radius R. For any fixed $0 < \delta_1 < 1$, Lemma 8.9 makes $(g_{R,s}, K_{R,s})$ a prepared exterior for all sufficiently small $s > 0$. Indeed it is the same smooth connected orientable one-ended exterior, complete with its compact boundary included and with compact complement of its coordinate end. It has finite ADM energy and a compactly supported tensor, so its ADM momentum is zero. On the end, $g_{R,s} - \delta = O_d(r^{-2})$ for every fixed derivative order d, its scalar curvature is $O(r^{-4-\delta_1})$, and its constraint densities are integrable. The boundary is strictly future trapped, and the global margin is

$$\mu_{R,s} - |J_{R,s}|_{g_{R,s}} \ge c_{R,s}\rho^{-4-\delta_1}, \qquad c_{R,s} > 0,$$

where ρ is a positive smooth extension of the rest-end radius. These are the geometric and decay conditions of Definition 2.1, with $n = 4$ and $\beta = \delta_1$.

The area condition follows along the same comparisons. In one end, the cut class of Definition 8.1 agrees with the prepared full-cut class: the outer domain is connected and closed, its interior lies in the open exterior, and its entire compact intrinsic boundary is counted, including every component and all coincidence with the original boundary. Lemma 8.3 gives $A_e(g) > 0$; the bound (8.38) and strictification give $A_e(g_{R,s}) \geq A_e(g_R) > 0$. The constraint and ADM normalizations also agree: $2\mu = R + (\operatorname{tr} K)^2 - |K|^2$, $J = \operatorname{div}(K - (\operatorname{tr} K)g)$, and the energy and momentum factors are $1/(6\omega_3)$ and $1/(3\omega_3)$, with the original scale unchanged.

Theorem 2.3 therefore yields, at every such fixed pair (R, s),

$$E_{R,s} \geq \frac{1}{2}\left(\frac{A_e(g_{R,s})}{\omega_3}\right)^{2/3}. \tag{8.39}$$

For this fixed prepared datum, the elliptic exhaustion and the limits $N \to \infty$ and $\epsilon \downarrow 0$ are part of Theorem 2.3. The resulting numerical inequality therefore holds at each fixed R and s, with estimates that may depend on both parameters.

Now let $s \downarrow 0$ with R fixed. Strictification gives both $E_{R,s} \to m_R$ and $A_e(g_{R,s}) \to A_e(g_R)$, so

$$m_R \geq \frac{1}{2}\left(\frac{A_e(g_R)}{\omega_3}\right)^{2/3} \geq \frac{1}{2}(1+\epsilon_R)^{-1}\left(\frac{A_e(g)}{\omega_3}\right)^{2/3}.$$

With strictification removed, letting $R \to \infty$ gives

$$\sqrt{E^2 - |P|^2} \geq \frac{1}{2}\left(\frac{A_e(g)}{\omega_3}\right)^{2/3} \qquad (E > |P|). \tag{8.40}$$

This passage uses the one-sided lower bound for $A_e(g_R)$.

Excluding nontimelike vectors. If $E \leq |P|$, take any sequence $a_j > 0$ tending to zero and set

$$\lambda_j = \frac{|P| + a_j - E}{2} > 0.$$

Lemma 8.5 gives, separately for each j, data of energy $|P| + a_j$, momentum P, and enclosing infimum at least $A_e(g)$. Their charges are future timelike, so (8.40) gives

$$\sqrt{(|P| + a_j)^2 - |P|^2} \geq \frac{1}{2}\left(\frac{A_e(g)}{\omega_3}\right)^{2/3} > 0.$$

The right side is fixed and positive by Lemma 8.3, while the left side tends to zero. This contradiction includes zero, negative, and null energy cases. The timelike numerical result is applied separately to each shell datum; its construction and constants may depend on that datum as the boost degenerates. Thus $E > |P|$ and (8.40) hold for every one-ended original datum in the theorem.

Several ends. Let the original manifold have finitely many ends and fix one end e. Truncate every other end at a sufficiently large sphere as in Lemma 8.3. The retained manifold is a complete connected one-ended exterior. Its added boundary components are strictly future trapped, its selected-end charges are still (E_e, P_e), and its full-cut infimum is at least $A_e(S)$. Applying the one-ended result gives

$$E_e \geq \sqrt{|P_e|^2 + \frac{1}{4}\left(\frac{A_e(S)}{\omega_3}\right)^{4/3}}.$$

Positivity of $A_e(S)$ was proved before any truncation. This gives both the asserted inequality and strict future timelikeness at e. □

The one-ended cases of Theorems 7.14 and 8.2 together prove Theorem 1.5.

8.8 The general maximal DEC energy inequality

For maximal data, the decay and integrability hypotheses already imply the existence of the momentum limit. The following end calculation supplies the limit needed to apply Theorem 8.2.

Lemma 8.11 (Existence of the momentum flux)**.** *On a smooth four-dimensional asymptotically flat end, suppose $g-\delta = O_2(r^{-q})$ and $K = O_1(r^{-1-q})$ for some $q>1$. If $|J|_g$ is integrable, where $J_i = \nabla^j(K_{ij} - (\operatorname{tr}_g K)g_{ij})$, then each ADM momentum limit in (8.2) exists and is finite.*

Proof. In the given coordinates set

$$\pi^g_{ij} = K_{ij} - (\operatorname{tr}_g K)g_{ij}, \qquad \pi^\delta_{ij} = K_{ij} - (\operatorname{tr}_\delta K)\delta_{ij}.$$

The differentiated decay assumptions give

$$\pi^g - \pi^\delta = O_1(r^{-1-2q}), \qquad \partial_j \pi^\delta_{ij} - J_i = O(r^{-2-2q}). \tag{8.41}$$

To verify the second bound, expand the covariant divergence as

$$J_i = g^{jk}(\partial_k \pi^g_{ij} - \Gamma^a_{ki}\pi^g_{aj} - \Gamma^a_{kj}\pi^g_{ia}).$$

Replacing g^{jk} by δ^{jk} produces an error of size $O(r^{-q})O(r^{-2-q})$; replacing π^g by π^δ produces the derivative of the error in the first bound in (8.41). Each connection term has size $O(r^{-1-q})O(r^{-1-q})$. These establish the asserted bounds.

Their Euclidean radial volume bound is Cr^{1-2q}, integrable because $q>1$. The asymptotic metric and Euclidean volume and covector norms are uniformly comparable, so $\partial_j\pi^\delta_{ij} \in L^1$. The Euclidean divergence theorem between two large spheres makes the flux $\int_{S_R} \pi^\delta_{ij} n^j_\delta \, \mathrm{d}A_\delta$ a Cauchy function of R, with finite limit. By the first bound in (8.41), its difference from the flux of π^g is $O(R^{2-2q}) \to 0$. This is the tensor appearing in (8.2), which proves the lemma. □

Corollary 8.12 (Maximal DEC energy inequality)**.** *Let (X^4, g_X, K_X) be smooth and connected, with X orientable, g_X complete as a metric space with its nonempty compact smooth boundary S_X included, and with exactly one end, which is asymptotically flat and diffeomorphic to the complement of a closed ball in $\mathbb{R}^4$. Let K_X be a smooth symmetric covariant two-tensor. Suppose*

$$\operatorname{tr}_{g_X} K_X = 0, \qquad \mu_X = \tfrac{1}{2}(R_{g_X} - |K_X|^2_{g_X}) \geq |\operatorname{div}_{g_X} K_X|_{g_X},$$
$$\mu_X, |\operatorname{div}_{g_X} K_X|_{g_X} \in L^1(X, \mathrm{d}V_{g_X}).$$

Assume for some $q>1$ that

$$g_X - \delta = O_2(r^{-q}), \qquad K_X = O_1(r^{-1-q}),$$

and that the ADM energy E_X in (8.1) is finite. On every component of S_X, require $H + \operatorname{tr}_{S_X} K_X \leq 0$ with the normal into X. Let e be the unique asymptotically flat end. For the enclosing-cut infimum $A_e(S_X)$ of Definition 8.1, computed with the metric g_X, one has

$$E_X \geq \frac{1}{2}\left(\frac{A_e(S_X)}{2\pi^2}\right)^{2/3}.$$

The boundary may have any topology and finitely many components.

Proof. Maximality identifies the momentum density with $J_X = \operatorname{div}_{g_X} K_X$. Lemma 8.11 supplies the finite ADM momentum. All hypotheses of Theorem 8.2 therefore hold for this one-ended exterior. Its invariant-mass inequality in particular gives the displayed energy bound. This deduction uses Theorem 2.3 through Theorem 8.2. □

References

[1] Shmuel Agmon, Avron Douglis, and Louis Nirenberg. Estimates near the boundary for solutions of elliptic partial differential equations satisfying general boundary conditions. I. *Communications on Pure and Applied Mathematics*, 12(4):623–727, November 1959. doi:10.1002/cpa.3160120405.

[2] Brian Allen, Edward Bryden, Demetre Kazaras, and Marcus Khuri. Proof of the spacetime Penrose inequality with suboptimal constant in the asymptotically flat and asymptotically hyperboloidal regimes. arXiv:2504.10641v1, 2025.

[3] Luigi Ambrosio, Nicola Fusco, and Diego Pallara. *Functions of Bounded Variation and Free Discontinuity Problems*. Oxford University Press, Oxford, 2000. doi:10.1093/oso/9780198502456.001.0001.

[4] Ishai Ben-Dov. Penrose inequality and apparent horizons. *Physical Review D*, 70(12):124031, 2004. arXiv:gr-qc/0408066. doi:10.1103/PhysRevD.70.124031.

[5] Yuchen Bi and Jintian Zhu. Riemannian Penrose inequality in all dimensions. arXiv:2605.00680v2, 2026.

[6] Hubert L. Bray. Proof of the Riemannian Penrose inequality using the positive mass theorem. *Journal of Differential Geometry*, 59(2):177–267, 2001. doi:10.4310/jdg/1090349428.

[7] Hubert L. Bray and Marcus A. Khuri. A Jang equation approach to the Penrose inequality. *Discrete and Continuous Dynamical Systems*, 27(2):741–766, 2010. arXiv:0910.4785. doi:10.3934/dcds.2010.27.741.

[8] Hubert L. Bray and Marcus A. Khuri. P.d.e.'s which imply the Penrose conjecture. *Asian Journal of Mathematics*, 15(4):559–612, 2011. arXiv:0905.2622. doi:10.4310/AJM.2011.v15.n4.a5.

[9] Hubert L. Bray and Dan A. Lee. On the Riemannian Penrose inequality in dimensions less than eight. *Duke Mathematical Journal*, 148(1):81–106, 2009. doi:10.1215/00127094-2009-020.

[10] Simon Brendle and Yipeng Wang. A dimension descent scheme for the positive mass theorem in arbitrary dimension. arXiv:2604.08473v2, 2026.

[11] Edward Bryden, Marcus Khuri, and Christina Sormani. Stability of the spacetime positive mass theorem in spherical symmetry. *The Journal of Geometric Analysis*, 31(4):4191–4239, 2021. doi:10.1007/s12220-020-00431-0.

[12] Alberto Carrasco and Marc Mars. A counter-example to a recent version of the Penrose conjecture. *Classical and Quantum Gravity*, 27:062001, 2010. arXiv:0911.0883. doi:10.1088/0264-9381/27/6/062001.

[13] Piotr T. Chruściel and Erwann Delay. On mapping properties of the general relativistic constraints operator in weighted function spaces, with applications. *Mémoires de la Société Mathématique de France*, 94:1–103, 2003. arXiv:gr-qc/0301073v2.

[14] Justin Corvino and Richard M. Schoen. On the asymptotics for the vacuum Einstein constraint equations. *Journal of Differential Geometry*, 73(2):185–217, 2006. arXiv:gr-qc/0301071. doi:10.4310/jdg/1146169910.

[15] Martin Costabel and Alan McIntosh. On Bogovskiĭ and regularized Poincaré integral operators for de Rham complexes on Lipschitz domains. *Mathematische Zeitschrift*, 265(2):297–320, 2010. arXiv:0808.2614v2. doi:10.1007/s00209-009-0517-8.

[16] Klaus Deimling. *Nonlinear Functional Analysis*. Springer-Verlag, Berlin, 1985. doi:10.1007/978-3-662-00547-7.

[17] Conghan Dong. The Penrose conjecture for initial data sets satisfying a 2-convexity condition. arXiv:2607.20741v2, 2026.

[18] Conghan Dong. The σ-inverse mean curvature flow and the generalized Penrose conjecture. arXiv:2605.26504v2, 2026.

[19] Ahmed Ellithy. The spacetime Penrose inequality under a quasi final state hypothesis. arXiv:2605.18730v1, 2026.

[20] Robert Geroch. Energy extraction. *Annals of the New York Academy of Sciences*, 224(1):108–117, 1973. doi:10.1111/j.1749-6632.1973.tb41445.x.

[21] David Gilbarg and Neil S. Trudinger. *Elliptic Partial Differential Equations of Second Order*. Classics in Mathematics. Springer, Berlin, second edition, 2001. Reprint of the 1998 edition. doi:10.1007/978-3-642-61798-0.

[22] Qing Han and Marcus Khuri. Existence and blow-up behavior for solutions of the generalized Jang equation. *Communications in Partial Differential Equations*, 38:2199–2237, 2013. arXiv:1206.0079v2.

[23] Sean A. Hayward. Gravitational energy in spherical symmetry. *Physical Review D*, 53(4):1938–1949, 1996. arXiv:gr-qc/9408002v2. doi:10.1103/PhysRevD.53.1938.

[24] Gerhard Huisken and Tom Ilmanen. The inverse mean curvature flow and the Riemannian Penrose inequality. *Journal of Differential Geometry*, 59(3):353–437, 2001. doi:10.4310/jdg/1090349447.

[25] Pong Soo Jang. On the positivity of energy in general relativity. *Journal of Mathematical Physics*, 19(5):1152–1155, 1978. doi:10.1063/1.523776.

[26] Pong Soo Jang and Robert M. Wald. The positive energy conjecture and the cosmic censor hypothesis. *Journal of Mathematical Physics*, 18(1):41–44, 1977. doi:10.1063/1.523134.

[27] Jaroslaw S. Jaracz. Nonexistence of solutions to the coupled generalized Jang equation/zero divergence system. *Classical and Quantum Gravity*, 40:195013, 2023. arXiv:2304.09332v1. doi:10.1088/1361-6382/acf17f.

[28] Jaroslaw S. Jaracz. Conformal deformations of initial data sets to the strict dominant energy condition using a spacetime Poisson equation. *International Journal of Geometric Methods in Modern Physics*, page 2550198, 2025. arXiv:2405.12249v1. doi:10.1142/S0219887825501981.

[29] Marcus Khuri and Hari Kunduri. The spacetime Penrose inequality for cohomogeneity one initial data. *Advances in Theoretical and Mathematical Physics*, 29(7):1905–1943, 2025. arXiv:2404.13247v1. doi:10.4310/ATMP.251120040338.

[30] Walter Littman, Guido Stampacchia, and Hans F. Weinberger. Regular points for elliptic equations with discontinuous coefficients. *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze*, 17(1–2):43–77, 1963.

[31] Edward Malec and Niall Ó Murchadha. Trapped surfaces and the Penrose inequality in spherically symmetric geometries. *Physical Review D*, 49(12):6931–6934, 1994. arXiv:gr-qc/9401014v1. doi:10.1103/PhysRevD.49.6931.

[32] Edward Malec and Niall Ó Murchadha. The Jang equation, apparent horizons, and the Penrose inequality. *Classical and Quantum Gravity*, 21:5777–5788, 2004. arXiv:gr-qc/0408044v2. doi:10.1088/0264-9381/21/24/007.

[33] Marc Mars. Present status of the Penrose inequality. *Classical and Quantum Gravity*, 26(19):193001, 2009. arXiv:0906.5566. doi:10.1088/0264-9381/26/19/193001.

[34] J. H. Michael and L. M. Simon. Sobolev and mean-value inequalities on generalized submanifolds of $\mathbb{R}^n$. *Communications on Pure and Applied Mathematics*, 26(3):361–379, 1973. doi:10.1002/cpa.3160260305.

[35] John Nash. The imbedding problem for Riemannian manifolds. *Annals of Mathematics*, 63(1):20–63, 1956. doi:10.2307/1969989.

[36] OpenAI. Boundary graph deformations for the spacetime Penrose inequality. Preprint, September 27, 2026, 2026.

[37] OpenAI. Conformal flow and the Riemannian Penrose inequality with minimizing frontiers. Preprint, September 27, 2026, 2026.

[38] OpenAI. Equality and rigidity in the spacetime Penrose inequality. Preprint, September 27, 2026, 2026.

[39] Roger Penrose. Naked singularities. *Annals of the New York Academy of Sciences*, 224(1):125–134, 1973. doi:10.1111/j.1749-6632.1973.tb41447.x.

[40] Richard Schoen and Shing-Tung Yau. On the proof of the positive mass conjecture in general relativity. *Communications in Mathematical Physics*, 65:45–76, 1979. doi:10.1007/BF01940959.

[41] Richard Schoen and Shing-Tung Yau. Proof of the positive mass theorem. II. *Communications in Mathematical Physics*, 79(2):231–260, 1981. doi:10.1007/BF01942062.

[42] Leon Simon. Introduction to geometric measure theory. `https://math.stanford.edu/~lms/ntu-gmt-text.pdf`, 2018. NTU lecture notes, version updated 7 March 2018.

[43] F. R. Tangherlini. Schwarzschild field in n dimensions and the dimensionality of space problem. *Il Nuovo Cimento*, 27:636–651, 1963. doi:10.1007/BF02784569.